# 复旦数学资料汇编
# Fourier 分析

来源：香蕉空间 Solution 资料库

2026 年 10 月 13 日

# 编者说明

本册按课程汇集原站讲义、笔记、习题及试卷，保持原资料内容。格式转换不构成数学证明审核。每篇保留来源网址；原站未建立的页面不编造正文。内容采用原站标注的 CC BY-SA 4.0 许可，署名与详细历史可由原网页查看。

# 目录

**第一章 Fourier 分析** **1**

1.1 解答: Fourier 分析上课习题 . . . . . 1

1.2 解答: GTM249 Chapter 1 Part I . . . . . 4

1.3 解答: GTM249 Chapter 1 Part II . . . . . 17

1.4 解答: GTM249 Chapter 2 Part I . . . . . 25

1.5 解答: GTM249 Chapter 2 Part II . . . . . 46

1.6 试卷: Fourier 分析 . . . . . 59

# 第一章　Fourier 分析

## 1.1　解答: Fourier 分析上课习题

原网页

1. 推广形式的 Hardy-Littlewood-Sobolev 不等式:
   设 $(X,\mu)$，$(Y,\nu)$ 为两个测度空间，$1<q<\infty$，$K(x,y)$ 为 $X\times Y$ 上的可测函数，且存在 $c>0$ s.t.

$$\|K(x,\cdot)\|_{L^{q,\infty}(Y,\nu)}\le c,\quad \text{a.e. } x\in X,$$

$$\|K(\cdot,y)\|_{L^{q,\infty}(X,\mu)}\le c,\quad \text{a.e. } y\in Y.$$

   若 $f\in L^p(Y,\nu)$（$1\le p<+\infty$），则

$$Tf(x)=\int_Y K(x,y)f(y)d\nu(y)\ \text{ 存在 a.e. } x\in X,$$

   且对于 $1<p<\infty$ 与 $1+\frac{1}{r}=\frac{1}{p}+\frac{1}{q}$，$\exists\, C(p,q,r,c)>0$ s.t.

$$\|Tf\|_{L^{r,\infty}(X,\mu)}\le C(p,q,r,c)\|f\|_{L^p(Y,\nu)},\ \ \forall f\in L^p(Y,\nu).$$

*Proof:* 分解 $K=K\chi_{|K|>\beta}+K\chi_{|K|\le\beta}:=K_1+K_2$，其中 $\beta>0$ 待定。
首先，估计

$$\|K_1(x,\cdot)\|_{L^1(Y,\nu)}=\int_0^\beta d_{K_1(x,\cdot)}(\alpha)d\alpha+\int_\beta^\infty d_{K_1(x,\cdot)}(\alpha)d\alpha\le\beta c^q\beta^{-q}+\int_\beta^\infty c^q\alpha^{-q}d\alpha=\frac{q}{q-1}c^q\beta^{1-q},$$

同理有

$$\|K_1(\cdot,y)\|_{L^1(X,\mu)}\le\frac{q}{q-1}c^q\beta^{1-q}.$$

于是

$$\begin{aligned}\Big|\int_Y K_1(x,y)f(y)d\nu\Big|&\le\int_Y|K_1(x,y)|^{\frac{1}{p'}}(|K_1(x,y)|^{\frac{1}{p}}|f(y)|)d\nu\quad \text{其中}\frac{1}{p}+\frac{1}{p'}=1\\&\le\Big(\int_Y|K_1(x,y)|d\nu\Big)^{\frac{1}{p'}}\Big(\int_Y|K_1(x,y)||f(y)|^pd\nu\Big)^{\frac{1}{p}}\\&\le\Big(\frac{q}{q-1}c^q\beta^{1-q}\Big)^{\frac{1}{p'}}\Big(\int_Y|K_1(x,y)||f(y)|^pd\nu\Big)^{\frac{1}{p}},\end{aligned}$$

由此得到

$$\begin{aligned}\Big\|\int_Y K_1(x,y)f(y)d\nu\Big\|_{L^p(X,\mu)} &\le \Big(\frac{q}{q-1}c^q\beta^{1-q}\Big)^{\frac{1}{p'}}\Big(\int_X\int_Y |K_1(x,y)||f(y)|^p d\nu d\mu\Big)^{\frac{1}{p}}\\ &= \Big(\frac{q}{q-1}c^q\beta^{1-q}\Big)^{\frac{1}{p'}}\Big(\int_Y\Big(\int_X |K_1(x,y)|d\mu\Big)|f(y)|^p d\nu\Big)^{\frac{1}{p}}\\ &\le \frac{q}{q-1}c^q\beta^{1-q}\|f\|_{L^p(Y,\nu)} \quad 因\frac{1}{p}+\frac{1}{p'}=1. \quad ①\end{aligned}$$

其次，估计

$$\|K_2(x,\cdot)\|^{p'}_{L^{p'}(Y,\nu)} = p'\int_0^\beta \alpha^{p'-1}d_{K_2(x,\cdot)}(\alpha)d\alpha \le p'\int_0^\beta \alpha^{p'-1-q}c^q d\alpha = \frac{p'}{p'-q}c^q\beta^{p'-q},$$

得到

$$\Big\|\int_Y K_2(x,y)f(y)d\nu\Big\|_{L^\infty(X,\mu)} \le \Big(\frac{p'}{p'-q}c^q\beta^{p'-q}\Big)^{\frac{1}{p'}}\|f\|_{L^p(Y,\nu)}. \quad ②$$

现在用 ①, ② 来估计 $d_{Tf}(\alpha)$。

由 $Tf(x)=\int_Y K_1(x,y)f(y)d\nu+\int_Y K_2(x,y)f(y)d\nu := T_1f(x)+T_2f(x)$ 得到

$$d_{Tf}(\alpha) \le d_{T_1f}(\alpha/2)+d_{T_2f}(\alpha/2).$$

取 $\beta$ s.t. $\big(\frac{p'}{p'-q}c^q\beta^{p'-q}\big)^{\frac{1}{p'}}\|f\|_{L^p}=\alpha/2$，则由 ② 得

$$d_{T_2f}(\alpha/2)=0.$$

于是

$$\begin{aligned}d_{Tf}(\alpha) &\le d_{T_1f}(\alpha/2)\\ &\le \big(\alpha/2\big)^{-p}\Big\|\int_Y K_1(x,y)f(y)d\nu\Big\|^p_{L^p(X,\mu)}\\ &\le \big(\alpha/2\big)^{-p}\Big(\frac{q}{q-1}c^q\beta^{1-q}\Big)^p\|f\|^p_{L^p} \quad 由①\\ &= C(p,q,c)\alpha^{-p}\big(\alpha\|f\|_{L^p}^{-1}\big)^{\frac{(1-q)pp'}{p'-q}}\|f\|^p_{L^p}\\ &= C(p,q,c)\big(\alpha^{-1}\|f\|_{L^p}\big)^r \quad 因 -p+\frac{(1-q)pp'}{p'-q}=-\frac{qp'}{p'-q}=-r,\end{aligned}$$

即

$$\|Tf\|_{L^{r,\infty}(X,\mu)} \le C(p,q,c)\|f\|_{L^p(Y,\nu)}.$$

注：更一般的结论见 Tao 的课程讲义 `https://terrytao.wordpress.com/2009/03/30/245c-notes-1-interpolation-of-lp-spaces/` 的 Lemma 43.

2. Let $1\le p_i\le q_i\le\infty$. Let $T$ be a quasi-linear operator defined on $L^{p_0}(X,\mu)+L^{p_1}(X,\mu)$ and taking values in the set of measurable functions on $(Y,\nu)$, where $K\ge 1$ is the constant that makes $|T(f+g)|\le K(|Tf|+|Tg|)$. Assume that $\|Tf\|_{L^{q_i,\infty}(Y,\nu)}\le A_i\|f\|_{L^{p_i}(X,\mu)}$ for some $A_i$, $i=0,1$. Fix $0<t<1$ and let
$$\frac{1}{p}=\frac{1-t}{p_0}+\frac{t}{p_1}, \quad \frac{1}{q}=\frac{1-t}{q_0}+\frac{t}{q_1}.$$
Then there exists a constant $C=C(p_0,p_1,q_0,q_1,K,A_0,A_1,t)$ such that
$$\|Tf\|_{L^q(Y,\nu)}\le C\|f\|_{L^p(X,\mu)}.$$

*Proof:* This is the lower triangular form of the Marcinkiewicz interpolation theorem. See Theorem 3.8 of the paper $Interpolation\ Theorems\ and\ Applications$ by Calista Bernard; or see Theorem 27 of Tao's note from the link given in the previous problem.

Note that in both of the references $T$ is assumed to be sublinear instead of quasi-linear, yet for the problem above, we have $d_{T(f+g)}(2K\alpha) \le d_{Tf}(\alpha) + d_{Tg}(\alpha)$ all the same.

3. A solution to the heat equation on $\mathbb{R}^n$

$$\begin{cases} (\frac{\partial}{\partial t} - \Delta)u(t,x) = 0, \\ \qquad\qquad u(0,x) = f(x) \end{cases}$$

is $e^{t\Delta}f = f * P_t$, where $P_t(x) = (4\pi t)^{-\frac{n}{2}} e^{-\frac{|x|^2}{4t}}$. Prove the following:

(a) $\|e^{t\Delta}\|_{L^p \to L^p} \le 1, \forall\ 1 \le p \le \infty$.

(b) $\|e^{t\Delta}\|_{L^1 \to L^\infty} \le C(n) t^{-\frac{n}{2}}, \forall\ t > 0$.

(c) $\|e^{t\Delta}\|_{L^p \to L^q} \le C(n) t^{-\frac{n}{2}(\frac{1}{p} - \frac{1}{q})}, \forall\ t > 0$.

*Proof:* (a) By Hölder's inequality,

$$\begin{aligned} |e^{t\Delta}f(x)| &= \Big| \int_{\mathbb{R}^n} f(x-y) P_t(y) \mathrm{d}y \Big| \\ &\le \big\| P_t^{\frac{p-1}{p}} \big\|_{L^{\frac{p}{p-1}}} \big\| f(x - \cdot) P_t(\cdot)^{\frac{1}{p}} \big\|_{L^p} = \Big( \int_{\mathbb{R}^n} |f(x-y)|^p P_t(y) \mathrm{d}y \Big)^{\frac{1}{p}} \end{aligned}$$

as $\|P_t\|_{L^1} = 1$. Then by Fubini's theorem,

$$\|e^{t\Delta}f(x)\|_{L^p} \le \Big( \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} |f(x-y)|^p P_t(y) \mathrm{d}y \mathrm{d}x \Big)^{\frac{1}{p}} = \Big( \int_{\mathbb{R}^n} \Big( \int_{\mathbb{R}^n} |f(x-y)|^p \mathrm{d}x \Big) P_t(y) \mathrm{d}y \Big)^{\frac{1}{p}} = \|f\|_{L^p}.$$

(b) For any $x$, we have $|e^{t\Delta}f| \le \|P_t\|_{L^\infty} \|f\|_{L^1} = (4\pi t)^{-\frac{n}{2}} \|f\|_{L^1}$.

(c) The result follows from combining part (a) and (b), using the Riesz-Thorin interpolation theorem. Still, we present a direct proof.

By Hölder's inequality,

$$\begin{aligned} |e^{t\Delta}f| &\le \int_{\mathbb{R}^n} |f(x-y)|^\alpha |f(x-y)|^{1-\alpha} P_t(y) \mathrm{d}y \\ &\le \big\| |f|^\alpha \big\|_{L^{\frac{p}{\alpha}}} \big\| |f(x-\cdot)|^{1-\alpha} P_t(\cdot) \big\|_{L^{\frac{p}{p-\alpha}}} = \|f\|_{L^p}^\alpha \Big( \int_{\mathbb{R}^n} \big( |f(x-y)|^{1-\alpha} P_t(y) \big)^{\frac{p}{p-\alpha}} \mathrm{d}y \Big)^{\frac{p-\alpha}{p}}. \end{aligned}$$

And by Hölder's inequality again,

$$\int_{\mathbb{R}^n} \big( |f(x-y)|^{1-\alpha} P_t(y) \big)^{\frac{p}{p-\alpha}} \mathrm{d}y \le \Big( \int_{\mathbb{R}^n} |f(x-y)|^{(1-\alpha)q} P_t(y)^{\frac{p}{p-\alpha}} \mathrm{d}y \Big)^{\frac{p}{(p-\alpha)q}} \Big( \int_{\mathbb{R}^n} P_t(y)^{\frac{p}{p-\alpha}} \mathrm{d}y \Big)^{1 - \frac{p}{(p-\alpha)q}}.$$

So we have

$$\begin{aligned} \|e^{t\Delta}f\|_{L^q} &\le \|f\|_{L^p}^\alpha \Big( \int_{\mathbb{R}^n} \Big( \int_{\mathbb{R}^n} \big( |f(x-y)|^{1-\alpha} P_t(y) \big)^{\frac{p}{p-\alpha}} \mathrm{d}y \Big)^{\frac{(p-\alpha)q}{p}} \mathrm{d}x \Big)^{\frac{1}{q}} \\ &\le \|f\|_{L^p}^\alpha \Big( \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} |f(x-y)|^{(1-\alpha)q} P_t(y)^{\frac{p}{p-\alpha}} \mathrm{d}y \mathrm{d}x \Big)^{\frac{1}{q}} \Big( \int_{\mathbb{R}^n} P_t(y)^{\frac{p}{p-\alpha}} \mathrm{d}y \Big)^{\frac{p-\alpha}{p} - \frac{1}{q}}. \end{aligned}$$

Taking $\alpha = 1 - \frac{p}{q}$, we see that

$$\|e^{t\Delta}f\|_{L^q} \le \|f\|_{L^p} \Big( \int_{\mathbb{R}^n} P_t(y)^{\frac{1}{1 - \frac{1}{p} + \frac{1}{q}}} \mathrm{d}y \Big)^{1 - \frac{1}{p} + \frac{1}{q}} = C(n) t^{-\frac{n}{2}(\frac{1}{p} - \frac{1}{q})} \|f\|_{L^p}.$$

4. For a semipositive matrix $A \in \mathbb{R}_{n\times n}$, define the function $F_A(x) = e^{-\langle Ax,x\rangle}$ on $\mathbb{R}^n$, which is bounded. Compute $\widehat{F_A}$.

*Proof:* For $\varepsilon > 0$, the matrix $A+\varepsilon I$ is positive, hence nondegenerate. Then for all $f \in \mathscr{S}(\mathbb{R}^n)$, we have

$$\begin{aligned}
&\langle \widehat{F_A}, f\rangle \\
&= \int_{\mathbb{R}^n} F_A(x)\hat{f}(x)\mathrm{d}x \\
&= \lim_{\varepsilon\to 0^+} \int_{\mathbb{R}^n} e^{-\langle (A+\varepsilon I)x,x\rangle}\hat{f}(x)\mathrm{d}x \\
&= \lim_{\varepsilon\to 0^+} \int_{\mathbb{R}^n} \left(e^{-\langle (A+\varepsilon I)x,x\rangle}\right)^{\wedge} f(x)\mathrm{d}x \\
&= \lim_{\varepsilon\to 0^+} \int_{\mathbb{R}^n} \pi^{\frac{n}{2}} \det(A+\varepsilon I)^{-\frac{1}{2}} e^{-\pi^2\langle (A+\varepsilon I)^{-1}x,x\rangle} f(x)\mathrm{d}x \\
&= \lim_{\varepsilon\to 0^+} \int_{\mathbb{R}^n} \pi^{\frac{n}{2}} e^{-\pi^2|x|^2} f((A+\varepsilon I)^{\frac{1}{2}}x)\mathrm{d}x \\
&= \int_{\mathbb{R}^n} \pi^{\frac{n}{2}} e^{-\pi^2|x|^2} f(A^{\frac{1}{2}}x)\mathrm{d}x.
\end{aligned}$$

The second equality above is from the Lebesgue dominated convergence theorem, and the fifth from substituting $x$ with $(A+\varepsilon I)^{\frac{1}{2}}x$. To verify the fourth, it suffices to consider the case when $A+\varepsilon I$ is a diagonal matrix after a rotation of $x \in \mathbb{R}^n$, reducing matters to the multiplication of the $\mathbb{R}^1$ case: to be specific, the computation is

$$\begin{aligned}
(e^{-\langle (A+\varepsilon I)x,x\rangle})^{\wedge} = (e^{-\langle D(Px),Px\rangle})^{\wedge} &= (e^{-\lambda_1 x_1^2-\cdots-\lambda_n x_n^2})^{\wedge}\circ P \\
&= \pi^{\frac{n}{2}}(\lambda_1\cdots\lambda_n)^{-\frac{1}{2}} e^{-\pi^2(\lambda_1^{-1}x_1^2+\cdots+\lambda_n^{-1}x_n^2)}\circ P \\
&= \pi^{\frac{n}{2}}\det(A+\varepsilon I)^{-\frac{1}{2}} e^{-\pi^2\langle D^{-1}x,x\rangle}\circ P \\
&= \pi^{\frac{n}{2}}\det(A+\varepsilon I)^{-\frac{1}{2}} e^{-\pi^2\langle (A+\varepsilon I)^{-1}x,x\rangle},
\end{aligned}$$

where $A+\varepsilon I = P^T D P$ is the decomposition such that $P$ is orthogonal and $D$ is the diagonal matrix $\mathrm{diag}(\lambda_1,\cdots,\lambda_n)$.
Thus $\widehat{F_A}(x) = \pi^{\frac{n}{2}} e^{-\pi^2x^2}\delta^{A^{\frac{1}{2}}}$, where $\delta^{A^{\frac{1}{2}}} f = f\circ A^{\frac{1}{2}}$.

## 1.2 解答: GTM249 Chapter 1 Part I

原网页

Errata for GTM 249, 250.

### 1.1 $L^p$ and Weak $L^p$

1.1.1 (a) For an arbitrary sequence $\{\alpha_n\}\downarrow\alpha$, we have the pointwise convergence $\chi_{|f|>\alpha_n}\downarrow\chi_{|f|>\alpha}$. If $d_f(\alpha)$ is unbounded, then the proposition holds automatically. Otherwise, $\chi_{|f|>\alpha_n}$ are all bounded by $\chi_{|f|>\alpha}$, thus

$$d_f(\alpha) = \int \chi_{|f|>\alpha}\,d\mu \overset{\text{DCT}}{=} \lim_{n\to\infty}\int \chi_{|f|>\alpha_n}d\mu = \lim_{n\to\infty} d_f(\alpha_n).$$

(b) Let $E = \{|f| > \alpha\}$ and $E_n = \{|f_n| > \alpha\}$. Thus by noting that $\mu(\bigcap_{n=m}^{\infty} E_n) \le \liminf_{n\to\infty} \mu(E_n)$ and $E \subset \bigcup_{m=1}^{\infty} \bigcap_{n=m}^{\infty} E_n$, $\mu$-a.e., we have $d_f \le \liminf_{n\to\infty} d_{f_n}$ by definition.

(c) We have $\limsup_{n\to\infty} d_{f_n} \le d_f$ by $|f_n| \uparrow |f|$. Now the result comes with (b) that

$$\limsup_{n\to\infty} d_{f_n} \le d_f \le \liminf_{n\to\infty} d_{f_n}.$$

1.1.2. (a) Use induction. Suppose this holds for the $k-1$ case, and consider the $k$ case. If $p_k = \infty$, then since $\sum_{j=1}^{k-1} \frac{1}{p_j} = \frac{1}{p}$, we have by pulling out the essential supremum of $|f_k|$ that

$$\begin{aligned} \|f_1 \cdots f_k\|_{L^p} &\le \|f_1 \cdots f_{k-1}\|_{L^p} \|f_k\|_{L^\infty} \\ &\le \|f_1\|_{L^{p_1}} \cdots \|f_{k-1}\|_{L^{p_{k-1}}} \|f_k\|_{L^\infty}. \end{aligned}$$

Otherwise, if $p_k < \infty$, then automatically $p < \infty$. Set $r = \dfrac{p_k}{p_k - p}$ and $q = \dfrac{p_k}{p}$. Then

$$\begin{aligned} \|f_1 \cdots f_k\|_{L^p} &= \| |f_1 \cdots f_k|^p \|_{L^1}^{\frac{1}{p}} \\ &\overset{\text{Hölder}}{\le} \| |f_1 \cdots f_{k-1}|^p \|_{L^r}^{\frac{1}{p}} \| |f_k|^p \|_{L^q}^{\frac{1}{p}} \\ &= \|f_1 \cdots f_{k-1}\|_{L^{pr}} \|f_k\|_{L^{pq}} \\ &\le \|f_1\|_{L^{p_1}} \cdots \|f_{k-1}\|_{L^{p_{k-1}}} \|f_k\|_{L^{p_k}}. \end{aligned}$$

Also, note that the $k = 2$ case is contained in the proof.

(b) The equality condition is the same as the one of Hölder's inequality in the $k = 2$ case.

(c) It suffices to show that $\|fg\|_{L^1} \|g^{-1}\|_{L^{\frac{q}{1-q}}} \ge \|f\|_{L^q}$, which is straightforward by (a).

1.1.3. (a) If $\|f\|_{L^\infty} = \infty$, then define $E_n = \{|f| > n\}$. Since $\mu(E_n) > 0$ for each $n$, we have $\|f\|_{L^p} \ge \|f\|_{L^p(E_n)} \ge n\mu(E_n)^{\frac{1}{p}}$. Let $p \to \infty$ to get $\liminf_{p\to\infty} \|f\|_{L^p} \ge n$, then let $n$ vary.

If $\|f\|_{L^\infty} < \infty$, now $\|f\|_{L^p} \le \|f\|_{L^\infty}^{\frac{p-p_0}{p}} \|f\|_{L^{p_0}}^{\frac{p_0}{p}}$. Let $p \to \infty$, we obtain $\limsup_{p\to\infty} \|f\|_{L^p} \le \|f\|_{L^\infty}$. For another, set $E_\gamma = \{|f| > \gamma \|f\|_{L^\infty}\}$ for arbitrary $\gamma \in (0,1)$. Then $\mu(E_\gamma) > 0$ and thus $\|f\|_{L^{p_0}(E_\gamma)} > 0$, and

$$\|f\|_{L^p} \ge (\gamma \|f\|_{L^\infty})^{\frac{p-p_0}{p}} \|f\|_{L^{p_0}(E_\gamma)}^{\frac{p_0}{p}}.$$

Again let $p \to \infty$, we get $\liminf_{p\to\infty} \|f\|_{L^p} \ge \gamma \|f\|_{L^\infty}$. Now let $\gamma \to 1$.

(b) By Jensen's inequality, we have

$$\int_X \log |f| \, d\mu \le \log \left( \int_X |f| \, d\mu \right).$$

This yields

$$\begin{aligned} \|f\|_{L^p} &= \left( \int_X |f|^p \, d\mu \right)^{\frac{1}{p}} \\ &\ge \left( \exp \left( \int_X p \log |f| \, d\mu \right) \right)^{\frac{1}{p}} \\ &= \exp \left( \int_X \log |f| \, d\mu \right). \end{aligned}$$

(c) Fix a sequence $\{p_n\} \downarrow 0$ bounded by $p_0$ and define

$$h_n(x) = \frac{1}{p_0}(|f(x)|^{p_0} - 1) - \frac{1}{p_n}(|f(x)|^{p_n} - 1).$$

Using that $\int_1^t s^{p-1} ds = \frac{1}{p}(t^p - 1) \downarrow \log t$ as $p \downarrow 0$, the Lebesgue monotone convergence theorem yields $\int_X h_n \, d\mu \uparrow \int_X h \, d\mu$, hence

$$\int_X \frac{1}{p_n}(|f(x)|^{p_n} - 1) \, d\mu \downarrow \int_X \log |f| \, d\mu.$$

Using part (b) and taking $t = \int_X |f|^{p_n} \, d\mu$ in the inequality $t \le e^{t-1}$, we get

$$\begin{aligned} \exp\left( \int_X \log |f| \, d\mu \right) &\le \left( \int_X |f|^{p_n} \, d\mu \right)^{\frac{1}{p_n}} \\ &\le \exp\left( \int_X \frac{1}{p_n}(|f|^{p_n} - 1) \, d\mu \right). \end{aligned}$$

Thus take $n \to \infty$ and we have $\lim\limits_{p \to 0} \|f\|_{L^p} = \exp\left( \int_X \log |f| \, d\mu \right)$.

1.1.4. (a) Use Jensen's inequality of integration to

$$f(x) = \sum_{k=1}^{\infty} a_k \chi_{[k-1,k)}$$

and $X = \mathbb{R} - (-\infty, 0]$, since $x^\theta$ is convex for $\theta \in (0, 1]$.

Another way is to assume that $1 = \left( \sum_{j=1}^{\infty} a_j \right)^\theta$ by homogeneity, which implies $a_j \le 1$, and subsequently

$$1 = \sum_{j=1}^{\infty} a_j \le \sum_{j=1}^{\infty} a_j^\theta.$$

(b) This is the same as the preceding question with a change in convexity. Or observe that

$$\left( \sum_{j=1}^{\infty} a_j^\theta \right)^{1/\theta} \le \sum_{j=1}^{\infty} (a_j^\theta)^{1/\theta} = \sum_{j=1}^{\infty} a_j.$$

(c) Use the discrete Jensen's inequality. The problem is the same as

$$\left( \frac{\sum_{j=1}^N a_j}{N} \right)^\theta \le \frac{1}{N} \sum_{j=1}^{N} a_j^\theta.$$

(d) The same as (c).

1.1.5. (a) Since $p \geq 1$, we have

$$\begin{aligned}
\|\sum_{j=1}^N f_j\|_{L^p}^p &= \int_X |\sum_{j=1}^N f_j|^p d\mu \\
&= \int_X |\sum_{j=1}^N f_j| |\sum_{j=1}^N f_j|^{p-1} d\mu \\
&\leq \sum_{i=1}^N \int_X |f_i| |\sum_{j=1}^N f_j|^{p-1} d\mu \\
&\overset{\text{Hölder}}{\leq} \left(\sum_{i=1}^N \|f_i\|_{L^p}\right) \left(\int_X |\sum_{j=1}^N f_i|^p d\mu\right)^{\frac{p-1}{p}} \\
&= \sum_{i=1}^N \|f_i\|_{L^p} \frac{\|\sum_{j=1}^N f_j\|_{L^p}^p}{\|\sum_{j=1}^N f_j\|_{L^p}}.
\end{aligned}$$

The rest is by multiplying both sides by $\|\sum_{j=1}^N f_j\|_{L^p}^{1-p}$.

(b) This is just the same with (a). Here the first "$\leq$" turns to "$=$" and the Hölder's inequality turns its direction.

(c) This is direct from 1.1.4 (a) pointwise on $X$, and subsequently 1.1.4 (c) where $a_j = \int_X |f_j|^p \, d\mu$.

(d) Take $n$ disjoint sets $M_j$ with the same measure $\alpha > 0$ and take $f_j = \chi_{M_j}$. Then $\|\sum_{j=1}^N f_j\|_{L^p} = N^{\frac{1}{p}} \alpha^{\frac{1}{p}}$ and $\sum_{j=1}^N \|f_j\|_{L^p} = N\alpha^{\frac{1}{p}}$.

1.1.6. (a) The $p = 1$ case is trivial. When $p > 1$, let $q$ be the conjugate number of $p$, then set $G(t) = \int_X F(x,t) \, d\mu(x)$, we have

$$\begin{aligned}
\|G\|_{L^p(\nu)}^p &= \int_T \left(\int_X F(x,t) \, d\mu(x)\right) G(t)^{p-1} d\nu(t) \\
&= \int_X \int_T F(x,t) G(t)^{p-1} d\nu(t) d\mu(x) \\
&\overset{\text{Hölder}}{\leq} \int_X \left(\int_T F(x,t)^p \, d\nu(t)\right)^{\frac{1}{p}} \|G\|_{L^p(\nu)}^{p-1} d\mu(x).
\end{aligned}$$

Now eliminating $\|G\|_{L^p(\nu)}^{p-1}$ on both sides finishes the proof when $\|G\|_{L^p(\nu)} < \infty$. Otherwise, choose ascending sequences of subsets $X_n \subset X$ and $T_n \subset T$, such that $\bigcup_n X_n = X$ and $\bigcup_n T_n = T$, and $\mu(X_n), \nu(T_n) < \infty$ for all $n$. Now take $F_k = \max(F, k)$. Thence

$$\left[\int_{T_m} \left(\int_{X_n} F_k(x,t) \, d\mu(x)\right)^p d\nu(t)\right]^{\frac{1}{p}} \leq \int_{X_n} \left[\int_{T_m} F(x,t)^p \, d\nu(t)\right]^{\frac{1}{p}} d\mu(x).$$

for all $m, n$ and $k \in \mathbb{N}$. By monotonicity, we obtain the desired result by taking $m, n \to \infty$ and then $k$.

(b) Let $(X, \mu)$ be a $\sigma$-finite measure space, and $(T, \nu)$ be a finite measure space. For every non-negative measurable function $F$ on the product space $(X, \mu) \times (T, \nu)$ we have

$$\left\| \int_X F(x,t) \, d\mu(x) \right\|_{L^\infty(T,\nu)} \leq \int_X \|F(x,t)\|_{L^\infty(T,\nu)} \, d\mu(x).$$

For the proof, take $p \to \infty$ in (a), and $\nu(T) < \infty$ guarantees that $\lim_{p\to\infty} \|\cdot\|_{L^p(T,\nu)} = \|\cdot\|_{L^\infty(T,\nu)}$.

(c) Since Hölder's inequality reverses when $0 < p < 1$.

(d) $(X, \mu)$ is not $\sigma$-finite now, and (a) as well as (b) no longer holds for the $F$. Indeed, $\int_X F(x,t)\,d\mu(x) = \mu(\{t\}) = 1$ for any $t \in T$, and therefore the left–hand side equals $\Big(\int_T d\nu(t)\Big)^{\frac{1}{p}} = 1$; $\int_T F(x,t)^p\,d\nu(t) = 1$ for any $x \in X$, and therefore the right–hand side equals $\int_X 0\,d\mu(x) = 0$.

1.1.7. (a) Since

$$\begin{aligned}\|\sum_{j=1}^N f_j\|_{L^{p,\infty}} &= \sup\{\gamma d_{f_1+\cdots+f_N}(\gamma)^{\frac{1}{p}} : \gamma > 0\}\\ &\le \sum_{j=1}^N \sup\{\gamma d_{f_j}\Big(\frac{\gamma}{N}\Big)^{\frac{1}{p}} : \gamma > 0\}\\ &= N\sum_{j=1}^N \sup\{\gamma\, d_{f_j}(\gamma)^{\frac{1}{p}} : \gamma > 0\} = N\sum_{j=1}^N \|f_j\|_{L^{p,\infty}}.\end{aligned}$$

(b) Since

$$\begin{aligned}\|\sum_{j=1}^N f_j\|_{L^{p,\infty}} &= \sup\{\gamma d_{f_1+\cdots+f_N}(\alpha)^{\frac{1}{p}} : \gamma > 0\}\\ &\le N^{\frac{1}{p}-1}\sum_{j=1}^N \sup\{\gamma d_{f_j}\Big(\frac{\gamma}{N}\Big)^{\frac{1}{p}} : \gamma > 0\}\\ &= N^{\frac{1}{p}}\sum_{j=1}^N \sup\{\gamma d_{f_j}(\gamma)^{\frac{1}{p}} : \gamma > 0\} = N^{\frac{1}{p}}\sum_{j=1}^N \|f_j\|_{L^{p,\infty}}.\end{aligned}$$

1.1.8. Suppose $\{f_n\}$ is an $L^p$ Cauchy sequence. Let $\{n_i\}$ be a sequence such that $\|f_{n_{i+1}} - f_{n_i}\|_{L^p} \le \dfrac{1}{2^i}$. Thus the series $f = f_{n_1} + \sum_{i=1}^\infty (f_{n_{i+1}} - f_{n_i})$ converges in $L^p$. The next thing is to verify that $f_n \xrightarrow{L^p} f$. This is because for all $\varepsilon > 0$, take $n, n_k$ sufficiently large such that $\|f_n - f_{n_k}\|_{L^p} < \varepsilon$, and then

$$\begin{aligned}\int_X |f_n - f|^p\,d\mu &= \int_X \liminf_{k\to\infty} |f_n - f_{n_k}|^p\,d\mu\\ &\overset{\text{Fatou}}{\le} \liminf_{k\to\infty}\int_X |f_n - f_{n_k}|^p\,d\mu\\ &= \liminf_{k\to\infty}\|f_n - f_{n_k}\|_{L^p}^p < \varepsilon^p,\end{aligned}$$

which implies $\|f_n - f\|_{L^p} < \varepsilon$. Thus we have shown the completeness.

1.1.9. Note that for $\varepsilon > 0$,

$$\{x \in X \mid f_n(x) \to f(x)\} \subseteqq \bigcup_{m=1}^\infty \bigcap_{n=m}^\infty \{x \in X \mid |f_n(x) - f(x)| < \varepsilon\} \subseteqq X,$$

in which both sides have measure $\mu(X)$. Since

$$\bigcap_{n=m}^\infty \{x \in X \mid |f_n(x) - f(x)| < \varepsilon\} \subseteqq \bigcap_{n=m+1}^\infty \{x \in X \mid |f_n(x) - f(x)| < \varepsilon\},$$

we have

$$\bigcap_{n=N}^\infty \{x \in X \mid |f_n(x) - f(x)| < \varepsilon\} > \mu(X) - \varepsilon$$

for some large $N$, thus

$$\mu(\{x \in X \mid |f_n(x) - f(x)| < \varepsilon\}) < \varepsilon$$

for $n > N$.

1.1.10. (a) This is due to

$$d_{f_\gamma}(\alpha) = \mu(\{|f| > \gamma\} \cap \{|f| > \alpha\}) = d_f(\max(\alpha, \gamma)).$$

$$d_{f^\gamma}(\alpha) = d_f(\alpha) - d_{f_\gamma}(\alpha) = \begin{cases} 0, & \alpha \geq \gamma \\ d_f(\alpha) - d_f(\gamma), & \alpha < \gamma \end{cases}.$$

(b) Using (a) we obtain

$$\begin{aligned} \|f_\gamma\|_{L^p}^p &= p \int_0^\infty \alpha^{p-1} d_f(\max(\alpha, \gamma))\, d\alpha \\ &= p\Big( \int_\gamma^\infty \alpha^{p-1} d_f(\alpha)\, d\alpha + \int_0^\gamma \alpha^{p-1} d_f(\gamma)\, d\alpha \Big) \\ &= p \int_\gamma^\infty \alpha^{p-1} d_f(\alpha)\, d\alpha + \gamma^p d_f(\gamma). \\ \|f^\gamma\|_{L^p}^p &= p \int_0^\gamma \alpha^{p-1} (d_f(\alpha) - d_f(\gamma))\, d\alpha \\ &= p \int_0^\gamma \alpha^{p-1} d_f(\alpha)\, d\alpha - \gamma^p d_f(\gamma). \end{aligned}$$

$$\begin{aligned} &\int_{\gamma < |f| \leq \delta} |f|^p d\mu = \|f^\delta\|_{L^p}^p - \|f^\gamma\|_{L^p}^p \\ &= p \int_\gamma^\delta \alpha^{p-1} d_f(\alpha)\, d\alpha - \delta^p d_f(\delta) + \gamma^p d_f(\gamma). \end{aligned}$$

(c) For $q > p$, from (b) and $q - p - 1 > -1$ we get

$$\begin{aligned} \|f^\gamma\|_{L^q}^q &= q \int_0^\gamma \alpha^{q-1} d_f(\alpha)\, d\alpha - \gamma^q d_f(\gamma) \\ &= q^p \int_0^\gamma \alpha^{q-p-1} \|f\|_{L^{p,\infty}}^p d\alpha - \gamma^q d_f(\gamma) < \infty. \end{aligned}$$

Plus, for $q < p$, from (b) and $q - p - 1 < -1$ we get

$$\begin{aligned} \|f_\gamma\|_{L^q}^q &= q \int_\gamma^\infty \alpha^{q-1} d_f(\alpha)\, d\alpha + \gamma^q d_f(\gamma) \\ &\leq q \int_\gamma^\infty \alpha^{q-p-1} \|f\|_{L^{p,\infty}}^p d\alpha + \gamma^q d_f(\gamma) < \infty. \end{aligned}$$

Finally, $L^{p,\infty} \subsetneqq L^{p_0} + L^{p_1}$ is a direct consequence of $f = f_\gamma + f^\gamma$.

1.1.11. (a) Take $A = \mu(E)^{-\frac{1}{p}} \|f\|_{L^{p,\infty}}$, then we have

$$\begin{aligned} \int_E |f(x)|^q\, d\mu(x) &= q \int_0^A \alpha^{q-1} d_f(\alpha)\, d\alpha + q \int_A^\infty \alpha^{q-1} d_f(\alpha)\, d\alpha \\ &\leq A^q \mu(E) + q \int_A^\infty \alpha^{q-p-1} \|f\|_{L^{p,\infty}}^p d\alpha \\ &= A^q \mu(E) + \frac{q}{p-q} A^{q-p} \|f\|_{L^{p,\infty}}^p \\ &= \frac{p}{p-q} \mu(E)^{1-\frac{q}{p}} \|f\|_{L^{p,\infty}}^q. \end{aligned}$$

(b) Take $E = X$ in the preceding problem, then this means $L^{p,\infty}(X,\mu) \subset L^q(X,\mu)$.

1.1.12. (a) By Exercise 1.1.11 (a), we have

$$\begin{aligned}|||f|||_{L^{p,\infty}} &= \sup_{0<\mu(E)<\infty} \mu(E)^{-\frac{1}{r}+\frac{1}{p}} \left( \int_E |f|^r d\mu \right)^{\frac{1}{r}} \\ &\leq \sup_{0<\mu(E)<\infty} \mu(E)^{-\frac{1}{r}+\frac{1}{p}} \left( \frac{p}{p-r} \right)^{\frac{1}{r}} \mu(E)^{\frac{1}{r}-\frac{1}{p}} \|f\|_{L^{p,\infty}} \leq \left( \frac{p}{p-r} \right)^{\frac{1}{r}} \|f\|_{L^{p,\infty}}.\end{aligned}$$

(b) Choose an ascending chain of subsets $X_n \subset X$ such that $\mu(X_n) < \infty$ and $\bigcup_n X_n = X$. Now set $E = \{|f| > \alpha\} \cap X_n$, we have

$$\begin{aligned}|||f|||_{L^{p,\infty}} &\geq \mu(E)^{-\frac{1}{r}+\frac{1}{p}} \left( \int_E |f|^r \, d\mu \right)^{\frac{1}{r}} \\ &\geq \mu(E)^{-\frac{1}{r}+\frac{1}{p}} (\alpha^r \mu(E))^{\frac{1}{r}} = \alpha d_f(\alpha)^{\frac{1}{p}}.\end{aligned}$$

Since $\alpha$ is arbitrary, we have $\|f\|_{L^{p,\infty}} \leq |||f|||_{L^{p,\infty}}$ for finite measure spaces $X_n$, and so for $X$ by definition.

(c) It's easy to verify that $|||f|||$ is a norm for $p > 1$ by Minkowski's inequality and taking proper $r \in (1,p)$. When it comes to $0 < p \leq 1$, we have $|||\cdot|||^r_{L^{p,\infty}}$ to be the desired metric (but not a norm) which induces the same topology by observing the inequality after taking the power of $r$ on all terms.

(d) Since

$$\begin{aligned}\big\| \liminf_{n\to\infty} |g_n| \big\|_{L^{p,\infty}} &\leq \sup_{0<m(E)<\infty} \mu(E)^{-\frac{1}{r}+\frac{1}{p}} \left( \int_E \liminf_{n\to\infty} |g_n|^r d\mu \right)^{\frac{1}{r}} \\ &\overset{\text{Fatou}}{\leq} \sup_{0<m(E)<\infty} \liminf_{n\to\infty} \mu(E)^{-\frac{1}{r}+\frac{1}{p}} \left( \int_E |g_n|^r d\mu \right)^{\frac{1}{r}} \\ &\leq \liminf_{n\to\infty} \sup_{0<m(E)<\infty} \mu(E)^{-\frac{1}{r}+\frac{1}{p}} \left( \int_E |g_n|^r d\mu \right)^{\frac{1}{r}} \\ &\leq \left( \frac{p}{p-r} \right)^{\frac{1}{r}} \liminf_{n\to\infty} \|g_n\|_{L^{p,\infty}}.\end{aligned}$$

1.1.13. (a) By definition, $\|f_\sigma\|_{L^{1,\infty}} = \sup_{\varepsilon>0} \mu\big(\big[\frac{\sigma^{-1}(1)-1}{N}, \frac{\sigma^{-1}(1)}{N}\big)\big)\Big(\frac{N}{1} - \varepsilon\Big)^1 = 1.$

(b) Since $\sum_{\sigma\in S_N} f_\sigma = N!\big(1 + \frac{1}{2} + \ldots + \frac{1}{N}\big)\chi_{[0,1)}$.

(c) If not, suppose $\|\cdot\|$ is an equivalent norm, that is, $C_1\|f\| \leq \|f\|_{L^{1,\infty}} \leq C_2\|f\|$ for all $f$ and some $C_1, C_2 > 0$. Then we have

$$\begin{aligned}&C_1 N!\Big(1 + \frac{1}{2} + \ldots + \frac{1}{N}\Big) \\ &= C_1 \| \sum_{\sigma\in S_N} f_\sigma \|_{L^{1,\infty}} \leq \| \sum_{\sigma\in S_N} f_\sigma \| \\ &\leq \sum_{\sigma\in S_N} \|f_\sigma\| \leq C_2 \sum_{\sigma\in S_N} \|f_\sigma\|_{L^{1,\infty}} = C_2 N!.\end{aligned}$$

This contradicts with the fact that $1 + \frac{1}{2} + \cdots + \frac{1}{N} \to \infty$ when $N \to \infty$.

1.1.14. (a) Since

$$\begin{aligned}\int_{|f|\le s}|f|^q\,d\mu &= q\int_0^s \alpha^{q-1}d_f(\alpha)\,d\alpha\\ &\le q\int_0^s \|f\|_{L^{p,\infty}}^p\alpha^{q-p-1}d\alpha \le \frac{q}{q-p}s^{q-p}\|f\|_{L^{p,\infty}}^p.\end{aligned}$$

(b) Since

$$\begin{aligned}\Big\|\max_{1\le j\le m}|f_j|\Big\|_{L^{p,\infty}}^p &= \sup\{\gamma^p d_{\max_{1\le j\le m}|f_j|}(\gamma)\}\\ &\le \sum_{j=1}^m \sup\{\gamma^p d_{f_j}(\gamma)\} \le \sum_{j=1}^m \|f_j\|_{L^{p,\infty}}^p.\end{aligned}$$

1.1.15. Assume WLOG that $\|f_j\|_{L^{p_j,\infty}}=1$ for all $j$. Now the estimate follows

$$\begin{aligned}d_{f_1\cdots f_k}(\alpha) &\le d_{f_1}\Big(\frac{\alpha}{s_1}\Big)+d_{f_2}\Big(\frac{s_1}{s_2}\Big)+\cdots+d_{f_k}\Big(\frac{s_{k-1}}{1}\Big)\\ &\le \Big(\frac{s_1}{\alpha}\Big)^{p_1}+\cdots+\Big(\frac{1}{s_{k-1}}\Big)^{p_k}.\end{aligned}$$

Take $x_1=\dfrac{s_1}{\alpha}, x_2=\dfrac{s_2}{s_1},\cdots,x_k=\dfrac{1}{s_k}$, then we can rewrite it as

$$d_{f_1\cdots f_k}(\alpha)\le x_1^{p_1}+\cdots+x_k^{p_k},$$

where $x_1\cdots x_k=\frac{1}{\alpha}$. Set $x_1^{p_1}=\frac{\lambda}{p_1},\cdots,x_k^{p_k}=\frac{\lambda}{p_k}$ for some constant $\lambda$ satisfying $\prod_j\big(\frac{\lambda}{p_j}\big)^{\frac{1}{p_j}}=\frac{1}{\alpha}$, and the right side becomes $\sum_j\frac{\lambda}{p_j}=p^{-1}\Big(\alpha^{-1}\prod_j p_j^{\frac{1}{p_j}}\Big)^p$. Hence $\|f_1\cdots f_k\|_{L^{p,\infty}}\le p^{-\frac{1}{p}}\prod_j p_j^{\frac{1}{p_j}}$.

1.1.16. We skip the cases of $\theta=0,1$ since there is nothing to prove.

(a) If $p_1\ne\infty$, then

$$\begin{aligned}\Big(\int_X|f|^p\,d\mu\Big)^{\frac{1}{p}} &= \Big(\int_X|f|^{p(1-\theta)}|f|^{p\theta}\,d\mu\Big)^{\frac{1}{p}}\\ &\le \Big(\int_X|f|^{p(1-\theta)\frac{p_0}{p(1-\theta)}}\,d\mu\Big)^{\frac{1-\theta}{p_0}}\Big(\int_X|f|^{p\theta\frac{p_1}{p\theta}}\,d\mu\Big)^{\frac{\theta}{p_1}}\\ &= \|f\|_{L^{p_0}}^{1-\theta}\|f\|_{L^{p_1}}^{\theta}.\end{aligned}$$

And if $p_1=\infty$, then $\frac{p_0}{p}=1-\theta$. Thus

$$\Big(\int_X|f|^p\,d\mu\Big)^{\frac{1}{p}}\le\|f^{p-p_0}\|_{L^\infty}^{\frac{1}{p}}\Big(\int_X|f|^{p_0}d\mu\Big)^{\frac{1}{p}}=\|f\|_{L^{p_0}}^{1-\theta}\|f\|_{L^\infty}^{\theta}.$$

(b) Since

$$t^p d_f(t)=\big(t^{\frac{(1-\eta)p}{\theta}}d_f(t)\big)^\theta\big(t^{\frac{\eta p}{1-\theta}}d_f(t)\big)^{1-\theta},$$

we have by definition that

$$\|f\|_{L^{p,\infty}}^p\le\|f\|_{L^{\frac{\eta p}{1-\theta},\infty}}^{\eta}\|f\|_{L^{\frac{(1-\eta)p}{\theta},\infty}}^{1-\theta},$$

where $p_0(1-\theta)=\eta p$ and $p_1\theta=(1-\eta)p$.

1.1.17. (a) The $n = 2$ case is nothing other than the $n = 2$ Hölder's inequality. For $n \geq 3$, suppose the proposition holds for $k \leq n-1$, then for $k = n$, set

$$P(x_1, \ldots, x_{n-1}) = \int_{\mathbb{R}} |f_1(\pi_1)(x) \cdots f_{n-1}(\pi_{n-1})(x)|\, dx_n,$$

then

$$\begin{aligned}\Lambda(f_1, \ldots, f_n) &\leq \int_{\mathbb{R}^{n-1}} P(x_1, \ldots, x_{n-1}) |f_n(\pi_n)(x)|\, dx_1 \cdots dx_{n-1} \\ &\leq \|P\|_{L^{\frac{n-1}{n-2}}(\mathbb{R}^{n-1})} \|f_n \circ \pi_n\|_{L^1(\mathbb{R}^{n-1})}.\end{aligned}$$

Note that by the induction hypothesis, we should have

$$\begin{aligned}\|P\|_{L^{\frac{n-1}{n-2}}(\mathbb{R}^{n-1})} &= \left( \int_{\mathbb{R}^{n-1}} \left( \int_{\mathbb{R}} |f_1(\pi_1)(x)| \cdots |f_{n-1}(\pi_{n-1})(x)|\, dx_n \right)^{\frac{n-1}{n-2}} dx_1 \cdots dx_{n-1} \right)^{\frac{n-2}{n-1}} \\ &\overset{\text{Hölder}}{\leq} \left( \int_{\mathbb{R}^{n-1}} \prod_{i=1}^{n-1} \left( \int_{\mathbb{R}} |f_i(\pi_i)(x)|^{n-1} dx_n \right)^{\frac{1}{n-2}} dx_1 \cdots dx_{n-1} \right)^{\frac{n-2}{n-1}} \\ &\overset{\text{induction}}{\leq} \prod_{i=1}^{n-1} \left( \int_{\mathbb{R}^n} |f_i(x)|^{n-1} dx_1 \cdots dx_n \right)^{\frac{1}{n-1}} = \prod_{i=1}^{n-1} \|f_i\|_{L^{n-1}(\mathbb{R}^{n-1})}.\end{aligned}$$

Thus,

$$\Lambda(f_1, \ldots, f_n) \leq \|P\|_{L^{\frac{n-1}{n-2}}(\mathbb{R}^{n-1})} \|f_n \circ \pi_n\|_{L^1(\mathbb{R}^{n-1})} \leq \prod_{i=1}^{n} \|f_i\|_{L^{n-1}(\mathbb{R}^{n-1})}.$$

(b) Take $f_j = \chi_{\pi_j[\Omega]}$, then the left–hand side is no less than

$$\begin{aligned}\int_{\mathbb{R}^n} \prod_{i=1}^{n} |f_i \circ \pi_i|\, dx &= \int_{\mathbb{R}^n} \chi_{\pi_1[\Omega]} \cdots \chi_{\pi_n[\Omega]}\, dx \\ &\geq \int_{\mathbb{R}^n} \chi_\Omega\, dx = |\Omega|.\end{aligned}$$

And the right–hand side is no more than

$$\begin{aligned}\prod_{i=1}^{n} \|f_j\|_{L^{n-1}(\mathbb{R}^{n-1})} &= \prod_{i=1}^{n} \left( \int_{\mathbb{R}^{n-1}} \chi_{\pi_i[\Omega]}\, dx_1 \cdots \widehat{dx_i} \cdots dx_n \right)^{\frac{1}{n-1}} = \prod_{i=1}^{n} |\pi_i[\Omega]|^{\frac{1}{n-1}} \\ &\leq \prod_{i=1}^{n} \left( \frac{1}{2} |\partial\Omega| \right)^{\frac{1}{n-1}} = 2^{\frac{n}{1-n}} |\partial\Omega|^{\frac{n}{n-1}}.\end{aligned}$$

## 1.2 Convolution and Approximate Identities

1.2.1. For $g \notin AB$, forbidding $h \in A$ along with $h^{-1}g \in B$, we have

$$\int_G f(h) g(h^{-1}g)\, dh = 0$$

since at least one of $f(h)$ and $g(h^{-1}g)$ vanishes.

1.2.2. Since

$$\begin{aligned}{}^t f * g(x) &= \int_G f(ty) g(y^{-1}x)\, dy = \int_G f(y) g(y^{-1}tx)\, d(ty) \\ &= \int_G f(y) g(y^{-1}tx)\, d(y) = {}^t (f * g)(x).\end{aligned}$$

and

$$f * g^t(x) = \int_G f(y)g(y^{-1}xt)\,dy = (f * g)^t(x).$$

1.2.3.

1.2.4. (a) Since $f\chi_{B(0,n)} \xrightarrow{L^p} f$.

(b) Let $\varphi \in C_c(\mathbb{R}^n)$ be a positive function supported in $B(0,1)$ with integral 1, then define $\varphi_k(x) = k^n\varphi(kx)$, which are compactly supported smooth functions with integral 1 as well. Now we show $\varphi_k * f \xrightarrow{L^p} f$, which is because

$$\begin{aligned}\|\varphi_k * f - f\|_{L^p}^p &= \int_{\mathbb{R}} \left| \int_{\mathbb{R}} \varphi_k(x-y)(f(x)-f(y))\,dy \right|^p dx \\ &\overset{\text{Jensen}}{\leq} \int_{\mathbb{R}} \int_{|y-x|<\frac{1}{k}} \varphi_k(x-y)|f(x)-f(y)|^p\,dydx \\ &\leq \int_{|t|<\frac{1}{k}} \left( \int_{\mathbb{R}} |f(x)-f(x-t)|^p\,dx \right) \varphi_k(-t)\,dt \\ &\leq \sup_{|t|<\frac{1}{k}} \int_{\mathbb{R}} |f(x)-f(x-t)|^p\,dx.\end{aligned}$$

Then apply the average continuity.

1.2.5. $\lambda$ is a Haar measure because for any $a > 0$,

$$\lambda(aA) = \int_{aA} \frac{dt}{t} = \int_A \frac{d(at)}{at} = \int_A \frac{dt}{t} = \lambda(A).$$

1.2.6. $\lambda$ is a left Haar measure because for any $(a,b) \in G$,

$$\begin{aligned}\lambda((a,b)A) &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \chi_{(a,b)A}(x,y) \frac{dxdy}{x^2} \\ &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \chi_{(a,b)A}((a,b)(x,y)) \det \begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix} \frac{dxdy}{(ax)^2} \\ &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \chi_A(x,y) \frac{dxdy}{x^2} = \lambda(A).\end{aligned}$$

$\rho$ is a right Haar measure because for any $(a,b) \in G$,

$$\begin{aligned}\rho(A(a,b)) &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \chi_{A(a,b)}(x,y) \frac{dxdy}{|x|} \\ &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \chi_{A(a,b)}((x,y)(a,b)) \left| \det \begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix} \right| \frac{dxdy}{|ax|} \\ &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \chi_A(x,y) \frac{dxdy}{|x|} = \rho(A).\end{aligned}$$

1.2.7. For the former, note that (here $G$ denotes the group $(\mathbb{R}^+, \frac{dt}{t})$)

$$\begin{aligned}&\| |f(x)|x^{\frac{1}{p}} * x^{-\frac{1}{p'}} \chi_{[1,\infty)} \|_{L^p(G)} \\ &= \left( \int_0^\infty \left( \int_0^\infty |f(t)|t^{\frac{1}{p}} (t^{-1}x)^{-\frac{1}{p'}} \chi_{[1,\infty)}(t^{-1}x) \frac{dt}{t} \right)^p \frac{dx}{x} \right)^{\frac{1}{p}} \\ &= \left( \int_0^\infty \left( \frac{1}{x} \int_0^x |f(t)|dt \right)^p dx \right)^{\frac{1}{p}}.\end{aligned}$$

And

$$\||f(x)|x^{\frac{1}{p}}\|_{L^p(G)} = \left(\int_0^\infty (|f(x)|x^{\frac{1}{p}})^p \frac{dx}{x}\right)^{\frac{1}{p}} = \|f\|_{L^p},$$
$$\|x^{-\frac{1}{p'}}\chi_{[1,\infty)}\|_{L^1(G)} = \int_1^\infty x^{-\frac{1}{p'}}\frac{dx}{x} = \frac{p}{p-1}.$$

Now combine with Minkowski's inequality,

$$\||f(x)|x^{\frac{1}{p}} * x^{-\frac{1}{p'}}\chi_{[1,\infty)}\|_{L^p(G)} \le \||f(x)|x^{\frac{1}{p}}\|_{L^1(G)}\|x^{-\frac{1}{p'}}\chi_{[1,\infty)}\|_{L^p(G)}.$$

The latter and Exercise 1.2.8 follow in a rather similar manner, so we omit the procedure.

1.2.9. By Young's inequality, it's quite straightforward to obtain $\|T\|_{L^p\to L^p} \le \|K\|_{L^1}$. To show the inverse, we use $K_R := K\chi_{B(0,R)}$ to approximate the result. Note that for $|x| \le R(1-\varepsilon)$, we have $B(0, R\varepsilon) \subset B(x, R)$. Thus

$$\int_{\mathbb{R}^n} \chi_R(x-y)K_{R\varepsilon}(y)\,dy = \int_{\mathbb{R}^n} K_{R\varepsilon}(y)\,dy = \|K_{R\varepsilon}\|_{L^1}.$$

We have $\|\chi_R\|_{L^p}^p = |B(0,R)|^p$, and

$$\begin{aligned}\|K*\chi_R\|_{L^p}^p &\ge \|K_{R\varepsilon}*\chi_R\|_{L^p(B(0,R(1-\varepsilon)))}^p \\ &= \int_{B(0,R(1-\varepsilon))}\left(\int_{\mathbb{R}^n} K_{R\varepsilon}(y)\chi_R(x-y)\,dy\right)^p dx \\ &= \int_{B(0,R(1-\varepsilon))} \|K_{R\varepsilon}\|_{L^1}^p dx = (1-\varepsilon)^n\|K_{R\varepsilon}\|_{L^1}^p.\end{aligned}$$

This means

$$\|T\|_{L^p\to L^p} \ge (1-\varepsilon)^{\frac{n}{p}}\|K_{R\varepsilon}\|_{L^1}$$

for each $\varepsilon \in (0,1)$ and $R > 0$. Let $R \to \infty$ and then $\varepsilon \to 0^+$, we get $\|T\|_{L^p\to L^p} \ge \|K\|_{L^1}$.

1.2.10. By Young's inequality, we obtain $\|T\|_{L^p\to L^p} \le \|K\|_{L^1}$. For the inverse, take $f_n = \chi_{[\frac{1}{n},n]}$ and $K_n = Kf_n$. Then for $n^{\varepsilon-1} \le x \le n^{1-\varepsilon}$, we have

$$f_n * K_{n^\varepsilon}(x) = \int_0^\infty K_{n^\varepsilon}(y)f_n(x-y)\frac{dy}{y} = \|K_{n^\varepsilon}\|_{L^1}.$$

Thus

$$\frac{\|K*f_n\|_{L^p}^p}{\|f_n\|_{L^p}^p} \ge \frac{\|K_{n^\varepsilon}*f_n\|_{L^p[n^{\varepsilon-1},n^{1-\varepsilon}]}^p}{2\log n} = \|K_{n^\varepsilon}\|_{L^1}^p\,(1-\varepsilon)\,.$$

Letting $n\to\infty$, we have $\|T\|_{L^p\to L^p}^p \ge \|K\|_{L^1}^p\,(1-\varepsilon)$; then $\varepsilon\to 0^+$, we get $\|T\|_{L^p\to L^p} \ge \|K\|_{L^1}$.

1.2.11. (a) Since

$$\begin{aligned}c_k &= \left(\int_1^1 (1-t^2)^k dt\right)^{-1} \\ &< \left(\int_{-\frac{1}{\sqrt{k}}}^{\frac{1}{\sqrt{k}}} (1-t^2)^k dt\right)^{-1} \\ &< \left(\int_{-\frac{1}{\sqrt{k}}}^{\frac{1}{\sqrt{k}}} (1-kt^2)\,dt\right)^{-1} = \frac{3}{4}\sqrt{k} < \sqrt{k}.\end{aligned}$$

Thanks to the adjustment given by $c_k$, we have

$$(Q_k * f - f)(x) = \int_{-1}^1 (f(x-t)-f(x))Q_k(t)\,dt.$$

Thus

$$\begin{aligned}\|Q_k * f - f\|_{L^1} &= \int_{\mathbb{R}} \left| \int_{-1}^{1} (f(x-t) - f(x))Q_k(t)\,dt \right| dx \\ &\le \int_{\mathbb{R}^2} |f(x) - f(x-t)|Q_k(t)\,dtdx \\ &\le \int_{\mathbb{R}} Q_k(t)\|f(x) - f(x-t)\|_{L^1} dt.\end{aligned}$$

Due to the average continuity, we have $\delta > 0$ for each $\varepsilon > 0$ such that $\|f(x) - f(x-t)\|_{L^1} < \varepsilon$ for $|t| < \delta$. Plus, we have $\lim\limits_{k\to\infty} \sqrt{k}(1-t^2)^k = 0$ uniformly for $\delta < |t| < 1$, thus take $k$ large so that $Q_k(x) < \varepsilon$ for $n > k$, we have

$$\begin{aligned}&\int_{\mathbb{R}} Q_k(t)\|f(x) - f(x-t)\|_{L^1} dt \\ &\le \int_{|t|\le\delta} Q_k(t)\|f(x) - f(x-t)\|_{L^1} dt \\ &\quad + \int_{|t|>\delta} Q_k(t)\|f(x) - f(x-t)\|_{L^1} dt \\ &\le \varepsilon\|Q_k\|_{L^1} + \int_{|t|>\delta} Q_k(t)(\|f(x)\|_{L^1} + \|f(x-t)\|_{L^1})\,dt \\ &\le (1 + 2\|f\|_{L^1})\varepsilon.\end{aligned}$$

This ends the proof.

(b) Due to the uniform continuity, we choose $\delta > 0$ so that $|f(x) - f(x-t)| < \varepsilon$ for $|t| \le \delta$. Now for large $k$, we have

$$\begin{aligned}|(Q_k * f - f)(x)| &= \left| \int_{-1}^{1} (f(x-t) - f(x))Q_k(t)\,dt \right| \\ &\le \int_{|t|\le\delta} |f(x) - f(x-t)|Q_k(t)\,dt \\ &\quad + \int_{|t|>\delta} Q_k(t)|f(x) - f(x-t)|\,dt \\ &\le \varepsilon\|Q_k\|_{L^1} + \varepsilon \sup_{x\in\mathbb{R}} f(x) \\ &= \varepsilon\big(1 + \sup_{x\in\mathbb{R}} f(x)\big),\end{aligned}$$

and note that the tending to 0 does not depend on the choice of $\delta$.

(c)&(d) Set $g(t) = f(t) - f(-1) - \dfrac{t+1}{2}(f(1) - f(-1))$, then $g(-1) = g(1) = 0$. $g$ can be viewed as satisfying the condition in (c) by zero extension outside $[-1, 1]$. Thus $g * Q_k \to g$ uniformly. However, also note that this means

$$g * Q_k + f(-1) + \frac{x+1}{2}(f(1) - f(-1)) \to f$$

uniformly on $[-1, 1]$, and $g * Q_k(x)$ is a polynomial in $x$, since

$$\begin{aligned}g * Q_k(x) &= \int_{-1}^{1} g(t)c_k(1 - (x-t)^2)^k\,dt \\ &= \sum_{j=0}^{2k} \left( \int_{-1}^{1} g(t)d_{j,k}t^j\,dt \right) x^j.\end{aligned}$$

1.2.12 Note that

$$\begin{aligned}|L(f)(x)|^2 &= \left|\int_0^\infty f(t)e^{-xt}\,dt\right|^2\\ &\le \int_0^\infty |f(t)|^2 e^{-xt}\sqrt{t}\,dt \int_0^\infty \frac{e^{-xt}}{\sqrt{t}}dt\\ &= \int_0^\infty |f(t)|^2 e^{-xt}\sqrt{t}\,dt \int_0^\infty \frac{e^{-u}}{\sqrt{u/x}}\frac{du}{x}\\ &= \sqrt{\frac{\pi}{x}}\int_0^\infty |f(t)|^2 e^{-xt}\sqrt{t}\,dt.\end{aligned}$$

Therefore

$$\begin{aligned}\|L(f)\|_{L^2}^2 &\le \sqrt{\pi}\int_0^\infty\int_0^\infty |f(t)|^2 e^{-xt}\sqrt{\frac{t}{x}}\,dtdx\\ &= \sqrt{\pi}\int_0^\infty\int_0^\infty |f(t)|^2 e^{-u}\frac{t}{\sqrt{u}}\frac{du}{t}\,dt = \pi\|f\|_{L^2}^2.\end{aligned}$$

Another solution using the hint of convolution:

$$\begin{aligned}L(f)(x) &= \int_0^\infty f(t)e^{-xt}dt = \int_0^\infty f\Big(\frac{1}{t}\Big)e^{-\frac{x}{t}}\frac{dt}{t^2}\\ &= \int_0^\infty \sqrt{\frac{x}{t}}e^{-\frac{x}{t}}\cdot\frac{1}{\sqrt{t}}f\Big(\frac{1}{t}\Big)\frac{dt}{t}\cdot\frac{1}{\sqrt{x}}\\ &= \left(\sqrt{t}e^{-t} * \frac{1}{\sqrt{t}}f\Big(\frac{1}{t}\Big)\right)(x)\cdot\frac{1}{\sqrt{x}}.\end{aligned}$$

Thus taking norm and using Minkowski's inequality we obtain

$$\begin{aligned}\|L(f)\|_{L^2(\mathbb{R}^+,dt)} &= \|\sqrt{t}e^{-t} * \frac{1}{\sqrt{t}}f\Big(\frac{1}{t}\Big)\|_{L^2(\mathbb{R}^+,\frac{dt}{t})}\\ &\le \|\sqrt{t}e^{-t}\|_{L^1(\mathbb{R}^+,\frac{dt}{t})}\|\frac{1}{\sqrt{t}}f\Big(\frac{1}{t}\Big)\|_{L^2(\mathbb{R}^+,\frac{dt}{t})}\\ &= \sqrt{\pi}\|f\|_{L^2(\mathbb{R}^+,dt)}.\end{aligned}$$

**证明** (Remark)**.** *The norm is exactly $\sqrt{\pi}$ indeed. In fact, consider a sequence of functions $f_n(t) = \frac{1}{\sqrt{t}}\chi_{[\frac{1}{n},n]}(t)$, we find $\|f_n\|_{L^2} = \sqrt{2\log(n)}$. Compute the norms of transformed functions:*

$$\begin{aligned}\|L(f_n)\|_2^2 &= \int_{\mathbb{R}^+}\left(\int_{\frac{1}{n}}^n \frac{1}{\sqrt{t}}e^{-st}dt\right)^2 ds\\ &= \int_{\mathbb{R}^+}\frac{1}{s}\left(\int_{\frac{s}{n}}^{ns}\frac{1}{\sqrt{t}}e^{-t}dt\right)^2 ds.\end{aligned}$$

*For arbitrary $\delta\in(0,1)$ fixed, we have*

$$\begin{aligned}&\int_{\mathbb{R}^+}\frac{1}{s}\left(\int_{\frac{s}{n}}^{ns}\frac{1}{\sqrt{t}}e^{-t}dt\right)^2 ds\\ &\ge \int_{(\frac{1}{n})^{1-\delta}}^{n^{1-\delta}}\frac{1}{s}\left(\int_{\frac{s}{n}}^{ns}\frac{1}{\sqrt{t}}e^{-t}dt\right)^2 ds\\ &\ge \int_{(\frac{1}{n})^{1-\delta}}^{n^{1-\delta}}\frac{1}{s}\left(\int_{(\frac{1}{n})^{\delta}}^{n^{\delta}}\frac{1}{\sqrt{t}}e^{-t}dt\right)^2 ds.\end{aligned}$$

*For $n$ sufficiently large, we have by the famous integral that*

$$\int_{(\frac{1}{n})^\delta}^{n^\delta} \frac{1}{\sqrt{t}} e^{-t} dt \geq \sqrt{\pi} - \varepsilon,$$

*thus,*

$$\begin{aligned} \|L(f_n)\|_{L^2}^2 &\geq \int_{(\frac{1}{n})^{1-\delta}}^{n^{1-\delta}} \frac{\sqrt{\pi}-\varepsilon}{s} ds \\ &= 2\log(n)(1-\delta)(\sqrt{\pi}-\varepsilon)^2 \\ &= \|f_n\|_{L^2}^2 (1-\delta)(\sqrt{\pi}-\varepsilon)^2. \end{aligned}$$

*Finally, take $\delta \to 0^+$ and $\varepsilon \to 0^+$.*

## 1.3 解答: GTM249 Chapter 1 Part II

原网页

## 1.3 Interpolation

1.3.1. This just requires some tiny changes in the proof of the original theorem.

1.3.2. Interpolating between $L^1$ and $L^r$ using Marcinkiewicz, we have

$$\begin{aligned} \|T\|_{L^{\frac{p+1}{2}} \to L^{\frac{p+1}{2}}} &\leq 2\left( \frac{\frac{p+1}{2}}{\frac{p+1}{2}-1} + \frac{\frac{p+1}{2}}{r-\frac{p+1}{2}} \right)^{\frac{1}{p}} A_0^{\frac{\frac{2}{p+1}-\frac{1}{r}}{1-\frac{1}{r}}} A_1^{\frac{1-\frac{2}{p+1}}{1-\frac{1}{r}}} \\ &= 2\left( \frac{p+1}{p-1} + \frac{p+1}{2r-p-1} \right)^{\frac{1}{p}} A_0^{\frac{2r-p-1}{(r-1)(p+1)}} A_1^{\frac{r(p-1)}{(p+1)(r-1)}}. \end{aligned}$$

Then using Riesz–Thorin to interpolate between $L^{\frac{p+1}{2}}$ and $L^r$, we have

$$\|T\|_{L^p \to L^p} \leq \left( 2\left( \frac{p+1}{p-1} + \frac{p+1}{2r-p-1} \right)^{\frac{1}{p}} A_0^{\frac{2r-p-1}{(r-1)(p+1)}} A_1^{\frac{r(p-1)}{(p+1)(r-1)}} \right)^{1-\theta} A_1^{\theta},$$

where $\theta = \frac{r(1-p)}{p(p+1-2r)}$ by solving $\frac{1}{p} = \frac{2(1-\theta)}{p+1} + \frac{\theta}{r}$. Thus

$$\|T\|_{L^p \to L^p} \leq 2^{\frac{(p-r)(1+p)}{p(p+1-2r)}} \left( \frac{p+1}{p-1} + \frac{p+1}{2r-p-1} \right)^{\frac{(r-p)(p+1)}{p^2(2r-p-1)}} A_0^{\frac{\frac{1}{p}-\frac{1}{r}}{1-\frac{1}{r}}} A_1^{\frac{1-\frac{1}{p}}{1-\frac{1}{r}}}.$$

Now we come to estimate the constant. Note that $p > 1$ implies $\frac{p}{p+1} > \frac{1}{2}$, we have then $\frac{r-p}{2r-p-1} < \frac{1}{2}$. Thus

$$\begin{aligned} \frac{(p-r)(1+p)}{p(p+1-2r)} &= \frac{1+p}{p} \frac{r-p}{2r-p-1} < 2 \cdot \frac{1}{2} = 1, \\ \frac{p+1}{p-1} + \frac{p+1}{2r-p-1} &< \frac{2(p+1)}{p-1}. \end{aligned}$$

With these estimates, we have

$$\begin{aligned} \|T\|_{L^p \to L^p} &< 2\left( \frac{2(p+1)}{p-1} \right)^{\frac{1}{p}} A_0^{\frac{\frac{1}{p}-\frac{1}{r}}{1-\frac{1}{r}}} A_1^{\frac{1-\frac{1}{p}}{1-\frac{1}{r}}} \\ &< 8(p-1)^{-\frac{1}{p}} A_0^{\frac{\frac{1}{p}-\frac{1}{r}}{1-\frac{1}{r}}} A_1^{\frac{1-\frac{1}{p}}{1-\frac{1}{r}}}, \end{aligned}$$

with the use of $2(p+1) = 4\big(\frac{p-1}{2}+1\big) < 4 \cdot e^{\frac{p-1}{2}} < 4 \cdot 4^{p-1} = 4^p$.

1.3.3. (a) It is a special case of (b). To be specific, take $p_0 = 1$ in (b), then the norm becomes

$$\begin{aligned}
&p^{1+\frac{1}{p}}\left[\frac{B(2,p-1)}{(p-1)^{p-1}}\right]^{\frac{1}{p}} A_0^{\frac{1}{p}} A_1^{1-\frac{1}{p}} \\
&= p^{1+\frac{1}{p}}\left[\frac{\Gamma(2)\Gamma(p-1)}{\Gamma(p+1)(p-1)^{p-1}}\right]^{\frac{1}{p}} A_0^{\frac{1}{p}} A_1^{1-\frac{1}{p}} \\
&= p^{1+\frac{1}{p}}\left[\frac{1}{p(p-1)^p}\right]^{\frac{1}{p}} A_0^{\frac{1}{p}} A_1^{1-\frac{1}{p}} = \frac{p}{p-1} A_0^{\frac{1}{p}} A_1^{1-\frac{1}{p}}.
\end{aligned}$$

(b) It suffices to study the case of $f \geq 0$ due to the assumption $|T(f)| \leq T(|f|)$. For $\alpha > 0$, write $f = f_0 + f_1$, where $f_0 = (f - \frac{\lambda\alpha}{A_1})\chi_{f \geq \frac{\lambda\alpha}{A_1}}$, then we have the relation

$$\{|T(f)| > \alpha\} \subseteqq \{|T(f_0)| > (1-\lambda)\alpha\}$$

since

$$\|T(f_1)\|_{L^\infty} \leq \|T\|_{L^\infty \to L^\infty} \|f_1\|_{L^\infty} \leq A_1 \frac{\lambda\alpha}{A_1} = \lambda\alpha.$$

Thus

$$\begin{aligned}
\|T(f)\|_{L^p}^p &= p\int_0^\infty \alpha^{p-1} d_{T(f)}(\alpha)\, d\alpha \\
&\leq p\int_0^\infty \alpha^{p-1} d_{T(f_0)}((1-\lambda)\alpha)\, d\alpha \\
&\leq \frac{pA_0^{p_0}}{(1-\lambda)^{p_0}} \int_0^\infty \alpha^{p-p_0-1} \|f_0\|_{L^{p_0}}^{p_0} d\alpha \\
&= \frac{pA_0^{p_0}}{(1-\lambda)^{p_0}} \int_0^\infty \alpha^{p-p_0-1} \int_{f \geq \frac{\lambda\alpha}{A_1}} \Big(f - \frac{\lambda\alpha}{A_1}\Big)^{p_0} d\mu d\alpha \\
&= \frac{pA_0^{p_0}}{(1-\lambda)^{p_0}} \int_X \int_0^{\frac{A_1 f(x)}{\lambda}} \alpha^{p-p_0-1} \Big(f - \frac{\lambda\alpha}{A_1}\Big)^{p_0} d\alpha d\mu \\
&= \frac{pA_0^{p_0} A_1^{p-p_0}}{(1-\lambda)^{p_0}\lambda^{p-p_0}} \int_X \int_0^{f(x)} \alpha^{p-p_0-1} (f-\alpha)^{p_0} d\alpha d\mu \\
&= \frac{pA_0^{p_0} A_1^{p-p_0}}{(1-\lambda)^{p_0}\lambda^{p-p_0}} B(p_0+1, p-p_0) \int_X |f|^p\, d\mu \\
&= \frac{pA_0^{p_0} A_1^{p-p_0}}{(1-\lambda)^{p_0}\lambda^{p-p_0}} B(p_0+1, p-p_0) \|f\|_{L^p}^p.
\end{aligned}$$

Now take $\lambda = 1 - \frac{p_0}{p}$ to end this proof.

(c) Again, write $f = f_0 + f_1$, where $f_0 = (f - \frac{\delta\alpha}{A_1})\chi_{f \geq \frac{\delta\alpha}{A_1}}$. Thus

$$\begin{aligned}
\|T(f)\|_{L^p}^p &= p\int_0^\infty \alpha^{p-1} d_{T(f)}(\alpha)\, d\alpha \\
&\leq p\int_0^\infty \alpha^{p-1} d_{T(f_0)}((1-\lambda)\alpha)\, d\alpha + p\int_0^\infty \alpha^{p-1} d_{T(f_1)}(\lambda\alpha)\, d\alpha \\
&= \frac{pA_0^{p_0}}{(1-\lambda)^{p_0}} \int_0^\infty a^{p-p_0-1} \|f_0\|_{L^{p_0}}^{p_0} d\alpha + \frac{pA_1^{p_1}}{\lambda^{p_1}} \int_0^\infty \alpha^{p-p_1-1} \|f_0\|_{L^{p_0}}^{p_0} d\alpha \\
&= \frac{pA_0^{p_0}}{(1-\lambda)^{p_0}} \int_0^\infty \alpha^{p-p_0-1} \int_{f \geq \delta\alpha} (f - \delta\alpha)^{p_0} d\alpha \\
&\quad + \frac{pA_1^{p_1}}{\lambda^{p_1}} \int_0^\infty \alpha^{p-p_1-1} \Big( \int_{f \geq \delta\alpha} \delta^{p_1} \alpha^{p_1} d\mu + \int_{f < \delta\alpha} f^{p_1} d\mu \Big) d\alpha \\
&= \frac{pA_0^{p_0}}{(1-\lambda)^{p_0}} \int_X \int_0^{\frac{f(x)}{\delta}} \alpha^{p-p_0-1} (f - \delta\alpha)^{p_0} d\alpha d\mu \\
&\quad + \frac{pA_1^{p_1}}{\lambda^{p_1}} \Big( \int_X \int_0^{\frac{f(x)}{\delta}} \alpha^{p-1} \delta^{p_1} d\alpha d\mu + \int_X \int_{\frac{f(x)}{\delta}}^\infty \alpha^{p-p_1-1} f^{p_1} d\alpha d\mu \Big) \\
&= \frac{pA_0^{p_0}}{(1-\lambda)^{p_0}} \int_X \int_0^1 \Big(\frac{f}{\delta}\alpha\Big)^{p-p_0-1} (f - f\alpha)^{p_0} d\Big(\frac{f}{\delta}\alpha\Big) d\mu \\
&\quad + \frac{pA_1^{p_1}}{\lambda^{p_1}} \Big( \delta^{p_1-p} \int_X f^p\, d\mu + \frac{\delta^{p_1-p}}{p_1 - p} \int_X f^p\, d\mu \Big) \\
&= \|f\|_{L^p}^p \Big( \frac{pA_0^{p_0}}{(1-\lambda)^{p_0}\delta^{p-p_0}} B(p - p_0, p_0 + 1) + \frac{pA_1^{p_1}\delta^{p_1-p}}{\lambda^{p_1}} \frac{p_1 - p - 1}{p_1 - p} \Big).
\end{aligned}$$

Now take $\delta$ such that $\frac{A_0^{p_0}}{\delta^{p-p_0}} = \frac{A_1^{p_1}}{\delta^{p-p_1}}$ and optimize over $\lambda$ to end the estimate.

1.3.7. Write

$$f = \sum_{k=0}^\infty f\chi_{S_k},$$

where $S_k = \{2^k \leq |f| < 2^{k+1}\}$ when $k > 1$ and $S_0 = \{|f| < 2\}$. Using Hölder's inequality and the hypotheses on $T$, we obtain that

$$\begin{aligned}
\int_Y |T(f\chi_{S_k})|\, d\nu &\leq \Big( \int_Y d\nu \Big)^{\frac{1}{k+1}} \Big( \int_Y |Tf|^{\frac{k+1}{k}} \chi_{S_k} d\nu \Big)^{\frac{k}{k+1}} \\
&\leq \nu(Y)^{\frac{1}{k+1}} A \Big( \frac{k+1}{k} - 1 \Big)^{-\alpha} \Big( \int_{S_k} |f|^{\frac{k+1}{k}} d\mu \Big)^{\frac{k}{k+1}} \\
&\leq 2^{k+1} \nu(Y)^{\frac{1}{k+1}} A k^\alpha \mu(S_k)^{\frac{k}{k+1}}.
\end{aligned}$$

If $\mu(S_k) \leq \frac{1}{3^{k+1}}$, then

$$2^{k+1} \nu(Y)^{\frac{1}{k+1}} A k^\alpha \mu(S_k)^{\frac{k}{k+1}} \leq 2\nu(Y)^{\frac{1}{k+1}} A k^\alpha \Big(\frac{2}{3}\Big)^k.$$

If $\mu(S_k) \geq \frac{1}{3^{k+1}}$, then

$$\begin{aligned}
\int_Y |T(f\chi_{S_k})|\, d\nu &\leq \nu(Y)^{\frac{1}{k+1}} A k^\alpha \mu(S_k)^{-\frac{1}{k+1}} \int_{S_k} |f|\, d\mu \\
&\leq 3\nu(Y)^{\frac{1}{k+1}} A \int_{S_k} |f| k^\alpha\, d\mu \\
&\leq 3\nu(Y)^{\frac{1}{k+1}} A \int_{S_k} |f| (\log_2^+ |f|)^\alpha\, d\mu.
\end{aligned}$$

For $S_0$, we have

$$\begin{aligned}\int_Y |T(f\chi_{S_0})|\,d\nu &\le \left(\int_Y d\nu\right)^{\frac{1}{2}}\left(\int_Y |T(f\chi_{S_0})|^2 d\nu\right)^{\frac{1}{2}}\\ &\le \nu(Y)^{\frac{1}{2}} A\cdot 2\mu(X)^{\frac{1}{2}}.\end{aligned}$$

Add the three parts together and use the countable subadditivity of $T$, we then conclude that

$$\begin{aligned}\int_Y |T(f)|\,d\nu &\le \sum_{k=0}^{\infty}\int_Y |T(f\chi_{S_k})|\,d\nu\\ &\le 6A\max(1,\nu(Y))^{\frac{1}{2}}\left[\int_X |f|(\log_2^+|f|)^\alpha d\mu + \sum_{k=1}^{\infty} k^\alpha\Big(\frac{2}{3}\Big)^k + \mu(X)^{\frac{1}{2}}\right]\\ &\le 6A(1+\nu(Y))^{\frac{1}{2}}\left[\int_X |f|(\log_2^+|f|)^\alpha d\mu + C_\alpha + \mu(X)^{\frac{1}{2}}\right].\end{aligned}$$

1.3.8. Use the change of variables $s = e^{\pi t}\in(0,\infty)$, and the first integral becomes

$$\begin{aligned}&\frac{\sin(\pi x)}{2}\int_{-\infty}^{+\infty}\frac{1}{\cosh(\pi t)+\cos(\pi x)}\,dt\\ =\ &\sin(\pi x)\int_0^\infty \frac{1}{s+s^{-1}+2\cos(\pi x)}\frac{ds}{\pi s}\\ =\ &\frac{\sin(\pi x)}{\pi}\int_0^\infty \frac{ds}{(s+\cos(\pi x))^2+\sin^2(\pi x)}\\ =\ &\frac{1}{\pi}\int_0^\infty \frac{d\big(\frac{s}{\sin(\pi x)}\big)}{\big(\frac{s}{\sin(\pi x)}+\cot(\pi x)\big)^2+1}\\ =\ &\frac{1}{\pi}\int_{\cot(\pi x)}^\infty \frac{du}{u^2+1} = x.\end{aligned}$$

For the second one,

$$\begin{aligned}&\frac{\sin(\pi x)}{2}\int_{-\infty}^{+\infty}\frac{1}{\cosh(\pi t)-\cos(\pi x)}\,dt\\ =\ &\sin(\pi x)\int_0^\infty \frac{1}{s+s^{-1}-2\cos(\pi x)}\frac{ds}{\pi s}\\ =\ &\frac{\sin(\pi x)}{\pi}\int_0^\infty \frac{ds}{(s-\cos(\pi x))^2+\sin^2(\pi x)}\\ =\ &\frac{1}{\pi}\int_0^\infty \frac{d\big(\frac{s}{\sin(\pi x)}\big)}{\big(\frac{s}{\sin(\pi x)}-\cot(\pi x)\big)^2+1}\\ =\ &\frac{1}{\pi}\int_{-\cot(\pi x)}^\infty \frac{du}{u^2+1} = 1-x.\end{aligned}$$

## 1.4 Lorentz Spaces

1.4.1. (a) Since

$$\int_A g\,d\mu = \int_X g\chi_A\,d\mu = \int_0^\infty (g\chi_A)^*(t)\,dt = \int_0^{\mu(A)} (g\chi_A)^*(t)\,dt \le \int_0^{\mu(A)} g^*(t)\,dt.$$

(b) Since

$$\begin{aligned}\int_X |fg|\,d\mu &= \int_X \int_0^\infty \int_0^\infty \chi_{\{|f|\geq s\}}\chi_{\{|g|\geq t\}}\,dtdsd\mu \\ &= \int_X \int_0^\infty \int_0^\infty \chi_{\{|f|\geq s\}\cap\{|g|\geq t\}}\,dtdsd\mu \\ &= \int_0^\infty \int_0^\infty \mu(\{|f|\geq s\}\cap\{|g|\geq t\})\,dtds \\ &\leq \int_0^\infty \int_0^\infty \min(\mu(\{|f|\geq s\}),\mu(\{|g|\geq t\}))\,dtds \\ &= \int_0^\infty \int_0^\infty \min(\mu(\{f^*\geq s\}),\mu(\{g^*\geq t\}))\,dtds \\ &= \int_0^\infty \int_0^\infty \int_0^\infty \chi_{\{f^*\geq s\}}(u)\chi_{\{g^*\geq t\}}(u)\,dudtds \\ &= \int_0^\infty f^*(u)g^*(u)\,du.\end{aligned}$$

1.4.2. We omit the proof of the $s=\infty$ case since $L^{p,\infty}(X)\subset L^{p,q}(X)$ always holds for $0<q<\infty$. Now for $q_1<\infty$ case, write $\gamma d_f(\gamma)^{\frac{1}{q_0}}\leq C_1$ and $\gamma d_f(\gamma)^{\frac{1}{q_1}}\leq C_2$. Since $q\in(q_0,q_1)$, we may always find $\delta_1\in(0,1)$ and $\delta_2>1$, such that

$$\frac{1}{q}=\frac{r_i}{q_0}+\frac{\delta_i-r_i}{q_1}$$

for each $i=0,1$ and some $r_i\in(0,\delta_i)$. Thence we may continue our estimate as follows:

$$\begin{aligned}\int_0^\infty (d_f(\gamma)^{\frac{1}{q}}\gamma)^s\frac{d\gamma}{\gamma} &= \int_0^1 (d_f(\gamma)^{\frac{1}{q}}\gamma)^s\frac{d\gamma}{\gamma}+\int_1^\infty (d_f(\gamma)^{\frac{1}{q}}\gamma)^s\frac{d\gamma}{\gamma} \\ &\leq \int_0^1 \left(\frac{C_1^{r_1}C_2^{\delta_1-r_1}}{\gamma^{\delta_1}}\gamma\right)^s\frac{d\gamma}{\gamma}+\int_1^\gamma \left(\frac{C_1^{r_2}C_2^{\delta_2-r_2}}{\gamma^{\delta_2}}\gamma\right)^s\frac{d\gamma}{\gamma} \\ &= \frac{C_1^{r_1}C_2^{\delta_1-r_1}}{s(1-\delta_1)}+\frac{C_1^{r_2}C_2^{\delta_2-r_2}}{s(\delta_2-1)}.\end{aligned}$$

Thus

$$\|f\|_{L^{q,s}}\leq q^{\frac{1}{s}}\left(\frac{C_1^{r_1}C_2^{\delta_1-r_1}}{s(1-\delta_1)}+\frac{C_1^{r_2}C_2^{\delta_2-r_2}}{s(\delta_2-1)}\right)^{\frac{1}{s}}.$$

For the $q_1=\infty$ case, suppose $|f|\leq C_2$ a.e., then $d_f(\gamma)=0$ for $\gamma>C_2$. Then

$$\begin{aligned}\int_0^\infty (d_f(\gamma)^{\frac{1}{q}}\gamma)^s\frac{d\gamma}{\gamma} &= \int_0^{C_2} (d_f(\gamma)^{\frac{1}{q}}\gamma)^s\frac{d\gamma}{\gamma} \\ &\leq \int_0^{C_2} (d_f(\gamma)^{\frac{1}{q}}\gamma^{\frac{q_0}{q}}\gamma^{\frac{q-q_0}{q}})^s\frac{d\gamma}{\gamma} \\ &\leq \int_0^{C_2} C_1^{\frac{s}{q}}\frac{d\gamma}{\gamma^{\frac{q(1-s)+q_0 s}{q}}} = \frac{q}{s(q-q_0)}C_1^{\frac{s}{q}}C_2^{\frac{s(q-q_0)}{q}}.\end{aligned}$$

Thus

$$\|f\|_{L^{q,s}}\leq\left(\frac{q^2}{s(q-q_0)}C_1^{\frac{s}{q}}C_2^{\frac{s(q-q_0)}{q}}\right)^{\frac{1}{s}}.$$

1.4.3. (a) For $t < \mu(X)$, we have

$$\begin{aligned}((f+g)^{**}(t))^r &= \sup_{\mu(E)\geq t} \frac{1}{\mu(E)} \int_E |f+g|^r d\mu \\ &\leq \sup_{\mu(E)\geq t} \frac{1}{\mu(E)} \int_E |f|^r d\mu + \sup_{\mu(E)\geq t} \frac{1}{\mu(E)} \int_E |g|^r d\mu \\ &= (f^{**}(t))^r + (g^{**}(t))^r\end{aligned}$$

since $r \leq 1$. Otherwise,

$$\begin{aligned}((f+g)^{**}(t))^r = \frac{1}{t}\int_X |f+g|^r &\leq \frac{1}{t}\int_X |f|^r + \frac{1}{t}\int_X |g|^r \\ &= (f^{**}(t))^r + (g^{**}(t))^r.\end{aligned}$$

For the second part, since $\frac{q}{r} > 1$, we have

$$\begin{aligned}|||f+g|||_{L^{p,q}}^r &= \left(\int_0^\infty (t^{\frac{1}{p}}(f+g)^{**}(t))^q \frac{dt}{t}\right)^{\frac{r}{q}} = \left(\int_0^\infty (t^{\frac{r}{p}-\frac{r}{q}}((f+g)^{**}(t))^r)^{\frac{q}{r}} \frac{dt}{t}\right)^{\frac{r}{q}} \\ &\leq \left(\int_0^\infty (t^{\frac{r}{p}-\frac{r}{q}}((f^{**}(t))^r + (g^{**}(t))^r))^{\frac{q}{r}} \frac{dt}{t}\right)^{\frac{r}{q}} \\ &\leq \left(\int_0^\infty (t^{\frac{r}{p}-\frac{r}{q}}(f^{**}(t))^r)^{\frac{q}{r}} \frac{dt}{t}\right)^{\frac{r}{q}} + \left(\int_0^\infty (t^{\frac{r}{p}-\frac{r}{q}}(g^{**}(t))^r)^{\frac{q}{r}} \frac{dt}{t}\right)^{\frac{r}{q}} \\ &= |||f|||_{L^{p,q}}^r + |||g|||_{L^{p,q}}^r.\end{aligned}$$

When $r = 1$, we have the property of scalar multiplication, whence $|||\cdot|||_{L^{p,q}}^r$ is a norm.

(b) The first inequality holds since $f^*(t) \leq f^{**}(t)$. This is quite obvious when $t \geq \mu(X)$, since $f^*(t) = 0$ in this case. For $t < \mu(X)$, take $s = f^*(t)$, then $d_f(s) \leq t$. Then $\forall\, \varepsilon > 0$, we have $d_f(s-\varepsilon) > t$ by definition. Set $\tilde{E} := \{|f| \geq s - \varepsilon\}$, we have

$$f^{**}(t) \geq \left(\frac{1}{\tilde{E}} \int_{\tilde{E}} |f|^r d\mu\right)^{\frac{1}{r}} \geq s - \varepsilon.$$

Since $\varepsilon$ is arbitrary, we have

$$f^{**}(t) \geq s = f^*(t).$$

Thus

$$|||f|||_{L^{p,q}} = \left(\int_0^\infty (t^{\frac{1}{p}} f^{**}(t))^q \frac{dt}{t}\right)^{\frac{1}{q}} \geq \left(\int_0^\infty (t^{\frac{1}{p}} f^*(t))^q \frac{dt}{t}\right)^{\frac{1}{q}} = \|f\|_{L^{p,q}}.$$

For the second inequality, we may seek for a proof for $f^{**}(t) \leq \left(\frac{p}{p-r}\right)^{\frac{1}{r}} f^*(t)$, but unfortunately this is untrue. (For $t > \mu(X)$, the right–hand side is definitely zero but unnecessarily for the left.) Note that the $f^{**}$ defined is controlled by $\left(\frac{1}{t}\int_0^t |f^*(s)|^r ds\right)^{\frac{1}{r}}$. We just show the

case of $0 \le t \le \mu(X)$. This is because

$$\begin{aligned}\sup_{\mu(E)\ge t}\left(\frac{1}{\mu(E)}\int_{\mu(E)}|f|^r d\mu\right)^{\frac{1}{r}} &= \sup_{\mu(E)\ge t}\left(\frac{1}{\mu(E)}\int_{\mu(E)}|f\chi_E|^r d\mu\right)^{\frac{1}{r}}\\ &= \sup_{\mu(E)\ge t}\left(\frac{1}{\mu(E)}\int_0^{\mu(E)}(f\chi_E)^*(s)^r ds\right)^{\frac{1}{r}}\\ &\le \sup_{\mu(E)\ge t}\left(\frac{1}{\mu(E)}\int_0^{\mu(E)}f^*(s)^r ds\right)^{\frac{1}{r}}\\ &\overset{\text{decreasing}}{\le}\left(\frac{1}{t}\int_0^t f^*(s)^r ds\right)^{\frac{1}{r}}.\end{aligned}$$

Thus,

$$\begin{aligned}|||f|||_{L^{p,q}} &\le \left(\int_0^\infty\left(t^{\frac{1}{p}}\left(\frac{1}{t}\int_0^t(f^*(t))^r ds\right)^{\frac{1}{r}}\right)^q\frac{dt}{t}\right)^{\frac{1}{q}}\\ &\le \left(\left(\int_0^\infty\left(\frac{1}{t}\int_0^t(f^*(t))^r ds\right)^{\frac{q}{r}}\right)^q t^{\frac{q}{p}-\frac{q}{r}}\frac{dt}{t}\right)^{\frac{r}{q}}\Bigg)^{\frac{1}{r}}\\ &\overset{\text{Hardy}}{\le}\left(\frac{\frac{q}{r}}{\frac{q}{p}-\frac{q}{r}}\right)^{\frac{1}{r}}\left(\int_0^\infty(t(f^*(t))^r\right)^{\frac{q}{r}}t^{\frac{q}{p}-\frac{q}{r}}\frac{dt}{t}\Bigg)^{\frac{1}{q}} = \left(\frac{p}{p-r}\right)^{\frac{1}{r}}\|f\|_{L^{p,q}}.\end{aligned}$$

(c) When $p, q > 1$, we may take $r = 1$, then $\|\cdot\|_{L^{p,q}}$ will be equivalent to the norm $|||\cdot|||_{L^{p,q}}$.

1.4.4. (a) Write $f = \sum_{n\in\mathbb{Z}} f\chi_{(2^n,2^{n+1}]}$ and $f_n = f\chi_{(2^n,2^{n+1}]}$. Since $\gamma d_f(\gamma)^{\frac{1}{p}} \le C < \infty$, we may fix a $q < \infty$, then

$$\begin{aligned}\|f_n\|_{L^{p,q}}^q &= p\int_0^\infty(\gamma d_{f_n}(\gamma)^{\frac{1}{p}})^q\frac{d\gamma}{\gamma}\\ &\le p\int_{2^n}^{2^{n+1}} C^q\frac{d\gamma}{\gamma} = pC^q\log 2.\end{aligned}$$

Thus, for any $f_n$, choose finitely combined simple functions $\phi_{n,\varepsilon}$ such that

$$\|f_n - \phi_{n,\varepsilon}\|_{L^{p,q}} \le \frac{\varepsilon}{2^{|n|+1}}\left(\frac{p}{q}\right)^{\frac{1}{q}}.$$

Then the estimate follows that (for $p > 1$, using the fact that $L^{p,\infty}$ is actually metrizable)

$$\begin{aligned}\|f - \sum_{n\in\mathbb{Z}}\phi_{n,\varepsilon}\|_{L^{p,\infty}} &\le \sum_{n\in\mathbb{Z}}\|f_n - \phi_{n,\varepsilon}\|_{L^{p,\infty}}\\ &\le \left(\frac{q}{p}\right)^{\frac{1}{q}}\sum_{n\in\mathbb{Z}}\frac{\varepsilon}{2^{|n|+1}}\left(\frac{p}{q}\right)^{\frac{1}{q}} < \varepsilon.\end{aligned}$$

For $0 < p \le 1$, just take the power of $r$ for some suitable $0 < r < p$ described in [Ex.1.1.12], and make some necessary changes to get the same estimate.

(b) Write $s = \sum_{i=1}^n a_i\chi_{A_i}$ and $A = \bigcup_{i=1}^n A_i$. Set $M = \max s = \max_{1\le i\le n} a_i$, then we have

$$\{|s-h| > \gamma\} \supseteq \{h > \gamma + s\} \supseteq \{h > \gamma + M\},$$

which implies

$$\big||s-h| > \gamma\big| \ge \big|h > \gamma + M\big| = (\gamma + M)^{-p}.$$

For this reason,

$$\|s-h\|_{L^{p,\infty}} = \sup_{\gamma>0}\gamma\big||s-h| > \gamma\big|^{1/p} \ge \sup_{\gamma>0}\gamma/(\gamma+M) = 1.$$

1.4.10. (a) Take $u_n = f_n\chi_{|f_n|\le\frac{\alpha}{2}}$, $v_n = f_n\chi_{|f_n|>\frac{\alpha}{2\lambda_n}}$ and $w_n = f_n\chi_{\frac{\alpha}{2}<|f_n|\le\frac{\alpha}{2\lambda_n}}$. Thus by setting $u = \sum_n \lambda_n u_n$, $v = \sum_n \lambda_n v_n$ and $w = \sum_n \lambda_n w_n$, we obtain a decomposition $f = u + v + w$. Clearly $|u| \le \frac{\alpha}{2}$. For $v$, we have

$$\begin{aligned}\mu(\{v \neq 0\}) &\le \sum_n \mu(|f_n| > \frac{\alpha}{2\lambda_n}) = \sum_n \mu(|\lambda_n f_n| > \frac{\alpha}{2})\\ &\le \sum_n \frac{2\lambda n}{\alpha} \le \frac{2}{\alpha}.\end{aligned}$$

Plus, for $\{|w| > \frac{\alpha}{2}\}$, we have

$$\begin{aligned}\frac{\alpha}{2}\mu(\{|w| > \frac{\alpha}{2}\}) &\le \int_{\{|w|>\frac{\alpha}{2}\}} |w|\, d\mu \le \int_X |w|\, d\mu\\ &\le \sum_n \int_X |f_n|\chi_{\frac{\alpha}{2}<|f_n|\le\frac{\alpha}{2\lambda_n}} d\mu\\ &\le \sum_n \lambda_n \bigg( \int_{\frac{\alpha}{2}}^{\frac{\alpha}{2\lambda_n}} d_{f_n}(t)\, dt + \int_0^{\frac{\alpha}{2}} d_{f_n}\Big(\frac{\alpha}{2}\Big)\, dt \bigg)\\ &\le \sum_n \lambda_n \bigg( \int_{\frac{\alpha}{2}}^{\frac{\alpha}{2\lambda_n}} \frac{dt}{t} + \int_0^{\frac{\alpha}{2}} \frac{2}{\alpha} dt \bigg)\\ &\le \sum_n \lambda_n(\log(\lambda_n) + 1) \le K + 1.\end{aligned}$$

Now combined together, we have

$$\begin{aligned}\mu(\{|f| > \alpha\}) &\le \mu(\{|u| > \frac{\alpha}{2}\}) + \mu(\{v \neq 0\}) + \mu(\{|w| > \frac{\alpha}{2}\})\\ &\le 0 + \frac{2}{\alpha} + \frac{2}{\alpha}(K+1) = \frac{2(K+2)}{\alpha},\end{aligned}$$

which is exactly $\|f\|_{L^{1,\infty}} \le 2(K+2)$.

(b) Since

$$\begin{aligned}\|Tf\|_{L^{1,\infty}} &= \|\sum_n \lambda_n T_n f\|_{L^{1,\infty}}\\ &\le 2(K+2)\sup_n \|T_n f\|_{L^{1,\infty}} \le 2B(K+2)\|f\|_{L^1}.\end{aligned}$$

(c) Since

$$\begin{aligned}\mu(\{\sum_{n=1}^{\infty} 2^{-\delta n} f_n > \alpha\}) &\le \sum_{n=1}^{\infty} \mu(\{2^{-\delta n} f_n > (2^\varepsilon - 1)2^{-\varepsilon n}\alpha\})\\ &\le \sum_{n=1}^{\infty} \frac{2^{-\delta n}}{(2^\varepsilon - 1)2^{-\varepsilon n}\alpha} = \sum_{n=1}^{\infty} \frac{2^{(\varepsilon-\delta)n}}{(2^\varepsilon - 1)\alpha}\\ &\le \frac{2^{\delta-\varepsilon} - 1}{(2^\varepsilon - 1)\alpha}.\end{aligned}$$

1.4.11. First we show

(a) (Fatou's Lemma in $L^{p,q}$)

(b) We know

$$\sup_{\alpha>0} \alpha d_{g_n - g}(\alpha)^{\frac{1}{s}} \le \Big(\frac{s}{q}\Big)^{\frac{1}{s}} \|g_n - g\|_{L^{q,s}}$$

Since $g_n \overset{L^{q,s}}{\to} g$, we have by this control that $g_n \to g$ in measure. Now choose a subsequence $g_{n_k}$ such that $g_{n_k} \to g$ a.e.

1.4.13. (Dyadic Decomposition) Set $c_n = 2^{\frac{n}{p}} f^*(2^n)$, $A_n := \{f^*(2^{n+1}) \le |f| \le f^*(2^n)\}$ and $f_n = c_n^{-1} f \chi_{A_n}$. Obviously, we have $f = \sum_{n\in\mathbb{Z}} f_n$. Since $f^*$ is decreasing, we have

$$|f_n| \le 2^{-\frac{n}{p}} f^*(2^n)^{-1} f^*(2^n) = 2^{-\frac{n}{p}}.$$

Clearly, support$(f_n) \subset A_n$, whose measure is $2^n$. For the last matter, set $E_n = [2^n, 2^{n+1}]$. Thus

$$\begin{aligned}\|f\|_{L^{p,q}}^q &= \int_0^\infty (t^{\frac{1}{p}} f^*(t))^{\frac{1}{q}} \frac{dt}{t} = \sum_{n\in\mathbb{Z}} \int_{E_n} (t^{\frac{1}{p}} f^*(t))^{\frac{1}{q}} \frac{dt}{t} \\ &\ge \sum_{n\in\mathbb{Z}} 2^{\frac{nq}{p}} f^*(2^{n+1})^q \int_{E_n} \frac{dt}{t} = \log 2 \sum_{n\in\mathbb{Z}} 2^{\frac{(n-1)q}{p}} f^*(2^n)^q \\ &= 2^{-\frac{q}{p}} \log 2 \|\{c_k\}_k\|_{l^q},\end{aligned}$$

which is enough to show the boundedness of $\|\{c_k\}_k\|_{l^q}$. Another direction follows from the estimate on each $E_n$ that $f^*(t) \le f^*(2^n)$ and $t \le 2^{n+1}$ and use a shift.

1.4.14. (a) Take $E' = E \cap \{|f| \le A(1-\gamma)^{-\frac{1}{p}} \mu(E)^{-\frac{1}{p}}\}$, then

$$d_f(A(1-\gamma)^{-\frac{1}{p}} \mu(E)^{-\frac{1}{p}}) \le (1-\gamma)\mu(E)$$

by the definition of $A$. Thus

$$\begin{aligned}\left|\int_{E'} f\, d\mu\right| &\le \int_{E'} |f|\, d\mu \\ &\le \mu(E) A(1-\gamma)^{-\frac{1}{p}} \mu(E)^{-\frac{1}{p}} \\ &= A(1-\gamma)^{-\frac{1}{p}} \mu(E)^{1-\frac{1}{p}}.\end{aligned}$$

(b) Since $\{|f| > \alpha\} \subset \{\mathrm{Re}\, f > \frac{\alpha}{\sqrt{2}}\} \cup \{\mathrm{Re}\, f < -\frac{\alpha}{\sqrt{2}}\} \cup \{\mathrm{Im}\, f > \frac{\alpha}{\sqrt{2}}\} \cup \{\mathrm{Im}\, f < -\frac{\alpha}{\sqrt{2}}\} = \bigcup_{i=1}^4 F_i$, set $E_{n,i} = X_n \cap F_i$, where $\{X_n\}$ is an ascending chain of finite measure subspace with union equal to $X$. Then by condition, we have

$$\left|\int_{E'_{n,i}} f\, d\mu\right| \ge \frac{\alpha}{\sqrt{2}} \gamma \mu(E_{n,i}).$$

Thus $\alpha\mu(E_n)^{\frac{1}{p}} \le B\sqrt{2}\gamma^{-1}$. Let $n \to \infty$, then $\alpha\mu(F_i)^{\frac{1}{p}} \le B\sqrt{2}\gamma^{-1}$. Thus,

$$\alpha^p \mu(\{|f| > \alpha\}) \le \sum_{i=1}^4 \alpha^p \mu(F_i) \le 4B^p \sqrt{2}^p \gamma^{-p},$$

which means $\|f\|_{L^{p,\infty}} \le B 4^{\frac{1}{p}} \sqrt{2}\gamma^{-1}$.

## 1.4 解答: GTM249 Chapter 2 Part I

原网页

## 2.1 Maximal Functions

2.1.1 (a) Similar to the proof of Theorem 2.1.6, except for the only difference

$$D(\mu)\sum_{r=1}^{l}\mu(B_{j_r})\geq\mu(\bigcup_{i=1}^{k}B_i).$$

(b)

$$\lim_{r\to 0}\frac{1}{\mu(B(x,r))}\int_{B(x,r)}f(y)\,dy=f(x).$$

2.1.2 (a) Select a subset $\mathscr{G}$ of $\mathscr{F}$ with the minimal cardinality such that $\bigcup_{J\in\mathscr{G}}J=\bigcup_{I\in\mathscr{F}}I$. Assume that $\bigcup_{J\in\mathscr{G}}J$ is connected, otherwise consider every connected component.

Write $\mathscr{G}=\{(a_i,b_i)\mid 1\leq i\leq n\}$. Let $\mathscr{G}_1=\{(a_i,b_i)\mid i \text{ is odd}\}$. We claim that $\mathscr{G}_1$ is a family of mutually disjoint intervals. Otherwise there exists $j$ such that $(a_j,b_j)\cap(a_{j+2},b_{j+2})\neq\varnothing$. If $b_{j+1}\leq b_{j+2}$, then $(a_{j+1},b_{j+1})\subset(a_j,b_{j+2})=(a_j,b_j)\cup(a_{j+2},b_{j+2})$, so $(a_{j+1},b_{j+1})$ can be removed; if $b_{j+1}>b_{j+2}$, then $(a_{j+2},b_{j+2})\subset(a_{j+1},b_{j+1})$, so $(a_{j+2},b_{j+2})$ can be removed. In both cases the minimality of $\mathscr{G}$ fails.

Similarly we can prove that $\mathscr{G}_2$ is a family of mutually disjoint intervals.

Similar to the proof of Theorem 2.1.6, except for the only difference

$$\mu(K)\leq\mu(\bigcup_{B_x\in\mathscr{F}}B_x)\leq\mu(\bigcup_{B_x\in\mathscr{G}_1}B_x)+\mu(\bigcup_{B_x\in\mathscr{G}_2}B_x)\leq\frac{2}{\alpha}\int_{E_\alpha}|f|\,d\mu.$$

(b) The first inequality is equivalent to

$$\int_X(|f|-\alpha)(\chi_A-\chi_{|f|>\alpha})\,d\mu\leq 0.$$

This inequality holds because the integrand is nonpositive whether $|f|>\alpha$ or $|f|\leq\alpha$.

Set $E_\alpha=\{M_\mu(f)>\alpha\}$, the following notation follows from the proof of part (a).

Let $F_1=\bigcup_{B_x\in\mathscr{G}_1}B_x\cap K, F_2=\bigcup_{B_x\in\mathscr{G}_2}B_x\cap K$, where $K$ is a compact subset of $E_\alpha$. We have

$$\begin{aligned}\mu(K)+\mu(F_1\cap F_2)=\mu(F_1)+\mu(F_2)&\leq\frac{1}{\alpha}\Big(\int_{F_1}|f|\,d\mu+\int_{F_2}|f|\,d\mu\Big)\\&=\frac{1}{\alpha}\Big(\int_K|f|\,d\mu+\int_{F_1\cap F_2}|f|\,d\mu\Big).\end{aligned}$$

Take $A=F_1\cap F_2$ in the first inequality and get

$$\frac{1}{\alpha}\int_{F_1\cap F_2}|f|\,d\mu-\mu(F_1\cap F_2)\leq\frac{1}{\alpha}\int_{\{|f|>\alpha\}}|f|\,d\mu-\mu(\{|f|>\alpha\}).$$

Add up the two inequalities, we obtain

$$\mu(\{|f|>\alpha\})+\mu(K)\leq\frac{1}{\alpha}\Big(\int_K|f|\,d\mu+\int_{\{|f|>\alpha\}}|f|\,d\mu\Big).$$

Take the supremum in $K$, and then the required conclusion follows.

When $\alpha=1$ and $f(x)=|x|^{-\frac{1}{p}}$, $M(f)(x)=a^{-\frac{1}{p}}$, where $a$ is the positive root of

$$-\frac{1}{p}a^{1-\frac{1}{p}}+|x|\Big(\Big(1-\frac{1}{p}\Big)a^{-\frac{1}{p}}-|x|^{-\frac{1}{p}}\Big)=0.$$

$$\{M(f)(x)>1\}=\Big\{-\frac{1}{p}+|x|\Big(\Big(1-\frac{1}{p}\Big)-|x|^{-\frac{1}{p}}\Big)<0\Big\}\triangleq(-A,A).$$

So we have

$$\int_{\{M(f)(x)>1\}}(|f|-1)=\frac{p}{p-1}A^{1-\frac{1}{p}}-A=-\frac{2}{p-1}=-\int_{\{|f|>1\}}(|f|-1).$$

(c) Multiplying the last inequality of part (b) by $p\alpha^{p-1}$, and integrating with respect to $\alpha$ from 0 to $\infty$ yields

$$\begin{aligned}\|f\|_{L^p}^p+\|M(f)\|_{L^p}^p&\le\frac{p}{p-1}\Big(\int_{\mathbb{R}}M(f)^{p-1}|f|\,d\mu+\|f\|_{L^p}^p\Big)\\&\le\frac{p}{p-1}(\|M(f)\|_{L^p}^{p-1}\|f\|_{L^p}+\|f\|_{L^p}^p).\end{aligned}$$

Let $t=\frac{\|M(f)\|_{L^p}}{\|f\|_{L^p}}$, then

$$(p-1)t^p-pt^{p-1}-1\le 0.$$

So $\|f\|_{L^p\to L^p}\le A_p$.

(d) According to part (b), $\frac{M(f)(x)}{|x|^{-\frac{1}{p}}}$ is the positive root of

$$(p-1)x^p-px^{p-1}-1,$$

which means $M(f)(x)=A_p$.

Consider $f_{\varepsilon,N}(x):=|x|^{-\frac{1}{p}+\varepsilon}\chi_N(|x|)$, where $\chi_N=\chi_{[0,N]}$.

It is easy to see that

$$\lim_{N\to\infty}\lim_{\varepsilon\to 0}\frac{\|M(f_{\varepsilon,N})\|_{L^p}}{\|f_{\varepsilon,N}\|_{L^p}}\ge A_p.$$

2.1.3 (a) Obviously, $\mathcal{M}(f)\le M(f)$.

For $B(z,r)$ containing $x$,

$$\frac{1}{|B(z,r)|}\int_{B(z,r)}|f|\le\frac{2^n}{|B(x,2r)|}\int_{B(x,2r)}|f|.$$

Take the supremum to get $M(f)\le 2^n\mathcal{M}(f)$.

For a ball $B$ containing $x$, $C_B$ denotes the minimal cube containing $B$, then

$$\frac{1}{|B|}\int_B|f|\le\frac{|C_B|}{|B|}\cdot\frac{1}{|C_B|}\int_{C_B}|f|=\frac{2^n}{\nu_n}\frac{1}{|C_B|}\int_{C_B}|f|.$$

Take the supremum to get $M(f)\le\frac{2^n}{\nu_n}M_c(f)$.

For a cube $C$ containing $x$, $B_C$ denotes the minimal ball containing $C$, then

$$\frac{1}{|C|}\int_C|f|\le\frac{|B_C|}{|C|}\cdot\frac{1}{|B_C|}\int_{B_C}|f|=\frac{n^{\frac{n}{2}}\nu_n}{2^n}\frac{1}{|B_C|}\int_{B_C}|f|.$$

Take the supremum to get $M_c(f)\le\frac{n^{\frac{n}{2}}\nu_n}{2^n}M(f)$.

Similar process can be applied to obtain the third inequality.

For $f\in L^1(\mathbb{R}^n)$,

$$\alpha d_{M_c(f)}(\alpha)\le\alpha d_{M(f)}\Big(\frac{2^n\alpha}{n^{\frac{n}{2}}\nu_n}\Big)\le\frac{n^{\frac{n}{2}}\nu_n}{2^n}\|M(f)\|_{L^{1,\infty}}.$$

So $M_c(f) \in L^{1,\infty}(\mathbb{R}^n)$.

For $f \in L^p(\mathbb{R}^n)$,

$$\|M_c(f)\|_{L^p} \le \frac{n^{\frac{n}{2}}\nu_n}{2^n}\|M(f)\|_{L^p}.$$

So $M_c(f) \in L^p(\mathbb{R}^n)$.

Similarly, $\mathcal{M}_c$ has the same property.

2.1.4 (a) Write $f = f\chi_{\{|f|>\alpha/2\}} + f\chi_{\{|f|\le\alpha/2\}} := f_1 + f_2$.

Notice that $M(f) \le M(f_1) + M(f_2) \le M(f_1) + \alpha/2$, so $\{M(f) > \alpha\} \subset \{M(f_1) > \alpha/2\}$.

From Theorem 2.1.6,

$$\mu(\{M(f) > \alpha\}) \le \mu(\{M(f_1) > \alpha/2\}) \le \frac{3^n}{\alpha/2}\int |f_1|\,dy = \frac{3^n}{\alpha/2}\int_{\{|f|>\alpha/2\}} |f|\,dy.$$

So

$$\alpha^p\mu(\{M(f) > \alpha\}) = 2^p \cdot 3^n(\alpha/2)^{p-1}\int_{\{|f|>\alpha/2\}} |f|\,dy \le 2^p \cdot 3^n\|f\|_{L^p}^p,$$

and hence

$$\|M(f)\|_{L^{p,\infty}} \le 2 \cdot 3^{n/p}\|f\|_{L^p}.$$

(b) Write $f = f\chi_{\{|f|>1/2\}} + f\chi_{\{|f|\le 1/2\}} := f_1 + f_2$.

Taking $p = 1$ in part (a) yields $\alpha d_{M(f_1)}(\alpha) \le 2 \cdot 3^n\|f_1\|_{L^1}$, so

$$\begin{aligned}
\|M(f_1)\|_{L^1(B)} &= \int_1^\infty d_{M(f_1)}(\alpha)\,d\alpha + \int_0^1 d_{M(f_1)}(\alpha)\,d\alpha \\
&\le \int_1^\infty \frac{3^n}{\alpha/2}\int_{\{|f|>1/2\}} |f|\,dyd\alpha + \int_0^1 |B|\,d\alpha \\
&= 2 \cdot 3^n\int_B \left(\int_1^{\max(1,2|f|)} \frac{d\alpha}{\alpha}\right)|f|\,dy + |B| \\
&= 2 \cdot 3^n\|f\log^+(2|f|)\|_{L^1(B)} + |B|.
\end{aligned}$$

Then $M(f)$ is also integrable over $B$, as $M(f) \le M(f_1) + M(f_2) \le M(f_1) + 1/2$.

(c) From Theorem 2.1.20, there exists a collection of disjoint open cubes $Q_j$ such that $\bigcup Q_j \subset \{M_c(f) > \alpha\}$ and $\bigcup_j Q_j \supset \{f > \alpha\}$, so

$$\int_{\{|f|>\alpha\}} |f| \le \sum\int_{Q_j} |f| \le 2^n\alpha\sum|Q_j|.$$

From Exercise 2.1.3,

$$|\{M(f) > c_n\alpha\}| \ge |\{M_c(f) > \alpha\}| \ge \sum|Q_j| \ge \frac{1}{2^n\alpha}\int_{\{|f|>\alpha\}} |f|.$$

(d) Let $x' = \rho^2|x|^{-2}x$ for some $\rho < |x| < 2\rho$. For $y \in B(0,\rho)$, we have

$$|x-y|^2 - |x'-y|^2 \ge (|x| - |x'|)(|x| + |x'| - 2|y|) > 0.$$

Consequently, for $R > |x| - \rho$, $B(x,R) \cap B(0,\rho) \subset B(x',R)$, and thus

$$\int_{B(x,R)} |f| = \int_{B(x,R)\cap B(0,\rho)} |f| \le \int_{B(x',R)} |f|,$$

which implies that

$$\mathcal{M}(f)(x) \le \mathcal{M}(f)(\rho^2|x|^{-2}x).$$

Using polar coordiantes, the determinant of the map $x \mapsto \rho^2|x|^{-2}x$ is $\pm(\rho/|x|)^{2n}$. Thus we have

$$\begin{aligned}\int_{B(0,2\rho)} \mathcal{M}(f)(x)\,dx &\le \int_{B(0,\rho)} \mathcal{M}(f)(x)\,dx + \int_{B(0,2\rho)\setminus B(0,\rho)} \mathcal{M}(f)(\rho^2|x|^{-2}x)\,dx \\ &= \int_{B(0,\rho)} \mathcal{M}(f)(x)\,dx + \int_{B(0,\rho)\setminus B(0,\rho/2)} \mathcal{M}(f)(y)\Big(\frac{|y|}{\rho}\Big)^{2n} dy \\ &\le (4^n+1)\int_{B(0,\rho)} \mathcal{M}(f)(x)\,dx.\end{aligned}$$

From Exercise 2.1.3, we conclude that

$$\int_{B(0,2\rho)} M(f)(x)\,dx \le 2^n(4^n+1)\int_{B(0,\rho)} M(f)(x)\,dx.$$

(e) For $x \notin 2B$,

$$M(f)(x) = \sup_{x\in\widetilde{B},\operatorname{diam}\widetilde{B}>\rho} \frac{1}{|\widetilde{B}|}\int_{\widetilde{B}} |f|\,dx < \frac{2^n}{|B|}\|f\|_{L^1} = \lambda_0.$$

Therefore we have

$$\begin{aligned}\int_{2B} M(f)(x)\,dx &\ge \int_{c_n^{-1}\lambda_0}^{\infty} |\{x\in 2B \mid M(f)(x) > c_n\alpha\}|\,d\alpha \\ &= \int_{c_n^{-1}\lambda_0}^{\infty} |\{M(f) > c_n\alpha\}|\,d\alpha \\ &\ge \int_{c_n^{-1}\lambda_0}^{\infty} \frac{2^{-n}}{\alpha}\int_{\{|f|>\alpha\}} |f(y)|\,dyd\alpha \\ &= 2^{-n}\int_{\{|f|>\alpha\}} \Big(\int_{c_n^{-1}\lambda_0}^{\max(|f|,c_n^{-1}\lambda_0)} \frac{d\alpha}{\alpha}\Big)|f(y)|\,dy \\ &= 2^{-n}\|f\log^+(\lambda_0^{-1}c_n|f|)\|_{L^1},\end{aligned}$$

where the second inequality is from part (c).

From part (d),

$$\int_{2B} M(f)(x)\,dx \le 2^n(4^n+1)\int_B M(f)(x)\,dx,$$

and then the required conclusion follows.

2.1.5 We have

$$\begin{aligned}&\int_A |S(f)|^q dx \\ &= \int_0^\infty q\alpha^{q-1}|A\cap\{S(f)>\alpha\}|\,d\alpha \\ &\le \int_0^{\frac{B}{|A|}\|f\|_{L^1}} q\alpha^{q-1}|A|\,d\alpha + \int_{\frac{B}{|A|}\|f\|_{L^1}}^\infty q\alpha^{q-1}\frac{B}{\alpha}\|f\|_{L^1}d\alpha \\ &\le \Big(\frac{B}{|A|}\|f\|_{L^1}\Big)^q|A| + \frac{q}{1-q}\Big(\frac{B}{|A|}\|f\|_{L^1}\Big)^{q-1}B\|f\|_{L^1} \\ &= \frac{1}{1-q}B^q\|f\|_{L^1}^q|A|^{1-q}.\end{aligned}$$

The second result follows from taking $S = M$, then $B = 3^n$.

2.1.6 (a) Define

$$M_1^{(i)} := \frac{1}{b_i - a_i} \sup_{(a_i,b_i)\ni x_i} \int_{a_i}^{b_i} |f(y_1, \cdots, y_n)|\, dy_i.$$

It is obvious that $M_s(f)(x) = M_1^{(1)} \circ \cdots \circ M_1^{(n)}(f)(x)$.

Thus,

$$\begin{aligned}\int_{\mathbb{R}^n} M_s(f)(x)^p dx &\le \int_{\mathbb{R}^n} M_1^{(1)} \circ \cdots \circ M_1^{(n)}(f)(x)^p dx_1 \cdots dx_n \\ &\le A_p^p \int_{\mathbb{R}^{n-1}} M_1^{(2)} \circ \cdots \circ M_1^{(n)}(f)(x)^p dx_2 \cdots dx_n \\ &\le \cdots \\ &\le A_p^{pn} \|f\|_{L^p}^p.\end{aligned}$$

(b) Consider $g_{\varepsilon,N}(x) := \prod_{j=1}^{n} f_{\varepsilon,N}(x_j)$ where $f_{\varepsilon,N}(x_j) := |x_j|^{-\frac{1}{p}+\varepsilon} \chi_N(|x_j|)$.

From Exercise 2.1.2 (c), $\|M_s\| \ge A_p^n$, so $\|M_s\| = A_p^n$.

(c) Consider $f_m = m^n \chi_{[0,\frac{1}{m}]^n}$, then $\|f\|_{L^1} = 1$.

For $x \in E_m := \left\{ \prod_{i=1}^{n} x_i \le 1, x_i \ge \frac{1}{m}, i = 1, \cdots, n \right\}$, write $R_x = \prod_{i=1}^{n} [0, x_i]$, we have

$$M_s(f_m)(x) \ge \frac{1}{|R_x|} \int_{R_x} f_m(x)\, dx = \frac{1}{|R_x|} \|f_m\|_{L^1} \ge 1.$$

Thus

$$\frac{\|M_s(f_m)\|_{L^{1,\infty}}}{\|f_m\|_{L^1}} \ge |\{M_s(f_m) \ge 1\}| \ge |E_m|.$$

Yet let $m \to \infty$, by the Lebesgue monotone convergence theorem and Tonelli's theorem,

$$\begin{aligned}\lim_{m\to\infty} |E_m| &= \int_{\prod_{i=1}^{n} x_i \le 1, x_i > 0, i=1,\cdots,n} \mathrm{d}x_1 \cdots \mathrm{d}x_n \\ &= \int_{\mathbb{R}^{n-1}} \frac{1}{x_2 \cdots x_n} \mathrm{d}x_2 \cdots \mathrm{d}x_n \left( \int_{\mathbb{R}} \frac{1}{x} \mathrm{d}x \right)^{n-1} = \infty.\end{aligned}$$

So $M_s$ is not of weak type (1,1).

2.1.7 Write $\psi(x) = (1 + |x|)^{-1-\varepsilon}$, $x \in \mathbb{R}$, and $T(f) := \sup_{t_1,\ldots,t_n>0} |f * \varphi_{t_1,\ldots,t_n}|$.

Notice that

$$\begin{aligned}|f * \varphi_{t_1,\ldots,t_n}| &\le A \left| f * ((1 + |x_1|)^{-1-\varepsilon} \cdots (1 + |x_n|)^{-1-\varepsilon}) \right| &(1.1)\\ &\le A\|\psi\|_{L^1} M^{(1)}(f * ((1 + |x_2|)^{-1-\varepsilon} \cdots (1 + |x_n|)^{-1-\varepsilon})) &(1.2)\\ &\le \cdots &(1.3)\\ &\le A\|\psi\|_{L^1}^n M^{(1)} \circ \cdots \circ M^{(n)}(f) = A\|\psi\|_{L^1}^n M_s(f). &(1.4)\end{aligned}$$

2.1.8 Let $f_0$ be the characteristic function of the unit ball $B$ in $\mathbb{R}^n$. Note that

$$M(f_0)(x) \ge \frac{1}{|B_x|} \int_{B_x} f_0(y)\, dy,$$

where $B_x = B\big(\frac{1}{2}(|x| - |x|^{-1})\frac{x}{|x|}, \frac{1}{2}(|x| + |x|^{-1})\big)$ for $|x| > 1$.
For $z \in B(0,1)$,

$$\begin{aligned} z \in B_x &\Leftrightarrow \big|z - \tfrac{1}{2}(|x| - |x|^{-1})\tfrac{x}{|x|}\big|^2 < \big|\tfrac{1}{2}\big(|x| + |x|^{-1}\big)\big|^2 \\ &\Leftrightarrow |z|^2 - (|x| - |x|^{-1})\tfrac{z\cdot x}{|x|} < \big|\tfrac{1}{2}\big(|x| + |x|^{-1}\big)\big|^2 - \big|\tfrac{1}{2}\big(|x| - |x|^{-1}\big)\big|^2 \\ &\Leftrightarrow (-|x| + |x|^{-1})\tfrac{z\cdot x}{|x|} < 1 - |z|^2 \\ &\Leftarrow z \cdot x > 0. \end{aligned}$$

This means half of the unit ball $\{z \in B \mid z \cdot x > 0\}$ is contained in $B_x$, implying that

$$M(f_0)(x) \geq \frac{|B|/2}{|B_x|} = \big(\tfrac{1}{2}(|x| + |x|^{-1})\big)^{-n}/2.$$

So

$$\begin{aligned} \frac{\|M(f_0)\|_{L^p}^p}{\|f_0\|_{L^p}^p} &\geq \frac{1}{|B|}\int_{|x|>1} \big[\big(\tfrac{1}{2}(|x| + |x|^{-1})\big)^{-n}/2\big]^p dx \\ &= \frac{n}{2^p}\int_1^\infty r^{n-1}\big(\tfrac{1}{2}(r + r^{-1})\big)^{-np} dr \end{aligned}$$

Let $f(r) = r\big(\frac{1}{2}(r + r^{-1})\big)^{-p}$. Then $f(1) = 1$ and $f'(1) = 1$, so $f(r) > 1$ on some interval $(1, r_0)$. Therefore,

$$\frac{\|M(f_0)\|_{L^p}^p}{\|f_0\|_{L^p}^p} \geq \frac{n}{2^p}\int_1^\infty \frac{f(r)^n}{r} dr > \frac{n}{2^p}\int_1^{r_0} \frac{dr}{r} \to \infty \quad \text{as } n \to \infty.$$

2.1.9 Consider the Besicovitch Set $E_\varepsilon$ (see Elias M. Stein, *Harmonic Analysis*, X.1, p.435), which satisfies (1) $E_\varepsilon$ is a union of $2^N$ rectangles $R_1, \cdots, R_{2^N}$, each having side lengths 1 and $2^{-N}$, and $|E_\varepsilon| < \varepsilon$, and
(2) the $\widetilde{R}_j$'s are mutually disjoint, $j = 1, \cdots, 2^N$. Here the $\widetilde{R}_j$ is defined to be the rectangle obtained by translating $R_j$ by two units along the longer side of $R_j$.
So $\|\chi_{E_\varepsilon}\|_{L^p}^p < \varepsilon$, and for $x \in \widetilde{R}_j$, $M_0(\chi_{E_\varepsilon})(x) \geq \frac{1}{3}$, which implies that

$$\|M_0(\chi_{E_\varepsilon})\|_{L^p}^p \geq \frac{1}{3}.$$

Let $\varepsilon \to 0$, and then we can see the unboundedness.
Notice that $M_0$ is weak type $(\infty, \infty)$, if $M_0$ is weak type $(1,1)$, from the Marcinkiewicz interpolation theorem, $M_0$ is strong type $(2,2)$. Contradiction!
(b) Lemma: Let $\{B_1, \cdots, B_k\}$ be a finite collection of open rectangles in $\mathbb{R}^2$, whose eccentricity is $b$. Then there exists a finite subcollection of pairwise disjoint rectangles $\{B_{j_1}, \cdots, B_{j_l}\}$ such that

$$16b^2 \sum_{r=1}^{l} |B_{j_r}| \geq |\bigcup_{i=1}^{k} B_i|.$$

Proof of lemma: First notice that if two rectangles with eccentricity $b$ intersect, we claim that the smaller one $B$ is contained in the bigger one $A$ scaled $4b$ times.

We place the center of $A$ at 0, the longer edge being parallel to the $x$-axis. Let the length of the edge of $A, B$ be $1, b; c, bc$, where $c < 1$.

So the abscissa of points in $B$ is no more than $\frac{1}{2} + \operatorname{diam} B \leq 4b \cdot \frac{1}{2}$, and the ordinate of points in $B$ is no more than $\frac{b}{2} + \operatorname{diam} B \leq 4b \cdot \frac{b}{2}$.

This implies $4bA \supset B$.

Then follows the proof of Lemma 2.1.5 to prove the lemma. Similar to the proof of Theorem 2.1.6, we see that

$$\|M_{00}(f)\|_{L^{1,\infty}} \leq 16b^2 \|f\|_{L^1}.$$

(c) Notice that for rectangles $A, B$ satisfying the assumption above, if $A$ and $B$ intersect, $|A| \geq |B|$, then $3A \subset B$. The coefficient 3 does not depend on the $\{a_2, \cdots, a_n\}$.

Then we can deduce the similar conclusion as in part (b).

2.1.10 (a) Define $O_f(y) = \limsup_{\varepsilon \to 0} |T_\varepsilon(f)(y)|$, then for $y \in D$,

$$O_f(y) \leq K\big((O_{f-g}(y) + O_g(y)\big) = KO_{f-g}(y).$$

We claim $\mu\{O_f(y) > \delta\} = 0, \forall \delta > 0$.

Note that $\forall \eta > 0, \exists g \in D, \|f - g\|_{L^p} < \eta$. So we have

$$\begin{aligned}\mu\{O_f(y) > \delta\} &\leq \mu\{O_{f-g}(y) > \delta/K\} \leq \mu\{T_*(f-g) > \delta/K\} \\ &\leq (K/\delta \|T_*(f-g)\|_{L^{q,\infty}})^q \leq (K/\delta \|T_*\|_{L^p \to L^{q,\infty}} \|f-g\|_{L^p})^q.\end{aligned}$$

Let $\eta \to 0$, we have proved our assertion, which implies that $\limsup_{\varepsilon \to 0} |T_\varepsilon(f)(y)| = 0$.

(b) Define

$$T_\varepsilon(f)(x) := \sup_{x \in B} \left( \frac{1}{|B|} \int_B |f(y) - f(x)|^p \, dy \right)^{\frac{1}{p}}.$$

By Minkowski's inequality, we have

$$\left( \int_B |f(y) - f(x) + g(y) - g(x)|^p \, dy \right)^{\frac{1}{p}} \leq \left( \int_B |f(y) - f(x)|^p \, dy \right)^{\frac{1}{p}} + \left( \int_B |g(y) - g(x)|^p \, dy \right)^{\frac{1}{p}}.$$

So we have $T_\varepsilon(f+g) \leq T_\varepsilon(f) + T_\varepsilon(g)$.

When $p \geq 1$, by Minkowski's inequality,

$$\left( \int_B |f(y) - f(x)|^p \, dy \right)^{\frac{1}{p}} \leq \left( \int_B |f(y)|^p \, dy \right)^{\frac{1}{p}} + |f(x)| \leq |f| + M(|f|^p)^{\frac{1}{p}};$$

when $p < 1$, by part (c) of Exercise 1.1.5,

$$\begin{aligned}\left( \int_B |f(y) - f(x)|^p \, dy \right)^{\frac{1}{p}} &\leq \left( \int_B |f(y)|^p + |f(x)|^p \, dy \right)^{\frac{1}{p}} \\ &\leq 2^{\frac{1}{p}-1} \left( \left( \int_B |f(y)|^p \, dy \right)^{\frac{1}{p}} + |f(x)| \right) \\ &\leq 2^{\frac{1}{p}-1} \big( |f| + M(|f|^p) \big)^{\frac{1}{p}}.\end{aligned}$$

Note that $M : L^1 \to L^{1,\infty}$ is bounded, we have

$$\delta^p \mu\{M(|f|^p)^{\frac{1}{p}}\} = \delta^p \mu\{M(|f|^p) > \delta^p\} \leq \|f^p\|_{L^1} \cdot \|M\| = \|f\|_{L^p}^p \cdot \|M\|.$$

So $f \mapsto M(|f|^p)^{\frac{1}{p}}$ is of weak type $(p,p)$, and the norm is less equal to $\|M\|^{\frac{1}{p}}$, which implies $T_*$ is of weak type $(p,p)$. For $f$ continuous, $T_*(f)=0$, and $\mathscr{C}(\mathbf{R}^n)$ is dense in $L^p(\mathbf{R}^n)$. Then we finish the proof by part (a).

(c) Set $Q=[0,1]^n$, then $f\chi_{Q+m}\in L^1(\mathbf{R}^n), m\in\mathbf{Z}^n$. Apply part (b) to $f\chi_{Q+m}$, we have $f(x)=0$, a.e. on every $m+Q$, therefore a.e. on $\mathbf{R}^n$.

2.1.11 (a) It's easy to see $E_\alpha := \{M_R(f)>\alpha\}$ is open.

Write $E_\alpha = \biguplus(a_i,b_i)$. For $x\in(a_j,b_j)$, let

$$N_x = \{s \mid \int_x^s |f|dt > \alpha(s-x)\}\cap(x,b_j].$$

The set $N_x$ is nonempty, and $\sup N_x = b_n$. Otherwise, there exists $r>0$ s.t. $\int_x^{x+r}|f(t)|dt>\alpha r$. Thus $x+r>b_n$, namely $r>b_n-x$. But as $b_n\notin E_\alpha$, $\int_{b_n}^{b_n+r'}|f(t)|dt\le\alpha r'$ for all $r'>0$. Take $r'=x+r-b_n$, then $\int_{b_n}^{x+r}|f(t)|dt\le\alpha(x+r-b_n)$. This leads to $\int_x^{b_n}|f(t)|dt>\alpha(b_n-x)$, whence $b_n\in N_x$, contradiction! Let $s_0=\sup N_x$, then $\forall\epsilon>0,\exists s_\epsilon$ such that $\int_x^{s_\epsilon}|f|>\alpha(s-x)$. If $s_0<b_j$, from the definition of $E_\alpha$, there exists $s_0'>s_0$, such that $\int_{s_0}^{s_0'}|f|>\alpha(s_0'-s_0)$.

Let $\epsilon\to0$, $\int_x^{s_0'}|f|>\alpha(s_0'-x)$, contradictory to the supremum. So $s_0=b_j$.

Let $x\to a_j$, $\int_{a_j}^{b_j}|f|\ge\alpha(b_j-a_j)$. If $\int_{a_j}^{b_j}|f|>\alpha(b_j-a_j)$, then $a_j\in E_\alpha$, a contradiction. Thus,

$$\int_{a_j}^{b_j}|f|=\alpha(b_j-a_j),$$

which implies that

$$|M_R(f)>\alpha| = \frac{1}{\alpha}\int_{\{M(f)>\alpha\}}|f|dt.$$

Similar for $M_L(f)$.

(b)

$$\begin{aligned}\|M_R(f)\|_{L^p}^p &= \int_0^\infty p\alpha^{p-1}\frac{1}{\alpha}\int_{\{M(f)>\alpha\}}|f|\,dtd\alpha\\ &= \frac{p}{p-1}\int_{\mathbb{R}^n}M(f)^{p-1}|f|\,dt\\ &\le \frac{p}{p-1}\|M_R(f)\|_{L^p}^{p-1}\|f\|_{L^p}.\end{aligned}$$

Here we use Hölder's inequality for the last inequality. Then we get

$$\|M_R(f)\|_{L^p}\le\frac{p}{p-1}\|f\|_{L^p}.$$

(c) Consider $f_{\varepsilon,N}:=x^{-\frac{1}{p}+\varepsilon}\chi_{(0,N)}$.

We have

$$\|f_{\varepsilon,N}\|_{L^p}^p=\int_0^N x^{-1+\varepsilon p}dx=\frac{1}{\varepsilon p}N^{\varepsilon p},\ M(f)(x)=\frac{1}{x}\int_0^x t^{-\frac{1}{p}+\varepsilon}dt=\frac{x^{-\frac{1}{p}+\varepsilon}}{1-\frac{1}{p}+\varepsilon}.$$

So

$$\|M(f_{\varepsilon,N})\|_{L^p}^p \geq \int_0^N \Big(\frac{x^{-\frac{1}{p}+\varepsilon}}{1-\frac{1}{p}+\varepsilon}\Big)^p dx = \Big(\frac{1}{1-\frac{1}{p}+\varepsilon}\Big)^p \cdot \frac{1}{\varepsilon p} N^{\varepsilon p}.$$

Let $\varepsilon \to 0$, then $\|M\| \geq \frac{p}{p-1}$.

(d)

$$\begin{aligned} M(f)(x) &\leq \sup_{x\in(a,b)} \int_a^b |f| \frac{1}{b-a} \\ &\leq \sup_{x\in(a,b)} \frac{\int_a^x |f|}{x-a}\frac{x-a}{b-a} + \frac{\int_x^b |f|}{b-x}\frac{b-x}{b-a} \\ &\leq \max(M_R, M_L)(f)(x). \end{aligned}$$

Let $a \to x^-$, then $\frac{\int_x^b |f|}{b-x} \leq M(f)(x)$, so $M_R(f) \leq M(f)$. Similarly we have $M_L(f) \leq M(f)$. We deduce from the three inequalities that $\max(M_R, M_L) = M(f)$.

(e)

$$\begin{aligned} \int_{\mathbb{R}} M(f)^p + N(f)^p dx &= \int_0^\infty p\alpha^{p-1}(|\{\max(M_L, M_R)(f) > \alpha\}| + |\{\min(M_L, M_R)(f) > \alpha\}|)\, d\alpha \\ &= \int_0^\infty p\alpha^{p-1} \frac{1}{\alpha}\Big(\int_{\{M_L(f)>\alpha\}} |f|\, dx + \int_{\{M_R(f)>\alpha\}} |f|\, dx\Big) d\alpha \\ &= p\int_{\mathbb{R}} |f|\Big(\int_0^{M_L(f)} \alpha^{p-2} d\alpha + \int_0^{M_R(f)} \alpha^{p-2} d\alpha\Big) dx \\ &= \frac{p}{p-1}\int_{\mathbb{R}} |f|\big(M_L(f)^{p-1} + M_R(f)^{p-1}\big)\, dx \\ &= \frac{p}{p-1}\int_{\mathbb{R}} |f|\big(M(f)^{p-1} + N(f)^{p-1}\big)\, dx. \end{aligned}$$

(f) Use part (e), we further get

$$\begin{aligned} \int_{\mathbb{R}} M(f)^p + N(f)^p dx &= \frac{p}{p-1}\int_{\mathbb{R}} |f|\big(M(f)^{p-1} + N(f)^{p-1}\big)\, dx \\ &\leq \frac{p}{p-1}\Big(\int_{\mathbb{R}} |f| M(f)^{p-1} + \int_{\mathbb{R}} |f|^p + \int_{\mathbb{R}} N(f)^p\Big). \end{aligned}$$

Here we used Hölder's inequality. Cancel the $\int N(f)^p$, then we obtain the required inequality.

2.1.12 (a) Firstly notice that $\mathscr{F}$, defined to be the set of all dyadic cubes, is countable. So any cover of dyadic cubes has a countable subcover, which is pairwise disjoint.

Let $E_\alpha = \{M_d(f) > \alpha\}$. $\forall x \in E_\alpha$, $\exists Q_x \in \mathscr{F}$ containing $x$, such that $\int_{Q_x} |f| > \alpha |Q_x|, Q_x \subset E_\alpha$. So $E_\alpha = \bigcup\limits_{x\in E_\alpha} Q_x = \bigcup\limits_{x\in I} Q_x$, where $I$ is countable, $\{Q_x\}_{x\in I}$ are mutually disjoint.

$$|E_\alpha| \leq |\bigcup_{x\in I} Q_x| = \sum_{x\in I} |Q_x| < \frac{1}{\alpha}\sum \int_{Q_x} |f| = \frac{1}{\alpha}\int_{E_\alpha} |f|.$$

(b) Apply the result of part (a), we have

$$\begin{aligned} \|M_d(f)\|_{L^p}^p &= p\int_0^\infty \alpha^{p-1} |\{M_d(f) > \alpha\}|\, d\alpha \\ &\leq p\int_0^\infty \alpha^{p-2} \int_{\{M_d(f)>\alpha\}} |f|\, dt d\alpha \end{aligned}$$

$$
\begin{aligned}
&= p\int_{\mathbb{R}^n}\int_0^{M_d(f)} \alpha^{p-2}d\alpha\,|f|\,dt\\
&= \frac{p}{p-1}\int_{\mathbb{R}^n} |M_d(f)|^{p-1}|f|\,dt\\
&\le \frac{p}{p-1}\|M_d(f)\|_{L^p}^{p-1}\|f\|_{L^p}.
\end{aligned}
$$

2.1.13 Use the result of Exercise 1.1.12 (a), we deduce

$$
\lambda|\{M(f)>\lambda\}|^{\frac{1}{p}} \le 3^n|M(f)>\lambda|^{-1+\frac{1}{p}}\int_{M(f)>\lambda}|f(y)|\,dy \le 3^n\left(\frac{p}{p-1}\right)\|f\|_{L^{p,\infty}}.
$$

Taking the supremum in $\lambda$ yields

$$
\|M(f)\|_{L^{p,\infty}} \le 3^n\left(\frac{p}{p-1}\right)\|f\|_{L^{p,\infty}}.
$$

2.1.14 It suffices to prove the inequality for $x=0$, as we may replace $f(t)$ by $f(t+x)$. Define $F, G$ as in Theorem 2.1.10, and let

$$
A_{\varepsilon_0} = \sup_{\varepsilon>\varepsilon_0}\frac{1}{\varepsilon^n}\int_{|y|\le\varepsilon}|f(y)|\,dy = \sup_{\varepsilon>\varepsilon_0}\frac{1}{\varepsilon^n}G(\varepsilon).
$$

Then for any $\varepsilon>\varepsilon_0$,

$$
\begin{aligned}
(f*K_\varepsilon)(0) &= \int_0^\infty G(r)d(-K_\varepsilon(re_1))\\
&\le \int_0^{\varepsilon_0}\varepsilon_0^n A_{\varepsilon_0}d(-K_\varepsilon(re_1)) + \int_{\varepsilon_0}^\infty r^n A_{\varepsilon_0}d(-K_\varepsilon(re_1))\\
&= \varepsilon_0^n A_{\varepsilon_0}\big(K_\varepsilon(0)-K_\varepsilon(\varepsilon_0 e_1)\big) + \varepsilon_0^n A_{\varepsilon_0}K_\varepsilon(\varepsilon_0 e_1) + \int_{\varepsilon_0}^\infty nr^{n-1}A_{\varepsilon_0}K_\varepsilon(re_1)\,dr\\
&= \big(\varepsilon_0^n K_\varepsilon(0) + \|K_\varepsilon(re_1)\|_{L^1(B(0,\varepsilon_0)^c)}/\nu_n\big)A_{\varepsilon_0}\\
&\le \big(\varepsilon_0^n/\varepsilon^n + \|K_\varepsilon\|_{L^1}/\nu_n\big)A_{\varepsilon_0}\\
&\le \big(1+\|K\|_{L^1}/\nu_n\big)A_{\varepsilon_0} \coloneqq C_{n,\delta}A_{\varepsilon_0}.
\end{aligned}
$$

## 2.2 The Schwartz Class and the Fourier Transform

2.2.1 (a) $g(x) = \begin{cases} e^{\frac{1}{|x|-1}} & |x|<1,\\ 0 & |x|\ge 1. \end{cases}$

(b) $h(x) = \begin{cases} 0 & |x|\le\frac{1}{2},\\ \frac{e^{\frac{1}{1-|x|}}}{e^{\frac{1}{|x|-\frac{1}{2}}}+e^{\frac{1}{1-|x|}}} & \frac{1}{2}<|x|<1,\\ 1 & 1\le|x|\le 2,\\ \frac{e^{\frac{1}{|x|-2}}}{e^{\frac{1}{4-|x|}}+e^{\frac{1}{|x|-2}}} & 2<|x|<4,\\ 0 & |x|\ge 4. \end{cases}$

(c) Take some $\phi\in\mathscr{S}$ such that $\phi^\vee$ is odd, real–valued and compactly supported.

For instance, $\phi^\vee(\xi)=\xi_1 h(\xi)$ where $\xi=(\xi_1,\cdots,\xi_n)$ and $h$ is defined as in part (b).

Then $\phi=(\phi^\vee)^\wedge$ takes purely imaginary values, so $\phi*\widetilde{\phi}$ takes real values.

Let

$$f = (\phi * \widetilde{\phi})^2 = ((\phi * \widetilde{\phi})^\wedge * (\phi * \widetilde{\phi})^\wedge)^\vee,$$

then

$$\hat{f} = (\phi * \widetilde{\phi})^\wedge * (\phi * \widetilde{\phi})^\wedge = (\hat{\phi}\phi^\vee) * (\hat{\phi}\phi^\vee) = (-\phi^{\vee 2}) * (-\phi^{\vee 2})$$

is nonnegative and compactly supported.

2.2.2 For all multi–indices $\alpha, \beta$,

$$\begin{aligned}\|x^\alpha(\partial^\beta(\hat{f}_k - \hat{f}))(x)\|_{L^\infty} &= \frac{(2\pi)^\beta}{(2\pi)^\alpha}\|(\partial^\alpha(x^\beta(f_k(x) - f(x))))^\wedge\|_{L^\infty} \\ &\leq \frac{(2\pi)^\beta}{(2\pi)^\alpha}\|\partial^\alpha(x^\beta(f_k(x) - f(x)))\|_{L^1} \to 0.\end{aligned}$$

The last step is because $f_k \to f$ in $\mathscr{S}$ implies $x^\beta f_k \to x^\beta f$ in $\mathscr{S}$, and subsequently $\partial^\alpha(x^\beta f_k) \to \partial^\alpha(x^\beta f)$ in $\mathscr{S}$, hence in $L^1$ by Proposition 2.2.6. And $\hat{\hat{f}}_k \to \hat{\hat{f}}$ gives $f_k^\vee \to f^\vee$.

2.2.3 If there exists nonzero functions $f$ such that $\hat{f} = \lambda f$, then

$$f = \tilde{\tilde{f}} = \hat{\hat{\hat{f}}} = \lambda^4 f,$$

so $\lambda^4 = 1$, then $\lambda = 1, -1, i, -i$. Now verify that $\{1, -1, i, -i\}$ is exactly the spectrum of the Fourier transform indeed:

$$(e^{-\pi x^2})^\wedge = e^{-\pi x^2}$$

shows that 1 is an eigenvalue;

$$(-2\pi x e^{-\pi x^2})^\wedge = (\partial(e^{-\pi x^2}))^\wedge = 2\pi i x e^{-\pi x^2}$$

shows that $-i$ is an eigenvalue;

$$\begin{aligned}(4\pi^2 x^2 e^{-\pi x^2} - \pi e^{-\pi x^2})^\wedge &= (4\pi^2 x^2 e^{-\pi x^2} - 2\pi e^{-\pi x^2})^\wedge + \pi(e^{-\pi x^2})^\wedge \\ &= (\partial^2(e^{-\pi x^2}))^\wedge + \pi e^{-\pi x^2} \\ &= -4\pi^2 x^2 e^{-\pi x^2} + \pi e^{-\pi x^2}\end{aligned}$$

shows that $-1$ is an eigenvalue;

$$\begin{aligned}(-8\pi^3 x^3 e^{-\pi x^2} + 6\pi^2 x e^{-\pi x^2})^\wedge &= (-8\pi^3 x^3 e^{-\pi x^2} + 12\pi^2 x e^{-\pi x^2})^\wedge + 3\pi(-2\pi x e^{-\pi x^2})^\wedge \\ &= (\partial^3(e^{-\pi x^2}))^\wedge + 6\pi^2 i x e^{-\pi x^2} \\ &= -8\pi^3 i x^3 e^{-\pi x^2} + 6\pi^2 i x e^{-\pi x^2}\end{aligned}$$

shows that $i$ is an eigenvalue.

2.2.4 Without loss of generality, assume $A = B = 1$, $M = N = L$, then the proposition becomes

$$\int_{\mathbb{R}^n} (1 + |x - y|)^{-L}(1 + |y|)^{-L} dy \leq C(1 + |x|)^{-L}.$$

This is because

$$\int_{\mathbb{R}^n} (1 + |x|)^{L}(1 + |x - y|)^{-L}(1 + |y|)^{-L} dy$$

$$\begin{aligned}
&\le \int_{\mathbb{R}^n} (1+|x-y|+1+|y|)^L (1+|x-y|)^{-L}(1+|y|)^{-L}dy \\
&\le \int_{\mathbb{R}^n} 2^{L-1}\big((1+|x-y|)^L + (1+|y|)^L\big)(1+|x-y|)^{-L}(1+|y|)^{-L}dy \\
&= \int_{\mathbb{R}^n} 2^{L-1}\big((1+|y|)^{-L} + (1+|x-y|)^{-L}\big)dy \\
&\le C.
\end{aligned}$$

2.2.5 Let $L_0^p = \{f \in L^p \mid f = 0 \text{ a.e. outside some compact set } K\}$.

When $1 \le p < \infty$, let $k_\varepsilon(x) = \varepsilon^{-n}k(\varepsilon^{-1}x)$ where $k \in \mathscr{C}_0^\infty$, then $\{k_\varepsilon\}_{\varepsilon>1}$ is a smooth approximate identity all the supports of which are contained in support $k$. For $f \in L_0^p$, we have $f * k_\varepsilon \in \mathscr{C}_0^\infty$ by the Lebesgue dominated convergence theorem, and $f * k_\varepsilon \xrightarrow{L^p} f$ by Theorem 1.2.19 (a), which shows that $\mathscr{C}_0^\infty$ is dense in $L_0^p$, hence dense in $L^p$.

When $p < 1$, by the $p = 1$ case $\mathscr{C}_0^\infty$ is dense in $L_0^1$ with respect to the $L^1$ norm and therefore also with respect to the $L^p$ norm because for $f \in L_0^1$ we have

$$\|f * k_\varepsilon - f\|_{L^p} \le |\operatorname{support} f + \operatorname{support} k|^{\frac{1}{p}-1}\|f * k_\varepsilon - f\|_{L^1}.$$

Moreover, $L_0^1$ is dense in $L_0^p$ since for $f \in L_0^p$ we have $f\chi_{|f|\le n} \in L_0^1$, hence dense in $L^p$.

When $p = \infty$, any sequence of $\mathscr{C}_0^\infty$ functions converges with respect to the $L^\infty$ norm to a continuous function, so $\mathscr{C}_0^\infty$ is not dense in $L^\infty$.

2.2.6 (a) $\|\hat{f} - \tau^y\hat{f}\|_{L^\infty}$

$= \|\hat{f} - (e^{2\pi ixy}f)^\wedge\|_{L^\infty}$ by Proposition 2.2.11 (6)

$\le \|f - e^{2\pi ixy}f\|_{L^1} \to 0$ as $y \to 0$ by the Lebesgue dominated convergence theorem.

(b)

$$\begin{aligned}
\int_{\mathbb{R}^n} f(x)\widehat{g}(x)\,dx &= \int_{\mathbb{R}^n} f(x)\bigg(\int_{\mathbb{R}^n} g(y)e^{-2\pi ixy}dy\bigg)dx \\
= \int_{\mathbb{R}^n}\bigg(\int_{\mathbb{R}^n} f(x)e^{-2\pi ixy}dx\bigg)g(y)\,dy &= \int_{\mathbb{R}^n} \widehat{f}(y)g(y)\,dy
\end{aligned}$$

by Fubini's theorem.

(c) Use (2.2.15). But now the left–hand side of (2.2.15) may not converge to $f(t)$ pointwise; only is known that the left–hand side converges to $f(t)$ with respect to the $L^1$ norm. Yet by the Riesz–Fischer theorem, there is a sequence of $\varepsilon_n \to 0$ as $n \to \infty$ such that the left–hand side converges to $f(t)$ a.e. if $\varepsilon$ is taken to be $\varepsilon_n$, and the right–hand side still converges to $(\hat{f})^\vee(t)$ after the restriction of $\varepsilon$, so we obtain $f = (\hat{f})^\vee$ a.e.

2.2.7 (a) Let $g(x) = e^{-\pi|\varepsilon x|^2}$ in Exercise 2.2.6 (b), then $\hat{g}(x) = \varepsilon^{-n}e^{-\pi|x/\varepsilon|^2}$, which is an approximate identity, so

$$\lim_{\varepsilon\to 0}\int_{\mathbb{R}^n} \hat{f}(x)e^{-\pi|\varepsilon x|^2}dx = \lim_{\varepsilon\to 0}\int_{\mathbb{R}^n} f(x)\varepsilon^{-n}e^{-\pi|x/\varepsilon|^2}dx = f(0)$$

because $f$ is continuous at 0 and $\varepsilon^{-n}e^{-\pi|x/\varepsilon|^2}$ is an approximate identity.

(b) Suppose that $\hat{f} \notin L^1$, then $\int_K \hat{f}(x)dx > f(0)$ for some bounded area $K$, so

$$\lim_{\varepsilon\to 0}\int_{\mathbb{R}^n} \hat{f}(x)e^{-\pi|\varepsilon x|^2}dx \ge \lim_{\varepsilon\to 0}\int_K \hat{f}(x)e^{-\pi|\varepsilon x|^2}dx = \int_K \hat{f}(x)\,dx > f(0),$$

contradicting the result of part (a). So $\hat{f} \in L^1$, and it follows that $\|\hat{f}\|_{L^1} = f(0)$ by the Lebesgue dominated convergence theorem.

2.2.8 Let $h = f * \bar{\tilde{f}}$, then $h \in L^1(\mathbb{R}^n) \cap L^\infty(\mathbb{R}^n)$ because

$$\|h\|_{L^1} \le \|f\|_{L^1}\|\bar{\tilde{f}}\|_{L^1} = \|f\|_{L^1}^2 < \infty$$
$$\text{and } \|h\|_{L^\infty} \le \|f\|_{L^2}\|\bar{\tilde{f}}\|_{L^2} = \|f\|_{L^2}^2 < \infty$$

by Theorem 1.2.12,

$$\hat{h} = \hat{f}\hat{\bar{\tilde{f}}} \overset{(5)}{=} \hat{f}\bar{\hat{\tilde{f}}} \overset{(4)}{=} \hat{f}\bar{\tilde{\hat{f}}} = \hat{f}\bar{\hat{f}} \ge 0$$

where (5) and (4) stands for Proposition 2.2.11 (5) and (4), and $f$ is continuous at 0 because

$$|h(y) - h(0)| = |((\tau^{-y}f - f) * \bar{\tilde{f}})(0)| \le \|(\tau^{-y}f - f) * \bar{\tilde{f}}\|_{L^\infty}$$
$$\le \|\tau^{-y}f - f\|_{L^2}\|\bar{\tilde{f}}\|_{L^2} \to 0 \quad \text{as } y \to 0.$$

Exercise 2.2.7 (b) yields

$$\|\hat{f}\|_{L^2}^2 = \|\hat{h}\|_{L^1} = h(0) = \int_{\mathbb{R}^n} f(x)\overline{\tilde{f}(-x)}\,dx = \|f\|_{L^2}^2.$$

2.2.9 (a) For $t \in [2n\pi, (2n+2)\pi]$ where $n \in \mathbb{N}$,

$$\begin{aligned}\int_0^t \frac{\sin(\xi)}{\xi}d\xi &\le \int_0^{(2n+1)\pi} \frac{\sin(\xi)}{\xi}\,d\xi \\ &= \int_0^\pi \frac{\sin(\xi)}{\xi}\,d\xi + \sum_{k=1}^n \int_0^\pi \left(\frac{\sin(\xi+(2k-1)\pi)}{\xi+(2k-1)\pi} + \frac{\sin(\xi+2k\pi)}{\xi+2k\pi}\right)d\xi \\ &= \int_0^\pi \frac{\sin(\xi)}{\xi}\,d\xi + \sum_{k=1}^n \int_0^\pi \left(-\frac{\sin(\xi)}{\xi+(2k-1)\pi} + \frac{\sin(\xi)}{\xi+2k\pi}\right)d\xi \\ &\le \int_0^\pi \frac{\sin(\xi)}{\xi}\,d\xi \le \pi.\end{aligned}$$

And for similar reasons,

$$\int_0^t \frac{\sin(\xi)}{\xi}\,d\xi \ge \sup_{n\in\mathbb{N}} \int_0^{2n\pi} \frac{\sin(\xi)}{\xi}\,d\xi \ge 0.$$

So

$$\left|\int_\varepsilon^t \frac{\sin(\xi)}{\xi}\,d\xi\right| \le \pi.$$

(b) $f$ is odd implies that $\hat{f}(\xi) = -i\int_{\mathbb{R}} f(x)\sin(2\pi x\xi)\,dx$, so by part (a)

$$\left|\int_\varepsilon^t \frac{\hat{f}(\xi)}{\xi}\,d\xi\right| = \left|\int_\varepsilon^t \int_{\mathbb{R}} f(x)\frac{\sin(2\pi x\xi)}{\xi}\,dx d\xi\right|$$
$$= \left|\int_{\mathbb{R}} f(x)\int_{2\pi\varepsilon\xi}^{2\pi t\xi} \frac{\sin(2\pi x\xi)}{2\pi x\xi}\,d(2\pi x\xi)dx\right| \le 4\|f\|_{L^1}.$$

(c) Suppose that such $f$ exists, then by part (b)

$$4\|f\|_{L^1} \ge \int_2^t \frac{g(\xi)}{\xi}\,d\xi = \int_2^t \frac{d\xi}{\xi\log(\xi)} = \log(\log(\xi))\Big|_2^t \to +\infty,$$

a contradiction.

2.2.10 By substituting $x$ with $-\frac{1}{x}$ which increases from 0 to $+\infty$ then from $-\infty$ to 0 as $x$ ranges from $-\infty$ to $+\infty$, we get

$$\begin{aligned}\int_{-\infty}^{+\infty} f\Big(x-\frac{1}{x}\Big)dx &= \frac{1}{2}\Big(\int_{-\infty}^{+\infty} f\Big(x-\frac{1}{x}\Big)dx + \int_{-\infty}^{+\infty} f\Big(-\frac{1}{x}+x\Big)d\Big(-\frac{1}{x}\Big)\Big)\\ &= \frac{1}{2}\int_{-\infty}^{+\infty} f\Big(x-\frac{1}{x}\Big)d\Big(x-\frac{1}{x}\Big).\end{aligned}$$

Then since $u = x - \frac{1}{x}$ increases from $-\infty$ to $+\infty$ twice as $x$ ranges from $-\infty$ to $+\infty$,

$$\frac{1}{2}\int_{-\infty}^{+\infty} f\Big(x-\frac{1}{x}\Big)d\Big(x-\frac{1}{x}\Big) = \int_{-\infty}^{+\infty} f(u)\,du.$$

2.2.11 (a) Use Exercise 2.2.10 with $f(x) = e^{-tx^2}$ to obtain that

$$\begin{aligned}\int_0^{\infty} e^{-y-\frac{t^2}{y}+2t}\frac{dy}{\sqrt{y}} &= 2\int_0^{\infty} f\Big(\sqrt{\frac{y}{t}}-\sqrt{\frac{t}{y}}\Big)d\Big(\sqrt{t}\sqrt{\frac{y}{t}}\Big)\\ &= \int_{-\infty}^{+\infty} f(u)\,d(\sqrt{t}u)\\ &= \int_{-\infty}^{+\infty} e^{-tu^2}d(\sqrt{t}u)\\ &= \sqrt{\pi}.\end{aligned}$$

(b) Set $t = \pi|x|$ in part (a) to get that

$$\begin{aligned}(e^{-2\pi|x|})^{\wedge}(\xi) &= \int_{\mathbb{R}^n} e^{-2\pi|x|}e^{-2\pi i\xi x}dx\\ &= \frac{1}{\sqrt{\pi}}\int_{\mathbb{R}^n}\int_0^{\infty} e^{-y-\frac{\pi^2|x|^2}{y}-2\pi i\xi x}\frac{dy}{\sqrt{y}}dx\\ &= \frac{1}{\sqrt{\pi}}\int_0^{\infty}\int_{\mathbb{R}^n} e^{-(1+|\xi|^2)y-\frac{\pi^2|x|^2}{y}-2\pi i\xi x+|\xi|^2y}dx\frac{dy}{\sqrt{y}}.\end{aligned}$$

For all $1 \le j \le n$, use a rectangle contour to lift the $x$-axis and get that

$$\int_{\mathbb{R}} e^{-\frac{\pi^2x_j^2}{y}-2\pi i\xi_jx_j+\xi_j^2y}dx_j = \frac{\sqrt{y}}{\sqrt{\pi}}\int_{\mathbb{R}} e^{-\pi\left(\frac{\sqrt{\pi}x_j}{\sqrt{y}}+\frac{i\xi_j\sqrt{y}}{\sqrt{\pi}}\right)^2}d\frac{\sqrt{\pi}x_j}{\sqrt{y}} = \frac{\sqrt{y}}{\sqrt{\pi}}$$

where $x = (x_1, \cdots, x_n)$ and $\xi = (\xi_1, \cdots, \xi_n)$, plugging into above

$$\begin{aligned}&= \frac{1}{\pi^{\frac{n+1}{2}}}\int_0^{\infty} e^{-(1+|\xi|^2)y}y^{\frac{n-1}{2}}dy\\ &= \frac{\Gamma(\frac{n+1}{2})}{\pi^{\frac{n+1}{2}}}\frac{1}{(1+|\xi|^2)^{\frac{n+1}{2}}}.\end{aligned}$$

2.2.12 (a) For all $x \in \mathbb{R}$,

$$|f(x)|^2 = \left|\int_{-\infty}^{x}(f^2)'(t)\,dt\right| = 2\left|\int_{-\infty}^{x} f(t)f'(t)\,dt\right| \le 2\|f\|_{L^p}\|f'\|_{L^{p'}}.$$

(b) For $x = (x_1, \cdots, x_n) \in \mathbb{R}^n$, write $\Omega = \{(t_1, \cdots, t_n) \in \mathbb{R}^n \mid t_1 \le x_1, \cdots, t_n \le x_n\}$. We have

$$\begin{aligned}|f(x)|^2 &= \left|\int_{-\infty}^{x_n}\cdots\int_{-\infty}^{x_1}\partial_1\cdots\partial_n(f^2)(t)\,dt_1\cdots dt_n\right| = \left|\int_{\Omega}\partial^{(1,\cdots,1)}(f^2)(t)\,dt\right|\\ &\le \|\sum_{\alpha+\beta=(1,\cdots,1)}(\partial^{\alpha}f)(\partial^{\beta}f)\|_{L^1} \le \sum_{\alpha+\beta=(1,\cdots,1)}\|\partial^{\alpha}f\|_{L^p}\|\partial^{\beta}f\|_{L^{p'}}.\end{aligned}$$

2.2.13 Let $y \in \mathbb{R}^n$. Integrate by parts and apply the Cauchy–Schwarz inequality and Plancherel's identity, we get

$$\begin{aligned}
\|f\|_{L^2}^2 &= \frac{1}{n}\int_{\mathbb{R}^n} f(x)\overline{f(x)}\sum_{j=1}^n \partial_j(x_j - y_j)\,dx \\
&= \frac{1}{n}\int_{\mathbb{R}^n} \sum_{j=1}^n \partial_j(f(x)\overline{f(x)})(x_j - y_j)\,dx \\
&= \frac{1}{n}\Big[\int_{\mathbb{R}^n} \sum_{j=1}^n (\partial_j f)(x)\overline{f(x)}(x_j - y_j)\,dx + \int_{\mathbb{R}^n} \sum_{j=1}^n \overline{(\partial_j f)(x)} f(x)(x_j - y_j)\,dx\Big] \\
&\le \frac{1}{n}\Big[\Big(\int_{\mathbb{R}^n} \sum_{j=1}^n |\partial_j f(x)|^2 dx\Big)^{\frac{1}{2}}\Big(\int_{\mathbb{R}^n} \sum_{j=1}^n |\overline{f(x)}(x_j - y_j)|^2 dx\Big)^{\frac{1}{2}} \\
&\quad + \Big(\int_{\mathbb{R}^n} \sum_{j=1}^n |\overline{(\partial_j f)(x)}|^2 dx\Big)^{\frac{1}{2}}\Big(\int_{\mathbb{R}^n} \sum_{j=1}^n |f(x)(x_j - y_j)|^2 dx\Big)^{\frac{1}{2}}\Big] \\
&= \frac{2}{n}\Big(\int_{\mathbb{R}^n} \sum_{j=1}^n |\partial_j f(x)|^2 dx\Big)^{\frac{1}{2}}\Big(\int_{\mathbb{R}^n} \sum_{j=1}^n |f(x)(x_j - y_j)|^2 dx\Big)^{\frac{1}{2}} \\
&= \frac{2}{n}\Big(\int_{\mathbb{R}^n} \sum_{j=1}^n |\widehat{\partial_j f}(\xi)|^2 d\xi\Big)^{\frac{1}{2}}\Big(\int_{\mathbb{R}^n} \sum_{j=1}^n |f(x)(x_j - y_j)|^2 dx\Big)^{\frac{1}{2}} \\
&= \frac{4\pi}{n}\Big(\int_{\mathbb{R}^n} \sum_{j=1}^n |\xi_j \hat{f}(\xi)|^2 dx\Big)^{\frac{1}{2}}\Big(\int_{\mathbb{R}^n} \sum_{j=1}^n |f(x)(x_j - y_j)|^2 dx\Big)^{\frac{1}{2}} \\
&= \frac{4\pi}{n}\Big(\int_{\mathbb{R}^n} |\xi|^2 |\hat{f}(\xi)|^2 d\xi\Big)^{\frac{1}{2}}\Big(\int_{\mathbb{R}^n} |f(x)|^2 |x - y|^2 dx\Big)^{\frac{1}{2}}.
\end{aligned}$$

Then replace $f(x)$ by $f(x)e^{2\pi i x z}$ in the inequality above and the desired inequality follows from

$$\sum_{j=1}^n |(\partial_j(f(\xi)e^{2\pi i\xi z})^\wedge|^2 = (2\pi)^2 \sum_{j=1}^n |(\xi_j - z_j)\hat{f}(\xi)|^2 = (2\pi)^2 |\xi - z|^2 |\hat{f}(\xi)|^2.$$

2.2.14 Let $B$ be the unit ball $\{x \in \mathbb{R}^n \mid |x| < 1\}$ and $B^c$ its complement. By the Cauchy–Schwarz inequality,

$$\begin{aligned}
&\|g\|_{L^1} = \|g\|_{L^1(B)} + \|g\|_{L^1(B^c)} \\
\le &\||x|^{-\alpha}\|_{L^2(B)} \||x|^\alpha g(x)\|_{L^2(B)} + \||x|^{-\beta}\|_{L^2(B^c)} \||x|^\beta g(x)\|_{L^2(B^c)} \\
\le &C(\||x|^\alpha g(x)\|_{L^2} + \||x|^\beta g(x)\|_{L^2}).
\end{aligned}$$

Then repalce $g(x)$ by $g(\lambda x)$ for $\lambda > 0$ and get

$$\begin{aligned}
&\|g\|_{L^1} = \lambda^n \|g(\lambda x)\|_{L^1} \\
\le &C\lambda^n(\||x|^\alpha g(\lambda x)\|_{L^2} + \||x|^\beta g(\lambda x)\|_{L^2}) \\
= &C(\lambda^{n/2-\alpha}\||x|^\alpha g(x)\|_{L^2} + \lambda^{n/2-\beta}\||x|^\beta g(x)\|_{L^2}) \\
= &C\||x|^\alpha g(x)\|_{L^2}^{\frac{\beta-n/2}{\beta-\alpha}} \||x|^\beta g(x)\|_{L^2}^{\frac{n/2-\alpha}{\beta-\alpha}} \text{ by taking suitable } \lambda.
\end{aligned}$$

## 2.3 The Class of Tempered Distributions

2.3.1 Such $\mu$ can be identified with the functional

$$f \mapsto \int_{\mathbb{R}^n} f\, d\mu$$

which is a tempered distribution because the condition of Proposition 2.3.4 (b) is satisfied: in fact,

$$\begin{aligned}\left|\int_{\mathbb{R}^n} f\, d\mu\right| &= \int_{\mathbb{R}^n} |f(x)|(1+|x|)^k \frac{d\mu}{(1+|x|)^k} \\ &\le M \sup_{x\in\mathbb{R}^n} |f(x)|(1+|x|)^k \\ &\le MC_k \sum_{|\alpha|\le k} \sup_{x\in\mathbb{R}^n} |x^\alpha f(x)| \quad \text{for all } f \in \mathscr{S},\end{aligned}$$

where the last inequality is due to (2.2.3).

The Lebesgue measure as a tempered distribution coincides with 1 since

$$\langle f, 1\rangle = \int_{\mathbb{R}^n} f(x)\, dx \quad \text{for all } f \text{ in } \mathscr{S}.$$

2.3.2 Pick any nonzero vector $v \in \mathbb{R}^n$, and it is easily shown that $(x+v)^{\alpha_1}\varphi_\varepsilon$ is $bv^{\alpha_1}$ times an approximate identity for any multi–index $\alpha_1$ as $\varphi \in \mathscr{S}$. Then for all multi–indices $\alpha, \beta$,

$$\begin{aligned}&\sup_{x\in\mathbb{R}^n} |x^\alpha \partial^\beta(\varphi_\varepsilon * f - bf)| \\ =\ &\sup_{x\in\mathbb{R}^n} |x^\alpha(\varphi_\varepsilon * \partial^\beta f - b\partial^\beta f)| \quad \text{by Proposition 2.2.7} \\ \le\ &\sum_{\alpha_1+\alpha_2=\alpha} \binom{\alpha}{\alpha_1} \sup_{x\in\mathbb{R}^n} |\big((x+v)^{\alpha_1}\varphi_\varepsilon\big) * \big((x-v)^{\alpha_2}\partial^\beta f\big) - bv^{\alpha_1}(x-v)^{\alpha_2}\partial^\beta f| \\ \to\ &0 \quad \text{by Proposition 1.2.21 (b).}\end{aligned}$$

In the last step $(x-v)^{\alpha_2}\partial^\beta f \in \mathscr{S}$ and is hence uniformly continuous on $\mathbb{R}^n$.

2.3.3 We have

$$\begin{aligned}&\langle (\delta^a f) * (\delta^a u), g\rangle \\ =\ &\langle \delta^a u, \widetilde{\delta^a f} * g\rangle \\ =\ &\langle u, a^{-n}\delta^{1/a}(\widetilde{\delta^a f} * g)\rangle \\ =\ &\langle u, a^{-n}\textstyle\int_{\mathbb{R}^n} f(-ay)g(a^{-1}x - y)\, dy\rangle \\ =\ &\langle u, a^{-n}\textstyle\int_{\mathbb{R}^n} f(-y)g(a^{-1}x - a^{-1}y)\, d(a^{-1}y)\rangle \\ =\ &\langle a^{-n}u, a^{-n}\tilde{f} * \delta^{1/a} g\rangle \\ =\ &\langle a^{-n} f * u, a^{-n}\delta^{1/a} g\rangle \\ =\ &\langle a^{-n}\delta^a(f * u), g\rangle \quad \text{for all } g \in \mathscr{S}.\end{aligned}$$

Or check that the two distributions have identical Fourier transforms:

$$\begin{aligned}\big((\delta^a f) * (\delta^a u)\big)^\wedge &= (\delta^a f)^\wedge(\delta^a u)^\wedge = a^{-n}\delta^{1/a}\hat{f}a^{-n}\delta^{1/a}\hat{u} = a^{-2n}\delta^{1/a}(\hat{f}\hat{u}), \\ \big(a^{-n}\delta^a(f * u)\big)^\wedge &= a^{-n}(a^{-n}\delta^{1/a}(f * u)^\wedge) = a^{-2n}\delta^{1/a}(\hat{f}\hat{u}).\end{aligned}$$

2.3.4 (a)

$$\langle \partial \chi_{[a,b]}, f\rangle = -\langle \chi_{[a,b]}, f'\rangle = -(f(b)-f(a)) = \langle \delta_a - \delta_b, f\rangle.$$

(b) To simplify notations, let $j=1$. Sure enough, everything is essentially the same for $j=2$.

$$\begin{aligned}\langle \partial_1 \chi_{B(0,1)}, f\rangle &= -\langle \chi_{B(0,1)}, \partial_1 f\rangle \\ &= -\int_{B(0,1)} \partial_1 f\, dx \\ &= -\int_{-1}^{1}\int_{-\sqrt{1-x_1^2}}^{\sqrt{1-x_1^2}} \partial_1 f\, dx_1 dx_2 \\ &= -\int_{-1}^{1} f(\sqrt{1-x^2},x) - f(-\sqrt{1-x^2},x)\, dx.\end{aligned}$$

(c)

$$\begin{aligned}\langle (\sin x)^\wedge, f\rangle = \langle \sin x, \hat f\rangle &= \langle (e^{ix}-e^{-ix})/2i, \hat f\rangle \\ &= \big((\hat f)^\vee(\tfrac{1}{2\pi}) - (\hat f)^\vee(-\tfrac{1}{2\pi})\big)/2i \\ &= \big(f(\tfrac{1}{2\pi}) - f(-\tfrac{1}{2\pi})\big)/2i \\ &= \langle (\delta_{\frac{1}{2\pi}} - \delta_{-\frac{1}{2\pi}})/2i, f\rangle.\end{aligned}$$

(d)

$$\begin{aligned}\langle \partial \log|x|, \varphi\rangle = -\langle \log|x|, \varphi'\rangle &= -\int_{\mathbb{R}} \log|x|\varphi'(x)\, dx \\ &= -\lim_{\varepsilon\to 0}\int_{\varepsilon\le|x|} \log|x|\varphi'(x)\, dx \\ &= \lim_{\varepsilon\to 0}\int_{\varepsilon\le|x|} \varphi(x)\frac{dx}{x} + \big(\varphi(-\varepsilon)-\varphi(\varepsilon)\big)\log|\varepsilon| \\ &= \lim_{\varepsilon\to 0}\int_{\varepsilon\le|x|} \varphi(x)\frac{dx}{x} \quad \text{since } \varphi(-\varepsilon)-\varphi(\varepsilon) = O(\varepsilon).\end{aligned}$$

2.3.5 (a) For all multi–indices $\alpha,\beta$ and $|h|<1$,

$$\begin{aligned}&\sup_{x\in\mathbb{R}^n}\Big|x^\alpha\partial^\beta\Big(\frac{\tau^{-he_j}f-f}{h} - \partial_j f\Big)\Big| \\ =&\sup_{x\in\mathbb{R}^n}\Big|x^\alpha\frac{(\partial^\beta f)(x+he_j)-(\partial^\beta f)(x)}{h} - \partial_j\partial^\beta f(x)\Big| \\ =&\sup_{x\in\mathbb{R}^n}|x^\alpha\partial_j\partial^\beta f(x+\theta he_j) - \partial_j\partial^\beta f(x)| \quad \text{where } \theta=\theta(x)\in(0,1) \\ \le& h\sup_{x\in\mathbb{R}^n}|x^\alpha\partial_j^2\partial^\beta f(x+\theta' he_j)| \quad \text{where } \theta'=\theta'(x)\in(0,\theta(x)) \\ \le& h\sup_{x\in\mathbb{R}^n}|\max\{(x+e_j)^\alpha,(x-e_j)^\alpha\}\partial_j^2\partial^\beta f(x)| \\ \to& 0 \quad \text{as } h\to 0,\end{aligned}$$

where the last step is because $|(x\pm e_j)^\alpha|$ are both dominated by a finite sum of some $|x^\gamma|$'s. In part (b) and (c), assume $f$ is identically equal to 1 in $B(0,r)$.

(b) For all multi–indices $\alpha, \beta$,

$$\begin{aligned}
&\sup_{x\in\mathbb{R}^n} |x^\alpha \partial^\beta(\varphi_k f - f)| \\
&= \sup_{x\in\mathbb{R}^n} |(\varphi_k - 1)x^\alpha \partial^\beta f + \sum_{\gamma\le\beta,\gamma\neq 0} x^\alpha \partial^\gamma \varphi_k \, \partial^{\beta-\gamma} f| \\
&\le \|\varphi_k - 1\|_{L^\infty} \sup_{|x|>kr} |x^\alpha \partial^\beta f| + \sum_{\gamma\le\beta,\gamma\neq 0} \frac{\|\partial^\gamma \varphi\|_{L^\infty}}{k^{|\gamma|}} \sup_{x\in\mathbb{R}^n} |x^\alpha \partial^{\beta-\gamma} f|.
\end{aligned}$$

Noticing that $|x_i| > kr/\sqrt{n}$ for some $1 \le i \le n$ when $|x| > kr$, we have

$$\sup_{|x|>kr} |x^\alpha \partial^\beta f| \le \max_{1\le i\le n} \frac{\sup_{|x|>kr} |x^{\alpha+e_i} \partial^\beta f|}{kr/\sqrt{n}}$$

where $e_i$ is the multi–index whose $i$-th component is 1 and the others 0, so $\sup_{x\in\mathbb{R}^n} |x^\alpha \partial^\beta(\varphi_k f - f)| \to 0$ as $k \to \infty$.

(c) By part (b) $(\varphi_k f)^\wedge \to \hat{f}$ in $\mathscr{S}$ since the Fourier transform is a homeomorphism (Corollary 2.2.15), which on one hand reduces the proposition to $\varphi_k(\varphi_k f)^\wedge - (\varphi_k f)^\wedge \to 0$ in $\mathscr{S}$, and on the other hand implies that for all $k$ and multi–indices $\alpha, \beta$,

$$\sup_{x\in\mathbb{R}^n} |x^\alpha \partial^\beta (\varphi_k f)^\wedge| \le C_{\alpha,\beta} < \infty.$$

Then for all multi–indices $\alpha, \beta$,

$$\begin{aligned}
&\sup_{x\in\mathbb{R}^n} |x^\alpha \partial^\beta \big((\varphi_k - 1)(\varphi_k f)^\wedge\big)| \\
&= \sup_{x\in\mathbb{R}^n} |x^\alpha (\varphi_k - 1)\partial^\beta \big((\varphi_k f)^\wedge\big) + \sum_{\gamma\le\beta,\gamma\neq 0} \partial^\gamma(\varphi_k - 1) x^\alpha \partial^{\beta-\gamma}\big((\varphi_k f)^\wedge\big)| \\
&\le \|\varphi_k - 1\|_{L^\infty} \sup_{|x|>kr} |x^\alpha \partial^\beta \big((\varphi_k f)^\wedge\big)| + \sum_{\gamma\le\beta,\gamma\neq 0} \frac{\|\partial^\gamma \varphi\|_{L^\infty}}{k^{|\gamma|}} C_{\alpha,\beta-\gamma}.
\end{aligned}$$

For the same reason as part (b),

$$\sup_{|x|>kr} |x^\alpha \partial^\beta \big((\varphi_k f)^\wedge\big)| \le \max_{1\le i\le n} \frac{C_{\alpha+1_i,\beta}}{kr/\sqrt{n}},$$

so $\sup_{x\in\mathbb{R}^n} |x^\alpha \partial^\beta \big((\varphi_k - 1)(\varphi_k f)^\wedge\big)| \to 0$ as $k \to \infty$.

2.3.6 The Fourier transform of a nonzero $\mathscr{C}_0^\infty$ function, as a distribution with compact support, is a real analytic function by Theorem 2.3.21, which cannot be a $\mathscr{C}_0^\infty$ function for it would otherwise be identically 0 due to the uniqueness of real analytic functions (the uniqueness is obtained by showing that the interior of the zero set of a real analytic function is both open and closed).

2.3.7 For all $h \in \mathscr{S}$,

$$\begin{aligned}
\langle g_N, h\rangle - \langle \hat{f}, h\rangle &= \langle (f\chi_{B(0,N)})^\wedge, h\rangle - \langle \hat{f}, h\rangle \\
&= \langle f\chi_{B(0,N)} - f, \hat{h}\rangle \\
&\le \|f\chi_{B(0,N)} - f\|_{L^p} \|\hat{h}\|_{L^{p'}} \to 0 \quad \text{as } N \to \infty.
\end{aligned}$$

2.3.8 Let

$$u_N = \sum_{|k|\le N} c_k\delta_k,\ u = \sum_{k\in\mathbb{Z}^n} c_k\delta_k.$$

Then $u_k \to u$ in $\mathscr{S}'$ because for all $f \in \mathscr{S}$,

$$\begin{aligned}
|\langle u-u_N, f\rangle| &= |\sum_{|k|>N} c_k f(k)| \\
&\le A\sum_{|k|>N}(1+|k|)^M|f(k)| \\
&\le A_N \sup_{|k|>N}(1+|k|)^{M+n+1}|f(k)| \\
&\le A_N \sup_{x\in\mathbb{R}^n}(1+|x|)^{M+n+1}|f(x)| \\
&\le A_N \sum_{|\beta|\le M} C_{n,M}\sup_{x\in\mathbb{R}^n}|x^\beta f(x)| \to 0 \quad \text{as } N\to\infty,
\end{aligned}$$

where $A_N = A\sum\limits_{|k|>N}(1+|k|)^{-n-1} \to 0$ as $N\to\infty$ and in the last inequality (2.2.3) is used.
Since $\widehat{u_N} = h_N$ (see Example 2.3.9), for all $f\in\mathscr{S}$,

$$\langle \hat{u}-h_N, f\rangle = \langle \hat{u}-\widehat{u_N}, f\rangle = \langle u-u_N, \hat{f}\rangle \to 0 \quad \text{as } N\to\infty.$$

2.3.9 (a) The usual definition for functions (still denoted by $u$) homogeneous of degree $\gamma\in\mathbb{C}$ is $\delta^\lambda u = \lambda^\gamma u$ for all $\lambda>0$, which agrees with the one given in the problem because for functions $u$,

$$\begin{aligned}
&\delta^\lambda u = \lambda^\gamma u && \text{for all } \lambda>0 \\
\Leftrightarrow\ &\langle \delta^\lambda u,\varphi\rangle = \langle \lambda^\gamma u,\varphi\rangle && \text{for all } \lambda>0,\ \varphi\in\mathscr{S} \\
\Leftrightarrow\ &\lambda^{-n}\langle u,\delta^{1/\lambda}\varphi\rangle = \langle \lambda^\gamma u,\varphi\rangle && \text{for all } \lambda>0,\ \varphi\in\mathscr{S} \\
\Leftrightarrow\ &\langle u,\delta^{1/\lambda}\varphi\rangle = (1/\lambda)^{-n-\gamma}\langle u,\varphi\rangle && \text{for all } \lambda>0,\ \varphi\in\mathscr{S}.
\end{aligned}$$

(b) This is because

$$\langle \delta_0, \delta^\lambda\varphi\rangle = \langle \delta_0,\varphi\rangle = \varphi(0).$$

(c) Since

$$\begin{aligned}
\langle \partial^\alpha u, \delta^\lambda\varphi\rangle &= (-1)^{|\alpha|}\langle u,\partial^\alpha(\delta^\lambda\varphi)\rangle = (-1)^{|\alpha|}\langle u,\lambda^{|\alpha|}\delta^\lambda(\partial^\alpha\varphi)\rangle = \lambda^{|\alpha|}(-1)^{|\alpha|}\langle u,\delta^\lambda(\partial^\alpha\varphi)\rangle \\
&= \lambda^{|\alpha|}(-1)^{|\alpha|}\lambda^{-n-\gamma}\langle u,\partial^\alpha\varphi\rangle = \lambda^{-n-(\gamma-|\alpha|)}\langle \partial^\alpha u,\varphi\rangle.
\end{aligned}$$

(d) Since

$$\langle \hat{u},\delta^\lambda\varphi\rangle = \langle u,\widehat{\delta^\lambda\varphi}\rangle = \langle u,\lambda^{-n}\delta^{\lambda^{-1}}\hat{\varphi}\rangle = \lambda^{-n}\langle u,\delta^{\lambda^{-1}}\hat{\varphi}\rangle = \lambda^{-n}\Big(\frac{1}{\lambda}\Big)^{-n-\gamma}\langle u,\hat{\varphi}\rangle = \lambda^{-n-(-n-\gamma)}\langle \hat{u},\varphi\rangle.$$

The converse follows similarly.

2.3.10 (a) $e^{inx}$ and $e^{-inx}$ converge to 0 in $\mathscr{S}'$ and $\mathscr{D}'$ as $n\to\infty$ as a consequence of Proposition 2.2.17, from which we know that the multiplication of distributions is not a continuous operation even when it is defined, for $(e^{inx}, e^{-inx})\to(0,0)$ in $\mathscr{S}'\times\mathscr{S}'$ yet $e^{inx}e^{-inx}=1$.
(b) $\sqrt{n}(1+n|x|^2)^{-1}\to\pi\delta_0$ in $\mathscr{D}'$.

This is because for all $f \in \mathscr{C}_0^\infty$, by the Lebesgue dominated convergence theorem,

$$\begin{aligned}
&\lim_{n\to\infty} \int_{\mathbb{R}} \frac{\sqrt{n}}{1+nx^2} f(x)\,dx \\
&= \lim_{n\to\infty} \int_{\mathbb{R}} \frac{1}{1+x^2} f\Big(\frac{x}{\sqrt{n}}\Big) dx \\
&= \int_{\mathbb{R}} \frac{1}{1+x^2} f(0)\,dx \\
&= \pi f(0).
\end{aligned}$$

2.3.11 Let $h$ be a Schwartz function in $\mathbb{R}^n$ whose Fourier transform is equal to 1 on the ball $B(0,1)$ and vanishes outside the ball $B(0,2)$ (which can be constructed as the inverse Fourier transform of a Schwartz function with the properties that the desired Fourier transform of $h$ is supposed to have). Since

$$(f * R^{-n}\delta^R h)^\wedge = \hat{f}(R^{-n}\delta^R h)^\wedge = \hat{f}\delta^{\frac{1}{R}}\hat{h} = \hat{f},$$

we establish the identity $f = f * R^{-n}\delta^R h$. Then

$$\begin{aligned}
\|\partial^\alpha f\|_{L^\infty} &= \|f * \partial^\alpha (R^{-n}\delta^R h)\|_{L^\infty} \\
&\le \|f\|_{L^\infty} \|\partial^\alpha (R^{-n}\delta^R h)\|_{L^1} \\
&= \|f\|_{L^\infty} R^{|\alpha|} \|\partial^\alpha h\|_{L^1}.
\end{aligned}$$

2.3.12 (a) $(\widehat{\Phi}(\xi/2^N) - 1)\hat{f} \to 0$ in $\mathscr{S}$ for $f \in \mathscr{S}$ as $N \to \infty$ because for all multi–indices $\alpha, \beta$,

$$\begin{aligned}
&\sup_{x\in\mathbb{R}^n} |x^\alpha \partial^\beta \big((\widehat{\Phi}(\xi/2^N) - 1)\hat{f}\big)| \\
&= \sup_{x\in\mathbb{R}^n} |\big(\widehat{\Phi}(\xi/2^N) - 1\big)x^\alpha \partial^\beta \hat{f} + \sum_{\gamma\le\beta,\gamma\neq 0} |\partial^\gamma \big((\widehat{\Phi}(\xi/2^N) - 1\big)x^\alpha \partial^{\beta-\gamma}\hat{f}| \\
&\le |\widehat{\Phi}\|_{L^\infty} \sup_{|x|>2^N} |x^\alpha \partial^\beta \hat{f}| + \sum_{\gamma\le\beta,\gamma\neq 0} \frac{\|\partial^\gamma \widehat{\Phi}\|_{L^\infty}}{2^{N|\gamma|}} \sup_{x\in\mathbb{R}^n} |x^\alpha \partial^{\beta-\gamma}\hat{f}|,
\end{aligned}$$

and for the same reason as Exercise 2.3.5,

$$\sup_{|x|>2^N} |x^\alpha \partial^\beta \hat{f}| \le \max_{1\le i\le n} \frac{\sup_{x\in\mathbb{R}^n} |x^{\alpha+1_i}\partial^\beta \hat{f}|}{2^N/\sqrt{n}}.$$

Thus for all $f \in \mathscr{S}$,

$$\begin{aligned}
|\langle (\widehat{\Phi}(\psi/2^N)\widehat{u})^\vee - u, f\rangle| &= |\langle \widehat{\Phi}(\psi/2^N)\widehat{u} - \widehat{u}, \hat{f}\rangle| \\
&= |\langle \widehat{u}, \big(\widehat{\Phi}(\psi/2^N) - 1\big)\hat{f}\rangle| \to 0 \quad \text{as } N \to \infty.
\end{aligned}$$

(b) Suppose that $\widehat{\Theta}$ is equal to 0 in $B(0,r)$, then $1 - \widehat{\Theta}$ is equal to 1 in $B(0,r)$. Apply part (a), which is also true when $B(0,1)$ is replaced by $B(0,r)$, and get

$$\Big(\big(1 - \widehat{\Theta}(\xi/2^N)\big)\widehat{u}\Big)^\vee \to u \quad \text{in } \mathscr{S}' \text{ as } N \to \infty,$$

namely,

$$\big(\widehat{\Theta}(\xi/2^N)\widehat{u}\big)^\vee \to 0 \quad \text{in } \mathscr{S}' \text{ as } N \to \infty.$$

2.3.13 Assume the converse. Then for all $f \in L^p(\mathbb{R}^n)$, $\hat{f}$ is locally integrable and hence the map $f \to \hat{f}$ is a well defined operator from $L^p(\mathbb{R}^n)$ to $L^1(B(0,M))$ for all $M > 0$ (i.e. $\|\hat{f}\|_{L^1(B(0,M))} < \infty$ for all $f \in L^p(\mathbb{R}^n)$, since $B(0,M)$ is covered by a finite number of neighborhoods of $x \in \overline{B(0,M)}$ on which $\hat{f}$ is integrable).

Then we shall use the closed graph theorem to deduce that the operator is bounded, or in other words, $\|\hat{f}\|_{L^1(B(0,M))} \leq C_M \|f\|_{L^p(\mathbb{R}^n)}$.

Suppose $(f_n, \hat{f}_n) \to (f, g)$ in $L^p(\mathbb{R}^n) \times L^1(B(0,M))$, then by Proposition 2.3.22, $f_n \to f$ in $\mathscr{S}'$, so $\hat{f}_n \to \hat{f}$ in $\mathscr{S}'$. Note that $\hat{f}_n \to g$ in $L^1(B(0,M))$ implies convergence in $\mathscr{S}'$, so we can conclude that $g$ is indeed $\hat{f}$ once we check that: if two $L^1$ functions are identical as tempered distributions, then they are the same as functions.

In fact, if for all $\varphi \in \mathscr{S}$, $\int_{\mathbb{R}^n} \hat{f}\varphi = \int_{\mathbb{R}^n} g\varphi$, then so will be true for $\varphi = \chi_{\{\hat{f} \geq g\}} - \chi_{\{\hat{f} < g\}}$, which is justified by finding a sequence of Schwartz functions all bounded by 1 converging pointwise to $\varphi$ and making use of the Lebesgue dominated convergence theorem — such Schwartz functions can be taken as a sequence of step functions $\chi_{A_n} - \chi_{B_n}$ mollified at the endpoints approximating $\varphi$, where the $A_n$'s and $B_n$'s are each a finite combination of intervals contained in $\{\hat{f} \geq g\}$ and $\{\hat{f} < g\}$ with $m(\{\hat{f} \geq g\} \backslash A_n) \to 0$ and $m(\{\hat{f} < g\} \backslash B_n) \to 0$. Then we get $\hat{f} = g$ in the usual sense of functions.

To violate the previous inequality $\|\hat{f}\|_{L^1(B(0,M))} \leq C_M \|f\|_{L^p(\mathbb{R}^n)}$ whenever $p > 2$, take

$$f_N(x) = t^{-n} e^{-\pi t^{-2}|x|^2},$$

where $t = (1+iN)^{1/2}$. Then for $\mathrm{Re}(t^{-2}) = \frac{1}{1+N^2} = |t|^{-4}$,

$$\|f_N\|_{L^p(\mathbb{R}^n)} = |t|^{-n} \left( \int_{\mathbb{R}^n} e^{-\pi(|t|^{-2}|x|)^{2p}} dx \right)^{\frac{1}{p}} = |t|^{(-1+\frac{2}{p})n} \|e^{-\pi|x|^2}\|_{L^p(\mathbb{R}^n)}.$$

And

$$\widehat{f_N}(\xi) = (t^{-n} \delta^{t^{-1}} (e^{-\pi|x|^2}))^\wedge(\xi) = (\delta^t f)(\xi) = e^{-\pi|\xi|^2(1+iN)}$$

using Proposition 2.2.11 and Example 2.2.9, so $\|\widehat{f_N}\|_{L^1(B(0,M))} = \|e^{-\pi|\xi|^2}\|_{L^1(B(0,M))}$ is a constant. Therefore the inequality cannot be true, for $|t|^{(-1+\frac{2}{p})n} \to 0$ as $N \to \infty$.

## 1.5 解答: GTM249 Chapter 2 Part II

原网页

## 2.4 More About Distributions and the Fourier Transform

2.4.1 Any sequence of functions

$$g(x) \frac{e^{-2\pi i x \cdot (\xi + h_n e_j)} - e^{-2\pi i x \cdot \xi}}{h_n}$$

where $e_j$ is the unit vector with 1 in the $j$th entry and $h_n \to 0$ as $n \to \infty$ converges pointwise to

$$g(x)(-2\pi i x_j) e^{-2\pi i x \cdot \xi},$$

and is bounded by (use $|e^{ia} - e^{ib}| \le |a - b|$)

$$|g(x)(-2\pi i x_j)|,$$

which is expected to be integrable. Under such assumption, it follows from the Lebesgue dominated convergence theorem that

$$\int_{\mathbb{R}^n} g(x) \frac{e^{-2\pi i x\cdot(\xi + h_n e_j)} - e^{-2\pi i x\cdot\xi}}{h_n}\, dx \to \int_{\mathbb{R}^n} g(x)(-2\pi i x_j) e^{-2\pi i x\cdot\xi}\, dx.$$

In other words,

$$\frac{\partial \widehat{g}}{\partial \xi_j}(\xi) = \int_{\mathbb{R}^n} g(x)(-2\pi i x_j) e^{-2\pi i x\cdot\xi}\, dx.$$

Take $g(x) = f(x)(-2\pi i x)^\alpha$ for all multi–indices $\alpha$ with sizes $|\alpha| = 0, \cdots, M-1$ consecutively, which all satisfy the integrability hypothesis since

$$|g(x)(-2\pi i x_j)| = |f(x)(-2\pi i x)^{\alpha + e_j}| \le \frac{(2\pi)^{|\alpha|+1} C}{(1+|x|)^{N-(|\alpha|+1)}}$$

where $N - (|\alpha| + 1) > (M + n) - M = n$.

Now we get the expression for $\widehat{f}^{(M)}$, and therefore $\widehat{f} \in \mathscr{C}^M$.

2.4.2 A bounded harmonic function, which is surely a tempered distribution, must be a polynomial by Corollary 2.4.3, and is a constant as all polynomials tend to $\infty$ as $x \to \infty$.

Suppose that $p(z)$ is a polynomial on $\mathbb{C}$ without any complex root. Then the entire function $\frac{1}{p(z)}$, which is a complex–valued harmonic function on $\mathbb{R}^2$, is bounded as it tends to 0 as $z \to \infty$, and is therefore a constant by Liouville's theorem for harmonic functions, yielding a contradiction.

2.4.3 $e^x$ is not in $\mathscr{S}'(\mathbb{R})$, as $e^{-x}\eta \in \mathscr{S}(\mathbb{R})$ where $\eta$ is a smooth function that equals 1 for $x > 0$ and vanishes for $x < -1$, but $\langle e^x, e^{-x}\eta\rangle = \int \eta$ does not converge.

For all $f \in \mathscr{S}(\mathbb{R})$,

$$|\langle e^x e^{ie^x}, f\rangle| = |\langle e^{ie^x}/i, f'\rangle| \le \|f'\|_{L^1} \le C \sum_{\alpha \le 3} \rho_{\alpha,1}(f)$$

by Proposition 2.2.6, so $e^x e^{ie^x}$ is in $\mathscr{S}'(\mathbb{R})$ by Proposition 2.3.4 (b).

2.4.4 Recall that $\operatorname{sech}(x) = \frac{1}{\cosh(x)} = \frac{2}{e^x + e^{-x}}$. We need to show that

$$I = \int_{\mathbb{R}} \frac{2}{e^{\pi x} + e^{-\pi x}} e^{-2\pi i x\xi}\, dx = \frac{2}{e^{\pi\xi} + e^{-\pi\xi}}.$$

Let

$$f(z) = \frac{2}{e^{\pi z} + e^{-\pi z}} e^{-2\pi i z\xi}.$$

Integrate $f(z)$ over the rectangle contour $\gamma_R$ with corners $(-R, 0)$, $(R, 0)$, $(-R, i)$ and $(R, i)$, using that $f(z+i) = -e^{2\pi\xi} f(z)$ and that the integral of $f$ over the two vertical sides goes to 0 as $R \to \infty$ since $|f(z)| \le \frac{2}{e^{\pi R}} e^{2\pi|\xi|}$ when $\operatorname{Re} z = \pm R$, we get

$$\big(1 - (-e^{2\pi\xi})\big) I = \lim_{R\to\infty} \int_{\gamma_R} f(z)\, dz = 2\pi i \operatorname{Res}\Big(f, \frac{i}{2}\Big),$$

and proceed our calculation as follows:

$$\begin{aligned} I &= \frac{1}{1+e^{2\pi\xi}}\, 2\pi i \lim_{z\to\frac{i}{2}} \frac{2(z-\frac{i}{2})}{e^{\pi z}+e^{-\pi z}} e^{-2\pi i\frac{i}{2}\xi} \\ &= \frac{2}{e^{\pi\xi}+e^{-\pi\xi}}\, 2\pi i \frac{1}{(e^{\pi z}+e^{-\pi z})'|_{z=\frac{i}{2}}} \\ &= \frac{2}{e^{\pi\xi}+e^{-\pi\xi}}. \end{aligned}$$

2.4.5 Let $w$ be a smooth function the support of whose Fourier transform is $[-1/2, 1/2]$ and let $\phi$ be a real-valued smooth nonconstant periodic function with period 1, which we can make further assumptions upon later. Then take a family of Schwartz functions

$$f_a(x) = w(x)e^{i\phi(x-a)} \quad \text{for } a \in (0, 1/3).$$

Clearly it is an uncountable family of Schwartz functions such that $|f_a| = |f_b|$ for all $a$ and $b$, and we will show that the $f_a$'s are linearly independent, which will be addressed later, and that $|\widehat{f_a}| = |\widehat{f_b}|$. To this end, we expand the function $e^{i\phi(x)}$ in the Fourier series

$$e^{i\phi(x)} = \sum_{k=-\infty}^{\infty} c_k e^{2\pi ikx}.$$

Then

$$f_a(x) = \sum_{k=-\infty}^{\infty} c_k e^{-2\pi ika} e^{2\pi ikx} w(x),$$

so

$$\widehat{f_a}(\xi) = \sum_{k=-\infty}^{\infty} c_k e^{-2\pi ika} \widehat{w}(\xi - k).$$

Since the functions $\widehat{w}(\xi - k)$ are supported in mutually disjoint intervals $[k - 1/2, k + 1/2)$, we have

$$|\widehat{f_a}(\xi)| = |c_k \widehat{w}(\xi - k(\xi))|,$$

where $k(\xi)$ is the integer such that $\xi - k(\xi) \in [-1/2, 1/2)$, which shows that $|\widehat{f_a}(\xi)|$ is independent of $a$.

Now we return to the linear independence of the $f_a$'s, under certain additional assumption on $\phi$. Suppose that

$$\sum_{k=1}^{n} d_k e^{i\phi(x-a_k)} = 0$$

for some distinct $a_k \in (0, 1/3)$. Let $\phi(x)$ coincide with $\frac{1}{x}$ on $(1/3, 1)$, then

$$\sum_{k=1}^{n} d_k e^{\frac{i}{z-a_k}} = 0$$

by the uniqueness of meromorphic functions as the left side equals 0 for $z \in (2/3, 1)$ (hence $z - a_k \in (1/3, 1)$). Taking $z = i\varepsilon + a_k$ where $\varepsilon \to 0^+$ gives $d_k = 0$.

2.4.6 We are going to verify that $(P_y * f)(x)$ satisfies the mean value property and is continuous on $\mathbb{R}^{n+1}_+$. The two properties together imply that $f$ is harmonic. (Sketch: On any ball consider the difference of such a function and the harmonic function solving the Dirichlet problem where the boundary–value is given by its confinement, which supposedly attains its nonzero maximum or minimum on a closed set, at the farthest point away from the center of the ball of which the mean value property fails.)

First establish the mean value property: for a ball $B$ in $\mathbb{R}^{n+1}_+$ centering at $(x_0, y_0)$,

$$\begin{aligned}
\fint_B (P_y * f)(x)\, d\bar{x} &= \fint_B \int_{\mathbb{R}^n} P_y(x-z) f(z)\, dz d\bar{x} \\
&= \int_{\mathbb{R}^n} \Big( \fint_B P_y(x-z)\, d\bar{x} \Big) f(z)\, dz \\
&= \int_{\mathbb{R}^n} P_{y_0}(x_0 - z) f(z)\, dz \\
&= (P_{y_0} * f)(x_0),
\end{aligned}$$

where $\bar{x} = (x, y)$ and a slash through an integral denotes an average.

Then check its continuity:

$$|(P_{y'} * f)(x') - (P_y * f)(x)| \le \|P_{y'}(x' - x + \cdot\,) - P_y(\cdot\,)\|_{L^{p'}} \|f\|_{L^p} \to 0$$

as $(x', y') \to (x, y)$, where $\frac{1}{p} + \frac{1}{p'} = 1$ and $\|P_{y'}(x' - x + \cdot\,) - P_y(\cdot\,)\|_{L^{p'}} \to 0$ is easily confirmed by, say, the Lebesgue dominated convergence theorem.

To show that $P_{y_1} * P_{y_2} = P_{y_1+y_2}$, it suffices to show that their Fourier transforms agree. Indeed, applying Proposition 2.2.11 (8) and Exercise 2.2.11,

$$\widehat{P_y} = (y^{-n} \delta^y P)^\wedge = \delta^y \widehat{P} = e^{-2\pi y|\xi|},$$

so

$$(P_{y_1} * P_{y_2})^\wedge = \widehat{P_{y_1} P_{y_2}} = e^{-2\pi y_1|\xi|} e^{-2\pi y_2|\xi|} = e^{-2\pi(y_1+y_2)|\xi|} = \widehat{P_{y_1+y_2}}.$$

2.4.7 (a) Observe that $v(x; x_0)$ is harmonic is equivalent to $v(x + x_0; x_0)$ is harmonic, where

$$\begin{aligned}
v(x + x_0; x_0) &= (|x_0|^2 - |x + x_0|^2)|x|^{-n} \\
&= -2x_0 \cdot x |x|^{-n} - |x|^{2-n}
\end{aligned}$$

is a linear combination of the harmonic function $|x|^{2-n}$ and its partial derivatives $\frac{\partial}{\partial x_j}|x|^{2-n} = (2-n)x_j|x|^{-n}$ for $n \ge 3$, and $\frac{\partial}{\partial x_j} \log|x| = x_j|x|^{-2}$ for $n = 2$.

Alternatively, one may calculate directly as follows:

$$\begin{aligned}
\Delta v(x + x_0; x_0) &= \Delta\big((1 - |x + x_0|^2) \cdot |x|^{-n}\big) \\
&= \Delta(1 - |x + x_0|^2)|x|^{-n} + 2\nabla(1 - |x + x_0|^2) \cdot \nabla(|x|^{-n}) + (1 - |x + x_0|^2)\Delta|x|^{-n} \\
&= (-2n)|x|^{-n} + 2(-2(x + x_0)) \cdot (-n|x|^{-n-2}x) + (1 - |x + x_0|^2)\, 2n|x|^{-n-2} \\
&= 2n\,|x|^{-n-2}\big(-|x|^2 + 2(x + x_0) \cdot x + |x_0|^2 - |x + x_0|^2\big) \\
&= 0,
\end{aligned}$$

in which the calculation of $\Delta|x|^{-n}$ deserves elaboration:

$$\begin{aligned}\Delta|x|^{-n} &= \operatorname{div}\big(-nx|x|^{-n-2}\big) = -n\big(\operatorname{div}(x)|x|^{-n-2} + x\cdot\nabla|x|^{-n-2}\big)\\ &= -n\Big(n|x|^{-n-2} + x\cdot\Big(\frac{-n-2}{2}|x|^{-n-4}2x\Big)\Big) = 2n|x|^{-n-2}.\end{aligned}$$

(b) For $\theta\in\mathbf{S}^{n-1}$ such that $|\theta-x_0|>\delta$, we have

$$v(rx_0;\theta) = \frac{1-r^2}{|rx_0-\theta|^n} = \frac{1-r^2}{|(r-1)x_0+(x_0-\theta)|^n} \le \frac{1-r^2}{(\delta-(1-r))^n}.$$

Then

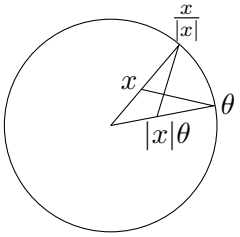
$$\int_{\substack{\theta\in\mathbf{S}^{n-1}\\ |\theta-x_0|>\delta}} v(rx_0;\theta)\,d\theta \le \omega_{n-1}\frac{1-r^2}{(\delta-(1-r))^n}$$

tends to 0 uniformly in $x_0$ as $r\uparrow 1$.

(c)

$$\frac{1}{\omega_{n-1}}\int_{\mathbf{S}^{n-1}}\frac{1-|x|^2}{|x-\theta|^n}\,d\theta = \fint_{\mathbf{S}^{n-1}}\frac{1-\big||x|\theta\big|^2}{\big||x|\theta-\frac{x}{|x|}\big|^n}\,d\theta = 1,$$

where the first equality is justified by the illustration below, and the second by the mean value property of the integrand which is the harmonic function $\theta\mapsto v(|x|\theta;\frac{x}{|x|})$ and $v(0;\frac{x}{|x|})=1$.




(d) The harmonicity of $u$ is easily shown by taking the Laplacian inside the integral; the continuity of $u$ at the boundary is a consequence of the observation in part (b) that the Poisson kernel $v(rx_0;\theta)$ is massed at $\theta$ as $r\uparrow 1$ — to put it more clearly, by part (c) we have

$$\begin{aligned}|u(rx_0)-f(x_0)| &= \left|\fint_{\mathbf{S}^{n-1}}\big(f(\theta)-f(x_0)\big)v(rx_0;\theta)\,d\theta\right|\\ &\le \sup_{|\theta-x_0|\le\delta}|f(\theta)-f(x_0)|\cdot\fint_{\mathbf{S}^{n-1}}v(rx_0;\theta)\,d\theta\\ &\quad+\sup_{\theta\in\mathbf{S}^{n-1}}|f(\theta)-f(x_0)|\cdot\frac{1}{\omega_{n-1}}\int_{\substack{\theta\in\mathbf{S}^{n-1}\\ |\theta-x_0|>\delta}}v(rx_0;\theta)\,d\theta,\end{aligned}$$

first choose $\delta$ small enough so that the first term is vanishingly small by the continuity of $f$, and then for the fixed $\delta$, let $r\uparrow 1$ so that the second term becomes vanishingly small as well by part (b).

2.4.8 (a) In Appendix D.4 the following identity is obtained:

$$\int_{\mathbf{S}^{n-1}}\frac{d\theta}{|\theta-e_1|^\alpha} = \frac{\omega_{n-2}}{2^{\frac{\alpha}{2}}}\int_{-1}^{+1}(1-s)^{\frac{n-3-\alpha}{2}}(1+s)^{\frac{n-3}{2}}ds.$$

Since $\mathbf{S}^n$ is invariant under the rotation that carries $\eta$ to $e_1$, we get

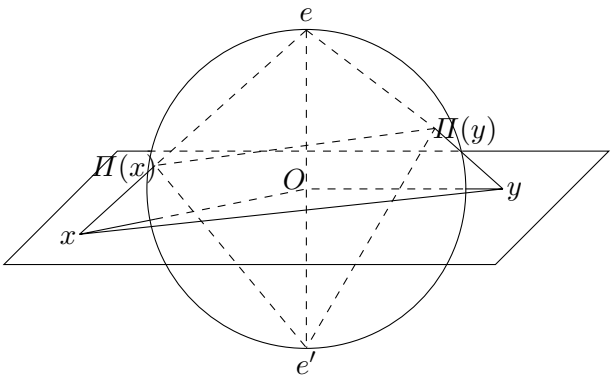
$$\int_{\mathbf{S}^n}|\xi-\eta|^{-\lambda}\,d\xi = \int_{\mathbf{S}^n}|\xi-e_1|^{-\lambda}\,d\xi$$

$$
\begin{aligned}
&= \frac{\omega_{n-1}}{2^{\frac{\lambda}{2}}} \int_{-1}^{+1} (1-s)^{\frac{n-\lambda}{2}-1}(1+s)^{\frac{n}{2}-1}\, ds \\
&= \frac{\omega_{n-1}}{2^{\frac{\lambda}{2}}} \int_{-1}^{+1} (2-2t)^{\frac{n-\lambda}{2}-1}(2t)^{\frac{n}{2}-1}\, d(2t-1) \\
&= \omega_{n-1} 2^{n-\lambda-1} \int_0^1 (1-t)^{\frac{n-\lambda}{2}-1} t^{\frac{n}{2}-1}\, dt \\
&= \omega_{n-1} 2^{n-\lambda-1} B\Big(\frac{n-\lambda}{2}, \frac{n}{2}\Big) \\
&= \frac{2\pi^{\frac{n}{2}}}{\Gamma(\frac{n}{2})} 2^{n-\lambda-1} \frac{\Gamma(\frac{n-\lambda}{2})\Gamma(\frac{n}{2})}{\Gamma(n-\frac{\lambda}{2})} \\
&= \pi^{\frac{n}{2}} 2^{n-\lambda} \frac{\Gamma(\frac{n-\lambda}{2})}{\Gamma(n-\frac{\lambda}{2})}.
\end{aligned}
$$

(b) Notice that

$$
\begin{aligned}
&\int_{\mathbb{R}^n} |x-y|^{-\lambda}(1+|x|^2)^{\frac{\lambda}{2}}(1+|y|^2)^{\frac{\lambda}{2}}(1+|x|^2)^{-n}\, dx \\
&= \int_{\mathbb{R}^n} (|\Pi(x)-\Pi(y)|/2)^{-\lambda} J_\Pi(x)/2^n\, dx \\
&= 2^{\lambda-n} \int_{\mathbf{S}^{n-1}} |\xi-\eta|^{-\lambda}\, d\xi \\
&= \frac{\pi^{\frac{n}{2}}\Gamma(\frac{n-\lambda}{2})}{\Gamma(n-\frac{\lambda}{2})},
\end{aligned}
$$

where $\Pi$ is the stereographic projection in Appendix D.6, in the first equality the so–called interesting formula $|\Pi(x)-\Pi(y)| = 2|x-y|(1+|x|^2)^{-1/2}(1+|y|^2)^{-1/2}$ is used, and the last equality is by (a), from which we deduce that the Jacobian of $\Pi$ is $J_\Pi(x) = \big(\frac{2}{1+|x|^2}\big)^n$ (instead of $J_\Pi(x) = \big(\frac{2}{1+|x|^2}\big)^{-n}$ as Appendix D.6 claims).
Now we give a proof of the formula for $|\Pi(x)-\Pi(y)|$.




Observe that $\triangle exy \backsim \triangle e\Pi(y)\Pi(x)$ with a scale denoted by $1:k$ as

$$|ex||e\Pi(x)| = |ey||e\Pi(y)| = |eO||ee'| = 2,$$

where $e = (0,\cdots,0,1)$ and $e' = (0,\cdots,0,-1)$ are the north pole and the south pole. Since

$$2 = |ex||e\Pi(x)| = k|ex||ey| = k(1+|x|^2)^{1/2}(1+|y|^2)^{1/2},$$

we get

$$|\Pi(x)\Pi(y)| = k|xy| = 2|xy|(1+|x|^2)^{-1/2}(1+|y|^2)^{-1/2}.$$

2.4.9 Theorem 2.4.6 establishes the formula

$$\pi^{\frac{z+n}{2}}\Gamma\Big(\frac{z+n}{2}\Big)^{-1}(|x|^z)^\wedge = \pi^{-\frac{z}{2}}\Gamma\Big(-\frac{z}{2}\Big)^{-1}|x|^{-n-z},$$

that is,

$$(|x|^z)^\wedge = \pi^{-z-\frac{n}{2}}\frac{\Gamma(\frac{z+n}{2})}{\Gamma(-\frac{z}{2})}|x|^{-n-z} \triangleq c_z|x|^{-n-z}.$$

Then

$$\begin{aligned}
&\int_{\mathbb{R}^n}\frac{dt}{|x-t|^{\alpha_1}|y-t|^{\alpha_2}}\\
&= (|t|^{-\alpha_1} * |t|^{-\alpha_2})(x-y)\\
&= ((|t|^{-\alpha_1})^\wedge(|t|^{-\alpha_2})^\wedge)^\vee(x-y)\\
&= c_{-\alpha_1}c_{-\alpha_2}(|t|^{-n+\alpha_1-n+\alpha_2})^\vee(x-y)\\
&= c_{-\alpha_1}c_{-\alpha_2}c_{-2n+\alpha_1+\alpha_2}|x-y|^{n-\alpha_1-\alpha_2},
\end{aligned}$$

where

$$\begin{aligned}
&c_{-\alpha_1}c_{-\alpha_2}c_{-2n+\alpha_1+\alpha_2}\\
&= \pi^{\alpha_1-\frac{n}{2}}\frac{\Gamma(\frac{n-\alpha_1}{2})}{\Gamma(\frac{\alpha_1}{2})}\pi^{\alpha_2-\frac{n}{2}}\frac{\Gamma(\frac{n-\alpha_2}{2})}{\Gamma(\frac{\alpha_2}{2})}\pi^{2n-\alpha_1-\alpha_2-\frac{n}{2}}\frac{\Gamma(\frac{n-2n+\alpha_1+\alpha_2}{2})}{\Gamma(-\frac{-2n+\alpha_1+\alpha_2}{2})}\\
&= \pi^{\frac{n}{2}}\frac{\Gamma(\frac{n-\alpha_1}{2})\Gamma(\frac{n-\alpha_2}{2})\Gamma(\frac{\alpha_1+\alpha_2-n}{2})}{\Gamma(\frac{\alpha_1}{2})\Gamma(\frac{\alpha_2}{2})\Gamma(n-\frac{\alpha_1+\alpha_2}{2})}.
\end{aligned}$$

2.4.10 (a) This is because for all rotations $A$ and $\xi \in \mathbb{R}^n$, by Proposition 2.2.11 (13),

$$\hat{f}(A\xi) = \widehat{f\circ A}(\xi) = \hat{f}(\xi).$$

(b) Now the identity above holds for all rotations $A$ which keep $e_1$ fixed.

2.4.11 The translation $(x,y)\mapsto(x+z,y+z)$ and a dilation with a rotation that carries $y$ to $e_1$ reduce matters to the case that $z=0$ and $y=e_1$. We prove the identity by showing that the Fourier transforms **in $x$** of both sides coincide.

First take the Fourier transform on the right side, setting aside its coefficient, and get

$$\begin{aligned}
&(|x-e_1|^{d_1-n}|x|^{d_3-n})^\wedge(\xi)\\
&= c_{d_1-n}c_{d_3-n}(|\xi|^{-d_1}e^{-2\pi i\xi e_1}) * |\xi|^{-d_3}\\
&= c_{d_1-n}c_{d_3-n}\int_{\mathbb{R}^n}|t|^{-d_1}e^{-2\pi it\cdot e_1}|\xi-t|^{-d_3}dt\\
&= c_{d_1-n}c_{d_3-n}|\xi|^{-d_1-d_3+n}\int_{\mathbb{R}^n}|t|^{-d_1}e^{-2\pi i|\xi|t\cdot e_1}|\xi/|\xi|-t|^{-d_3}dt.
\end{aligned}$$

Let $A_\xi$ be a rotation that transforms $e_1$ to $\xi/|\xi|$, then the expression above is equal to

$$\begin{aligned}
&= c_{d_1-n}c_{d_3-n}|\xi|^{-d_1-d_3+n}\int_{\mathbb{R}^n}|A_\xi^{-1}t|^{-d_1}e^{-2\pi i|\xi|A_\xi^{-1}t\cdot A_\xi^{-1}e_1}|e_1-A_\xi^{-1}t|^{-d_3}d(A_\xi^{-1}t)\\
&= c_{d_1-n}c_{d_3-n}|\xi|^{-d_1-d_3+n}\hat{h}(A_\xi^{-1}e_1),
\end{aligned}$$

where $h(t) = |t|^{-d_1}|t-e_1|^{-d_3}$.

Next take the Fourier transform on the left side, and get

$$(h * |t|^{-d_2})^\wedge(\xi) = c_{-d_2}|\xi|^{n-d_2}\hat{h}(\xi).$$

Since $-d_1 - d_3 + n = n - d_2$ and

$$\begin{aligned} & c_{-d_2}/(c_{d_1-n}c_{d_3-n}) \\ &= \pi^{d_2-\frac{n}{2}}\frac{\Gamma(\frac{n-d_2}{2})}{\Gamma(\frac{d_2}{2})}\Bigg/\left(\pi^{\frac{n}{2}-d_1}\frac{\Gamma(\frac{d_1}{2})}{\Gamma(\frac{n-d_1}{2})}\pi^{\frac{n}{2}-d_3}\frac{\Gamma(\frac{d_3}{2})}{\Gamma(\frac{n-d_3}{2})}\right) \\ &= \pi^{\frac{n}{2}}\prod_{j=1}^{3}\frac{\Gamma(\frac{n-d_j}{2})}{\Gamma(\frac{d_j}{2})}, \end{aligned}$$

it remains to show that $\hat{h}(\xi) = \hat{h}(A_\xi^{-1}e_1)$, which explains what part (b) of the preceding exercise is for.

2.4.12 (a) On $P_2$ we have $\left|e^{i(Re^{i\theta})^2}\right| = e^{-R^2\sin(2\theta)} \le e^{-\frac{4}{\pi}R^2\theta}$, and hence

$$\left|\int_{P_2} e^{iz^2}dz\right| \le \int_0^{\frac{\pi}{4}} e^{-\frac{4}{\pi}R^2\theta}|Re^{i\theta}d\theta| \le \frac{1}{R}\int_0^{\infty} e^{-\frac{4}{\pi}R^2\theta}d(R^2\theta) \to 0 \quad \text{as } R\to\infty.$$

Then we get from $\int_{P_1\cup P_2\cup P_3} e^{iz^2}dz = 0$ that

$$\begin{aligned} \lim_{R\to\infty}\int_0^R e^{ix^2}dx &= \lim_{R\to\infty}\int_{P_1} e^{iz^2}dz = -\lim_{R\to\infty}\int_{P_3} e^{iz^2}dz \\ &= -\lim_{R\to\infty}\int_0^R e^{-r^2}e^{i\frac{\pi}{4}}dr = e^{i\frac{\pi}{4}}\frac{\sqrt{\pi}}{2} = \frac{\sqrt{2\pi}}{4}(1+i). \end{aligned}$$

(b) Enough to deal with the case of $n = 1$, and the general case is its mere multiplication. When $n = 1$,

$$\begin{aligned} \int_{\mathbb{R}} e^{i\pi x^2}e^{-2\pi ix\xi}\,dx &= e^{-i\pi\xi^2}\int_{\mathbb{R}} e^{i\pi(x-\xi)^2}dx \\ &= e^{-i\pi\xi^2}2\int_0^{\infty} e^{i\pi x^2}dx \\ &= e^{-i\pi\xi^2}e^{i\frac{\pi}{4}}. \end{aligned}$$

## 2.5 Convolution Operators on $L^p$ Spaces and Multipliers

2.5.1 By the Lebesgue dominated convergence theorem, $\forall\varepsilon > 0, \exists N$ s.t.

$$\|f - f\chi_{B(0,N)}\|_{L^q} < \varepsilon.$$

Let $g = f\chi_{B(0,N)}$. Then for $|h| > 2N$, the supports of $\tau^h(g)$ and $g$ are disjoint, so

$$\begin{aligned} \left|\|\tau^h(f)+f\|_{L^q} - 2^{\frac{1}{q}}\|f\|_{L^q}\right| &= \left|\|\tau^h(f)+f\|_{L^q} - \|\tau^h(g)+g\|_{L^q} - 2^{\frac{1}{q}}(\|f\|_{L^q} - \|g\|_{L^q})\right| \\ &\le \|\tau^h(f-g)+(f-g)\|_{L^q} + 2^{\frac{1}{q}}\left|\|f\|_{L^q} - \|g\|_{L^q}\right| \\ &\le \|\tau^h(f-g)\|_{L^q} + \|(f-g)\|_{L^q} + 2^{\frac{1}{q}}\|f-g\|_{L^q} \\ &\le (2+2^{\frac{1}{q}})\varepsilon. \end{aligned}$$

Let $\varepsilon$ tend to 0 to come to the required conclusion.

2.5.2 To verify Proposition 2.5.14, we have

$$\|\tau^x(m)\|_{\mathscr{M}_p} = \sup \frac{\|(\tau^x(\tau^{-x}(\hat{f})m))^\vee\|_{L^p}}{\|f\|_{L^p}} = \sup \frac{\|(\widehat{e^{-2\pi i(\,\cdot\,)\cdot x}fm})^\vee\|_{L^p}}{\|e^{-2\pi i(\,\cdot\,)\cdot x}f\|_{L^p}} = \sup \frac{\|(\hat{f}m)^\vee\|_{L^p}}{\|f\|_{L^p}} = \|m\|_{\mathscr{M}_p}.$$

$$\|\delta^h(m)\|_{\mathscr{M}_p} = \sup \frac{\|(\delta^h(\delta^{h^{-1}}(\hat{f})m))^\vee\|_{L^p}}{\|f\|_{L^p}} = \sup \frac{\|(\widehat{\delta^h f}m)^\vee\|_{L^p}}{\|\delta^h f\|_{L^p}} = \sup \frac{\|(\hat{f}m)^\vee\|_{L^p}}{\|f\|_{L^p}} = \|m\|_{\mathscr{M}_p}.$$

$$\|\widetilde{m}\|_{\mathscr{M}_p} = \sup \frac{\|(\hat{f}\widetilde{m})^\vee\|_{L^p}}{\|f\|_{L^p}} = \sup \frac{\|(\hat{\tilde{f}}m)^\vee\|_{L^p}}{\|\tilde{f}\|_{L^p}} = \sup \frac{\|(\hat{f}m)^\vee\|_{L^p}}{\|f\|_{L^p}} = \|m\|_{\mathscr{M}_p}.$$

$$\|e^{2\pi i(\,\cdot\,)\cdot x}m\|_{\mathscr{M}_p} = \sup \frac{\|(\widehat{f}e^{2\pi i(\,\cdot\,)\cdot x}m)^\vee\|_{L^p}}{\|f\|_{L^p}} = \sup \frac{\|(\widehat{\tau^{-x}f}m)^\vee\|_{L^p}}{\|\tau^{-x}f\|_{L^p}} = \sup \frac{\|(\hat{f}m)^\vee\|_{L^p}}{\|f\|_{L^p}} = \|m\|_{\mathscr{M}_p}.$$

$$\|m\circ A\|_{\mathscr{M}_p} = \sup \frac{\|(\widehat{f\circ A^{-1}}\circ A\cdot m\circ A)^\vee\|_{L^p}}{\|f\|_{L^p}} = \sup \frac{\|(\widehat{f\circ A^{-1}}m)^\vee\circ A\|_{L^p}}{\|f\circ A^{-1}\|_{L^p}} = \sup \frac{\|(\hat{f}m)^\vee\|_{L^p}}{\|f\|_{L^p}} = \|m\|_{\mathscr{M}_p}.$$

Note that $\delta_j^n\hat{f}(\xi) = \frac{1}{h}\widehat{\delta_j^{h^{-1}}}$, then the proof is similar with that of $\delta^h$.

2.5.3 (a)

$$\|\psi m\|_{\mathscr{M}_p} = \sup \frac{\|(\hat{f}\psi m)^\vee\|_{L^p}}{\|f\|_{L^p}} = \sup \frac{\|(\hat{f}m)^\vee * \psi^\vee\|_{L^p}}{\|f\|_{L^p}}.$$

By Young's inequality,

$$\|\psi m\|_{\mathscr{M}_p} \le \sup \frac{\|(\hat{f}m)^\vee\|_{L^p}}{\|f\|_{L^p}}\|\psi^\vee\|_{L^1} = \|m\|_{\mathscr{M}_p}\|\psi^\vee\|_{L^1}.$$

(b) Since $m$ is essentially bounded, by Fubini's theorem,

$$\begin{aligned} T_{\psi * m}(f)(x) &= \int_{\mathbb{R}^n}\int_{\mathbb{R}^n} \psi(y)m(\xi-y)\,\mathrm{d}y\,\hat{f}(\xi)e^{2\pi i x\cdot\xi}\,\mathrm{d}\xi \\ &= \int_{\mathbb{R}^n}\psi(y)\int_{\mathbb{R}^n} m(\xi-y)\hat{f}(\xi)e^{2\pi i x\cdot\xi}\,\mathrm{d}\xi\mathrm{d}y \\ &= \int_{\mathbb{R}^n}\psi(y)\big(m(\,\cdot\,-y)\hat{f}(\,\cdot\,)\big)^\vee(x)\,\mathrm{d}y. \end{aligned}$$

Therefore

$$\begin{aligned} \|T_{\psi*m}(f)\|_{L^p} &\le \int_{\mathbb{R}^n}|\psi(y)|\|(\tau^y m\hat{f})^\vee\|_{L^p}\,\mathrm{d}y \\ &\le \int_{\mathbb{R}^n}|\psi(y)|\,\|m\|_{\mathscr{M}_p}\|f\|_{L^p}\,\mathrm{d}y \\ &= \|\psi\|_{L^1}\|m\|_{\mathscr{M}_p}\|f\|_{L^p}, \end{aligned}$$

where in the last inequality the first property of Proposition 2.5.14 is used.

2.5.4 (a) Notice that $\widehat{T(f)} = \widehat{f * \partial^\gamma\delta_0} = \hat{f}\widehat{\partial^\gamma\delta_0} = \hat{f}(2\pi i x)^\gamma$, which shows that the operator $\mathscr{F}\circ T$ is continuous as it is bounded. But $\mathscr{F}$ is a homeomorphism on $\mathscr{S}$, which implies that $T$ is continuous.

(b) When $p > q$, $\mathscr{M}_{p,q} = 0$, implying that there is no nontrivial element.

When $q \ge p$, for $f \in \mathscr{S}$,

$$\frac{\|T(\delta^\lambda f)\|_{L^q}}{\|\delta^\lambda f\|_{L^p}} = \frac{\|\delta^\lambda f * \partial^\gamma\delta_0\|_{L^q}}{\|\delta^\lambda f\|_{L^q}} = \frac{\|\partial^\gamma\delta^\lambda f * \delta_0\|_{L^q}}{\|\delta^\lambda f\|_{L^q}} = \frac{|\lambda|^{|\gamma|}|\lambda|^{\frac{n}{q}}}{|\lambda|^{\frac{n}{p}}}\frac{\|T(f)\|_{L^q}}{\|f\|_{L^p}} = |\lambda|^{|\gamma|+\frac{n}{q}-\frac{n}{p}}\frac{\|T(f)\|_{L^q}}{\|f\|_{L^p}}.$$

By the condition, the coefficient $|\lambda|^{|\gamma|+\frac{n}{q}-\frac{n}{p}}$ is unbounded when $\lambda$ varies.

2.5.5 Let $r = \frac{n}{n-\gamma}$. From Theorem 1.4.25,

$$\frac{\|f * K_\gamma\|_{L^q}}{\|f\|_{L^p}} \le B\|K_\gamma\|_{L^{r,\infty}} = B\nu_n.$$

2.5.6 (a) We only need to consider the case of $1 \le p \le 2$. Note that $\|m_j\|_{L^\infty} \le \|m_j\|_{\mathscr{M}_p} \le C$. By the Lebsegue dominated convergence theorem,

$$T_{m_j}(f)(x) = \int_{\mathbb{R}^n} \hat{f}(\xi) m_j(\xi) e^{2\pi i x\cdot\xi}\, d\xi \to \int_{\mathbb{R}^n} \hat{f}(\xi) m(\xi) e^{2\pi i x\cdot\xi}\, d\xi = T_m(f)(x).$$

By Fatou's lemma,

$$\int_{\mathbb{R}^n} |T_m(f)|^p\, dx \le \liminf_{j\to\infty} \int_{\mathbb{R}^n} |T_{m_j}(f)|^p\, dx \le \liminf_{j\to\infty} \|m_j\|_{\mathscr{M}_p}^p \|f\|_{L^p}^p.$$

Therefore, $\|m\|_{\mathscr{M}_p} \le \liminf_{j\to\infty} \|m_j\|_{\mathscr{M}_p} \le C$.

(b) Let $m_j = \delta^{\frac{1}{j}} m$, $j \ge 1$, then $m_j \to m_0$.

Note that $\|m_j\|_{\mathscr{M}_p} = \|m\|_{\mathscr{M}_p}$, so by part (a), $\|m_0\|_{\mathscr{M}_p} \le \|m\|_{\mathscr{M}_p}$.

(c) Obviously, $m_0(x) = \begin{cases} m(0+), x > 0, \\ m(0-), x < 0. \end{cases}$ Therefore,

$$\|m_0\|_{\mathscr{M}_p} \ge \|m_0\|_{\mathscr{M}_2} = \|m_0\|_{L^\infty} = \max(|m(0+)|, |m(0-)|).$$

(d) By Fubini's theorem and Minkowski's inequality, we see that

$$\|m\|_{\mathscr{M}_p} = \sup \frac{\big\| \int_0^\infty (m_t \hat{f})^\vee \frac{dt}{t} \big\|_{L^p}}{\|f\|_{L^p}} \le \int_0^\infty \|m_t\|_{\mathscr{M}_p} \frac{dt}{t}.$$

2.5.7

$$\|f\|_{L^p} = \|(\hat{f}m^{-1}m)^\vee\|_{L^p} \le \|m\|_{\mathscr{M}_p} \|(\hat{f}m^{-1})^\vee\|_{L^p} = \|m\|_{\mathscr{M}_p} \|T(f)\|_{L^p}.$$

The lower bound for $|m|$ is to guarantee that $\hat{f}m^{-1} \in L^1(\mathbb{R}^n)$ so that $(\hat{f}m^{-1})^\vee$ is indeed a function.

2.5.8 (a) Using Exercise 1.2.9, we have

$$\|m\|_{\mathscr{M}_p} = \sup \frac{\|f * m^\vee\|_{L^p}}{\|f\|_{L^p}} = \|m^\vee\|_{L^1}.$$

(b)

$$(m_2 m_1^{-1})^\vee(\xi) = \int_{-1}^1 (1-|x|) e^{-2\pi i x\xi}\, dx = \frac{1}{2(\pi\xi)^2}(\cos(2\pi\xi) - 1) \le 0.$$

By part (a),

$$\|m_2 m_1^{-1}\|_{\mathscr{M}_p} = \|(m_2 m_1^{-1})^\vee\|_{L^1} = \int_{\mathbb{R}} \frac{1-\cos x}{\pi x^2}\, dx.$$

Consider the integral of $f(z) = \frac{e^{iz}-1}{z^2}$ along the path below, which is the boundary of half of an annulus. Since $e^{iz} = 1 + iz + O(|z|^2)$ near 0,

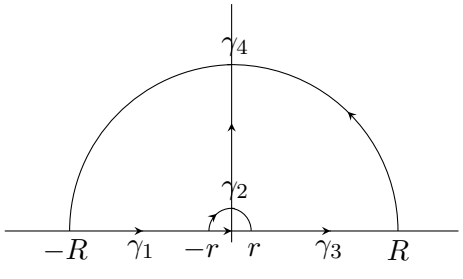
$$\int_{\gamma_2} \frac{e^{iz}-1}{\pi z^2}\, dz = \int_{\gamma_2} \frac{i}{\pi z} + O(1)\, dz = \frac{i}{\pi}(-i\pi) + O(r) = 1 + O(r).$$

Since $|e^{iz}| = e^{-\operatorname{Im} z} \le 1$ on the upper half–plane,

$$\left| \int_{\gamma_4} \frac{e^{iz}-1}{\pi z^2}\, dz \right| \le \int_{\gamma_4} \frac{|e^{iz}|+1}{\pi|z|^2}\, |dz| \le \int_{\gamma_4} \frac{2}{\pi R^2}\, |dz| = \frac{2}{R}.$$

As $f$ has no singularity in the domain enclosed, we have

$$\int_{\mathbb{R}} \frac{1-\cos x}{\pi x^2}\,dx = -\operatorname{Re}\lim_{\substack{r\to 0,\\ R\to\infty}} \int_{\gamma_1\cup\gamma_3} \frac{e^{iz}-1}{\pi z^2}\,dz = \operatorname{Re}\lim_{\substack{r\to 0,\\ R\to\infty}} \int_{\gamma_2\cup\gamma_4} \frac{e^{iz}-1}{\pi z^2}\,dz = 1.$$




Hence $\|m_2 m_1^{-1}\|_{\mathscr{M}_p} = 1$. Then

$$\|m_2\|_{\mathscr{M}_p} \le \|m_2 m_1^{-1}\|_{\mathscr{M}_p}\|m_1\|_{\mathscr{M}_p} = \|m_1\|_{\mathscr{M}_p}.$$

Note that $m_1(\xi) = \lim_{R\to\infty} m_2\big(\frac{\xi}{R}\big)$, then by Exercise 2.5.6 (b),

$$\|m_1\|_{\mathscr{M}_p} \le \|m_2\|_{\mathscr{M}_p}.$$

2.5.9

$$\Big\|\Big(\sum_m a_m e^{-2\pi i m x}\hat{f}(x)\Big)^{\vee}\Big\|_{L^p} = \Big\|\Big(\sum_m a_m \widehat{\tau^m f}\Big)^{\vee}\Big\|_{L^p} = \Big\|\sum_m a_m \tau^m f\Big\|_{L^p} = \Big\|\sum_m a_m f(x-m)\Big\|_{L^p},$$

from which we know (a) is equivalent to (b).

For a sequence $\{a_m\}$ that satisfies (c), it is easy to see $\{a_m\} \in l^p(\mathbb{Z}^n)$. Let $K$ be the operator

$$L^p(\mathbb{R}^n) \to L^p(\mathbb{R}^n), \quad f \mapsto \sum_m a_m f(x-m).$$

Write the cube $Q = \big[-\frac{1}{2}, \frac{1}{2}\big]^n$, $|Q| = 1$. Note that

$$\|f\|_{L^p}^p = \int_Q \sum_n |f(x+n)|^p\,dx,$$

so $\{f(x-m)\}_{m\in\mathbb{Z}^n} \in l^p(\mathbb{Z}^n)$, a.e. $x$, so $Kf$ is defined almost everywhere.

$$\int_{\mathbb{R}^n} |Kf|^p\,dx = \int_Q \sum_n \Big|\sum_m a_m f(a+n-m)\Big|^p\,dx \le \int_Q A^p \sum_n |f(x+n)|^p\,dx = A^p\|f\|_{L^p}^p.$$

This implies $\|K\| \le A$.

For a sequence $\{c_m\}$, take $f(x) = \sum_m c_m \chi_Q(x-m)$. Then

$$\frac{\Big(\sum_n \Big|\sum_m a_m c_{n-m}\Big|^p\Big)^{\frac{1}{p}}}{\Big(\sum_n |c_n|^p\Big)^{\frac{1}{p}}} = \frac{\Big(\sum_n \Big|\int_{n+Q}\sum_m a_m f(x-m)\,dx\Big|^p\Big)^{\frac{1}{p}}}{\Big(\sum_n |c_n|^p\Big)^{\frac{1}{p}}}$$

$$\le \frac{\Big(\sum_n \int_{n+Q}\Big|\sum_m a_m f(x-m)\Big|^p\,dx\Big)^{\frac{1}{p}}}{\|f\|_{L^p}} = \frac{\Big\|\sum_m a_m f(x-m)\Big\|_{L^p}}{\|f\|_{L^p}} \le \|K\|.$$

Here we use Hölder's inequality. This implies $A \le \|K\|$. So (a) is equivalent to (c).

2.5.10 Let $Q = \left[-\frac{1}{2}, \frac{1}{2}\right]^n$, by the invariance of norm under translation, we can assume that $m$ is supported in $Q$. Since $m$ is in $L^\infty$ and is compactly supported, it lies in $L^2$ and hence it has a Fourier expansion. Write

$$m(\xi) = \sum_{j\in\mathbb{Z}^n} a_j e^{2\pi i j\cdot\xi}, \quad \xi \in [0,1]^n.$$

Then

$$M(\xi) = \sum_{j\in\mathbb{Z}^n} a_j e^{2\pi i j\cdot\xi}, \quad \xi \in \mathbb{R}^n.$$

Let $h \in C^\infty$ to be determined later. Define the operators

$$E(\{c_j\}_j) = \sum_{k\in\mathbb{Z}^n} c_k \chi_Q(x+k),$$
$$R(f)(j) = \int_{-j+Q} f(x)\,\mathrm{d}x.$$

Consider

$$\begin{aligned} RT_{mh}E(\{c_k\}_k)(j) &= RT_{mh}\Big(\sum_{k\in\mathbb{Z}^n} c_k\chi_Q(\cdot+k)\Big)(j) \\ &= R\Big(\Big(mh\sum_{k\in\mathbb{Z}^n} c_k\widehat{\chi_Q}e^{2\pi i(\cdot)\cdot k}\Big)^\vee\Big)(j) \\ &= \sum_{k\in\mathbb{Z}^n} c_k q_{j-k}, \end{aligned}$$

where

$$\begin{aligned} q_j &= \int_Q (mh\widehat{\chi_Q})^\vee(x-j)\,\mathrm{d}x \\ &= \int_Q\int_Q m(\xi)h(\xi)\widehat{\chi_Q}(\xi)e^{2\pi i(x-j)\cdot\xi}\,\mathrm{d}\xi\mathrm{d}x \\ &= \sum_{k\in\mathbb{Z}^n} a_k \int_Q (\widehat{\chi_Q})^2(\xi)h(\xi)e^{2\pi i(k-j)\cdot\xi}\,\mathrm{d}\xi. \end{aligned}$$

(The exchange of sum and integral is from the convergence of the series in $L^2$.) Thus $q_j = a_j$ if we pick a $C_c^\infty(\mathbb{R}^n)$ function $h$ equal to

$$\begin{aligned} (\widehat{\chi_Q})^{-2}(\xi) &= \prod_{j=1}^n \Big(\int_{-1/2}^{1/2} e^{-2\pi i x_j\xi_j}\,dx_j\Big)^{-2} \\ &= \prod_{j=1}^n \Big(\frac{e^{-\pi i\xi_j} - e^{\pi i\xi_j}}{-2\pi i\xi_j}\Big)^{-2} \\ &= \prod_{j=1}^n \pi^2\xi_j^2\sin^{-2}(\pi\xi_j) \end{aligned}$$

on $Q$. Therefore the operator $RT_{mh}E$ is given by convolution with the sequence $\{a_j\}_j$.
It is obvious that $\|E\|_{l^p\to L^p} \le 1$, and by Hölder's inequality, $\|R\|_{L^p\to l^p} \le 1$. So we have

$$\|RT_{mh}E\|_{l^p\to l^p} \le \|T_{mh}\|_{L^p\to L^p} \le \|h^\vee\|_{L^1}\|m\|_{\mathscr{M}_p}.$$

The last inequality comes from Exercise 2.5.3, and the required conclusion comes from Exercise 2.5.9.

2.5.11 By Proposition 2.4.8, $\hat{u} \in C^\infty(\mathbb{R}^n\backslash\{0\})$, and $\hat{u}$ is homogenous of degree $-i\tau$. Let $m = \hat{u}$, then

$$\|m\|_{L^\infty(\mathbb{R}^n\backslash\{0\})} = \|m\|_{L^\infty(\partial B(0,1))} < \infty.$$

So

$$\frac{\|f * u\|_{L^2}}{\|f\|_{L^2}} = \frac{\|(\hat{f}m)^\vee\|_{L^2}}{\|f\|_{L^2}} \leq \|m\|_{\mathscr{M}_2} = \|m\|_{L^\infty} < \infty.$$

2.5.12 Consider $T : (m_1, m_2, f) \mapsto \big((m_1 * m_2)\hat{f}\big)^\vee$. Note that

$$\|T(m_1, m_2, f)\|_{L^2} = \|(m_1 * m_2)f^\vee\|_{L^2} \leq \|m_1 * m_2\|_{L^\infty}\|f\|_{L^2} \leq \|m_1\|_{L^\infty}\|m_2\|_{L^1}\|f\|_{L^2}.$$

So $T$ maps $L^\infty \times L^1 \times L^2$ to $L^2$ with norm no more than 1.

$$\begin{aligned}\|T(m_1, m_2, f)\|_{L^1} &\leq \|(m_1 * m_2)^\vee\|_{L^1}\|f\|_{L^1} = \|m_1^\vee m_2^\vee\|_{L^1}\|f\|_{L^1}\\ &\leq \|f\|_{L^1}\|m_1^\vee\|_{L^2}\|m_2^\vee\|_{L^2} = \|f\|_{L^1}\|m_1\|_{L^2}\|m_2\|_{L^2}.\end{aligned}$$

So $T$ maps $L^2 \times L^2 \times L^1$ with norm no more than 1.

Observe that $\begin{cases}\frac{1}{r} = \frac{1}{2}\frac{2}{r}\\ \frac{1}{r'} = \frac{1}{2}\frac{2}{r} + 1\big(1 - \frac{2}{r}\big)\\ \frac{1}{p} = \frac{2}{r} + \frac{1}{2}\big(1 - \frac{2}{r}\big)\end{cases}$. By the multilinear interpolation theorem (see GTM250 pp.516-7), we have

$$\|T(m_1, m_2, f)\|_{L^p} \leq \|m_1\|_{L^r}\|m_2\|_{L^{r'}}\|f\|_{L^p},$$

which implies that $m_1 * m_2 \in \mathscr{M}_p(\mathbb{R}^n)$.

2.5.13 By Exercise 2.4.12, $(e^{i|\xi|^2})^\vee = e^{i\frac{\pi n}{4}}e^{-i\pi|x|^2} \in L^\infty$. Let $m(\xi) = e^{i|\xi|^2}$, then

$$\|T_m(f)\|_{L^\infty} = \|f * m^\vee\|_{L^\infty} \leq \|f\|_{L^1}\|m^\vee\|_{L^\infty},$$

which implies $T_m$ is of strong $(1, \infty)$ type. Plus,

$$\|T_m(f)\|_{L^2} = \|\hat{f}(\xi)e^{i|\xi|^2}\|_{L^2} = \|\hat{f}(\xi)\|_{L^2} = \|f\|_{L^2},$$

which implies $T_m$ is of strong $(2, 2)$ type.

Consequently by the Riesz–Thorin interpolation theorem, $T_m$ is of strong $(p, p')$ type.

If $T_m$ were of strong $(p, p)$ type for some $p \neq 2$, for $\|m\|_{\mathscr{M}_p} = \|m\|_{\mathscr{M}_{p'}}$, we could assume that $1 \leq p < 2$. Then by Riesz–Thorin again, $T_m$ is of strong $(p, 2)$ type with

$$\|f\|_{L^2} = \|T_m(f)\|_{L^2} \leq \|f\|_{L^p}.$$

This is surely not true.

## 2.6 Oscillatory Integrals

2.6.1 Since $u^{(k)}$ is continuous, it can assumed that $u^{(k)}$ is positive. Replacing $\frac{u}{c_0}, c_0\lambda$ with $u, \lambda$ in Proposition 2.6.7 leads to the required conclusion.

2.6.2 Let $u(t)=\begin{cases} t+m(k-1), & t\in[2m,2m+1) \\ kt+(m+1)(1-k), & t\in[2m+1,2m+2) \end{cases}$, $m\in\mathbb{N}$.

It is easy to see

$$\left|\int_0^{2n} e^{i\lambda u(t)}\,\mathrm{d}t\right| = \left|\left(\sum_{m=0}^{n-1} e^{i\lambda(k+1)m}\right)\cdot\frac{1}{\lambda}\left(e^{i\lambda}-1+\frac{e^{i(k+1)\lambda}-e^{i\lambda}}{k}\right)\right| = \left|\frac{1-e^{i\lambda(k+1)n}}{1-e^{i\lambda(k+1)}}\frac{1}{\lambda}\left(e^{i\lambda}-1+\frac{e^{i(k+1)\lambda}-e^{i\lambda}}{k}\right.\right.$$

Let $\lambda=\pi$, for any $\varepsilon>0$, by Kroncker's lemma (4.2.2), there exists $k>2, k\in\mathbb{Z}$ such that $\left|1-e^{i\lambda(k+1)}\right|<\varepsilon$, and choose $n\in\mathbb{Z}$ such that $\left|1-e^{i\lambda(k+1)n}\right|>1$.

Modify in the $\delta$-neighborhood of $\mathbb{Z}$ to make $u$ smooth, choose $\varepsilon$ sufficiently small, we get the counterexample.

2.6.3

2.6.4 (a) Apply Exercise 2.6.1 to the intervals $[a,c-\delta]$ and $[c+\delta,b]$, and notice that $\left|\int_{c-\delta}^{c+\delta} e^{i\lambda u(t)}\right|\le 2\delta$, we reach the conclusion.

(b) Similar to part (a).

(c) Choose $\delta=\lambda^{-\frac{1}{k+1}}$.

2.6.5 (a) Notice that

$$\int_{\varepsilon\le|t|\le 1} e^{i\lambda t}\,\frac{\mathrm{d}t}{t} = \int_{\varepsilon\le|t|\le 1} 2i\sin(\lambda t)\,\frac{\mathrm{d}t}{t} = \int_{\lambda\varepsilon\le|t|\le\lambda} 2i\sin t\,\frac{\mathrm{d}t}{t}.$$

By Exercise 2.2.9, we come to the required conclusion.

(b)

$$\left|\int_{\varepsilon\le|t|\le 1} e^{i(\lambda t\pm t^k)}\frac{\mathrm{d}t}{t}\right| = \left|\int_{\varepsilon\le|t|\le 1} e^{i\lambda t}\frac{e^{\pm it^k}-1}{t}+\frac{e^{i\lambda t}}{t}\,\mathrm{d}t\right| \le \left|\int_{\varepsilon\le|t|\le 1}\frac{e^{i\lambda t}}{t}\,\mathrm{d}t\right|+\int_{\varepsilon\le|t|\le 1}|t|^{k-1}\,\mathrm{d}t.$$

By part (a), we come to the required conclusion.

(c) When $\xi_1=\xi_2=0$, it is trivial. When $\xi_2=0$, it follows from the same argument in part (a). When $\xi_2\neq 0$, let $t=|\xi_2|^{\frac{1}{k}}s$, and split the domain of integration into the sets $|t|\ge 1$ and $|t|\le 1$. When $|t|\le 1$ use part (b) and in the case $|t|\ge 1$ use Corollary 2.6.8, noting that

$$\frac{\mathrm{d}^k}{\mathrm{d}t}(\xi_1|\xi_2|^{-\frac{1}{k}}t+t^k)=k!\ge 1.$$

2.6.6 (a)

$$\left|\int_0^{a\lambda} e^{i\lambda\log t}\,\mathrm{d}t\right| = \left|\frac{1}{i\lambda+1}a\lambda e^{i\lambda\log(a\lambda)}\right|\le a.$$

(b) Split $(0,b)$ into $(0,1)$ and $(1,b)$. Apply Proposition 2.6.7 with $k=1$ to the former one, and $k=2$ to the other.

2.6.7 We can assume that $\lambda=1$, for otherwise let $s=\lambda^{\frac{1}{\gamma}}t$. On the interval $(0,\delta)$ apply Proposition 2.6.7 with $k=[\gamma]+1$, and on the interval $(\delta,b)$ with $k=[\gamma]$.

## 1.6 试卷: Fourier 分析

原网页

# 2022 学年期末试卷

一、（15 分）习题 1.3.7.

二、（30 分）习题 2.1.11.

三、（10 分）习题 2.3.10.

四、（15 分）习题 2.2.12.

五、（30 分） $m_\alpha(x) := (1+4\pi^2|x|^2)^{-\alpha}, \alpha \in (0,1)$，求 $m_\alpha$ 的 Fourier 变换（不必给出显式表达），并以此证明 $m_\alpha(x)$ 是一个 $L^p$ Fourier 乘子，$1 \leqslant p \leqslant \infty$.

comment| 可以直接把作业纸交上去；可以提前交卷。 comment| 第五题这样做：由 GTM250 Proposition 1.2.5，$m_\alpha^\vee > 0$，且 $\|m_\alpha^\vee\|_{L^1} = 1$。由 GTM249 Exercise 2.5.8，$\|m_\alpha\|_{\mathscr{M}_p} = 1$。

# 2023 学年期末试卷

一、（20 分）习题 1.1.12.

二、（10 分）习题 1.3.7.

三、（10 分）习题 2.2.14.

四、（5+5 分）习题 2.3.12.

五、（6+9 分）习题 2.5.6.

六、（15 分）对半正定阵 $A$，求 $\exp(-\langle Ax, x\rangle)$ 的 Fourier 变换.

七、（5+15 分）对 $f \in \mathcal{S}$, 以及缓增分布 $u$, 有 $(I-\Delta)u = f$, 证明：$u \in \mathcal{S}$。

对 $1 < p < \infty$, 证明：存在常数 $A_p$, 使得 $\forall f \in \mathcal{C}_c^\infty, \forall 1 \leq i,j \leq n$, 都有 $\|\partial_i\partial_j u\|_{L^p} \leq A_p\|f\|_{L^p}$ 成立.

comment| 第六题见课上习题栏目。